

# Space–Time Algebra

A compact development of paravectors and their physical applications

Dr. Luke A. Wendt

## Introduction: results at a glance

**One complex scalar and one complex three-vector provide a common language for geometry, electromagnetism, and quantum mechanics.** Their product contains both dot and cross products; their determinant gives the spacetime interval. Quaternions generate rotations, the same transformation law gives Lorentz boosts, and four Maxwell equations fit into one line. The Dirac equation factors the free Klein–Gordon equation; their low-energy limits recover Pauli and Schrödinger, including spin and its magnetic interaction. Projectors give both spectral functions and spin measurement probabilities.

This introduction presents the results together, from the algebra to the physical equations and their limits. Links point to background material and the detailed development later in the paper. Natural units  $\hbar = c = 1$ , rationalized electromagnetic units, and signature  $(+, -, -, -)$  are used.

## One complex paravector

$$\begin{array}{ll} \sigma_n^2 = 1, & \sigma_n \sigma_m = -\sigma_m \sigma_n \quad (n \neq m), \\ \mathbf{i} := \sigma_1 \sigma_2 \sigma_3, & \mathbf{i}^2 = -1, \\ \boldsymbol{\sigma} := \{\sigma_1, \sigma_2, \sigma_3\}, & \mathbf{i}\boldsymbol{\sigma} = \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\ Z := (a + b\mathbf{i}) + (\mathbf{A} + \mathbf{i}\mathbf{B}) \cdot \boldsymbol{\sigma}, & a, b \in \mathbb{R}, \quad \mathbf{A}, \mathbf{B} \in \mathbb{R}^3. \end{array}$$

Every element has eight real components: one complex scalar and one complex three-vector.

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma} = X + \mathbf{i}Y, \quad X := a + \mathbf{A} \cdot \boldsymbol{\sigma}, \quad Y := b + \mathbf{B} \cdot \boldsymbol{\sigma}.$$

The component extractions are

$$\begin{aligned} \text{re}(Z) &:= X, \\ \text{im}(Z) &:= Y, \\ \text{sc}(Z) &:= S = a + \mathbf{i}b, \\ \text{vec}(Z) &:= \mathbf{V} = \mathbf{A} + \mathbf{i}\mathbf{B}. \end{aligned}$$

## Dot and cross products in one multiplication

The product carries both the dot and cross products:

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.$$

Throughout,  $\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}$ .

The commutator is

$$[A, B] := AB - BA.$$

## Determinant, inverse, and conjugations

The scalar and vector parts give the determinant, trace, norm, adjugate, and inverse directly.

Write  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ , where  $S \in \mathbb{C}$  is a complex scalar and  $\mathbf{V} \in \mathbb{C}^3$  is a complex vector.

Sigma conjugation  $(\cdot)^*$  complex conjugates scalar coefficients and reverses the sign of  $\boldsymbol{\sigma}$ . Hermitian conjugation  $(\cdot)^H$  reverses the order of products of the generators  $\sigma_k$ .

$$\begin{aligned} Z^* &= S^* - \mathbf{V}^* \cdot \boldsymbol{\sigma}, & Z^H &= S^* + \mathbf{V}^* \cdot \boldsymbol{\sigma}, \\ \text{adj } Z &:= S - \mathbf{V} \cdot \boldsymbol{\sigma} = Z^{*H}, & \det Z &:= S^2 - \mathbf{V}^2 = Z \text{ adj } Z, \\ \text{tr } Z &:= 2S = Z + \text{adj } Z, & \|Z\|^2 &:= SS^* + \mathbf{V} \cdot \mathbf{V}^* = \text{sc}(ZZ^H). \\ Z^{-1} &= \frac{\text{adj } Z}{\det Z} & \text{if } \det Z &\neq 0. \end{aligned}$$

## Quaternions inside the same algebra

In this representation, quaternions have a real scalar part and a purely imaginary vector part:

$$\begin{aligned} q_k &:= -i\sigma_k, & q_1^2 &= q_2^2 = q_3^2 = q_1 q_2 q_3 = -1, \\ Q &:= a + \mathbf{A} \cdot \mathbf{q} = a - i\mathbf{A} \cdot \boldsymbol{\sigma}, & a &\in \mathbb{R}, \quad \mathbf{A} \in \mathbb{R}^3, \\ Q^H &= a + i\mathbf{A} \cdot \boldsymbol{\sigma}, \\ QQ^H &= \det Q = a^2 + \mathbf{A}^2. \end{aligned}$$

## Rotations and boosts in the same form

Represent a spacetime event by  $X$  and take  $\mathbf{u}$  to be a real unit axis:

$$X := t + \mathbf{r} \cdot \boldsymbol{\sigma}, \quad X' := TXT^H = t' + \mathbf{r}' \cdot \boldsymbol{\sigma}.$$

Choosing  $T = R$  gives Rodrigues' rotation through  $\theta$ ;  $R$  is a unit quaternion in the basis  $q_k = -i\sigma_k$ .

$$\begin{aligned} R &:= e^{-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2} = \cos \frac{\theta}{2} - i\mathbf{u} \cdot \boldsymbol{\sigma} \sin \frac{\theta}{2}, \\ t' &= t, \\ \mathbf{r}' &= \mathbf{r} \cos \theta + (\mathbf{u} \times \mathbf{r}) \sin \theta + (\mathbf{u} \cdot \mathbf{r})(1 - \cos \theta)\mathbf{u}. \end{aligned}$$

Choosing  $T = L$  gives a Lorentz boost to a frame moving at  $+\beta\mathbf{u}$ , with rapidity  $\theta$ ,  $\beta := \tanh \theta$ , and Lorentz factor  $\gamma := \cosh \theta$ .

Define the parallel and perpendicular components:

$$\begin{aligned} \mathbf{r}_{\parallel} &:= (\mathbf{u} \cdot \mathbf{r})\mathbf{u}, & \mathbf{r}_{\perp} &:= \mathbf{r} - \mathbf{r}_{\parallel}. \\ L &:= e^{-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2} = \cosh \frac{\theta}{2} - \mathbf{u} \cdot \boldsymbol{\sigma} \sinh \frac{\theta}{2} = L^H, \\ t' &= \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \\ \mathbf{r}' &= \mathbf{r}_{\perp} + \gamma(\mathbf{r}_{\parallel} - \beta t \mathbf{u}). \end{aligned}$$

## Projectors reduce functions to scalar evaluations

$$Z = a + b\mathbf{u} \cdot \boldsymbol{\sigma}, \quad a, b \in \mathbb{C}, \quad \mathbf{u} \in \mathbb{C}^3, \quad \mathbf{u}^2 = 1.$$

The eigenvalues and complementary projectors are

$$\begin{aligned} \lambda_{\pm} &:= a \pm b, & \Pi_{\pm} &:= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi_{\pm}^2 &= \Pi_{\pm}, & \Pi_+ \Pi_- &= 0, & \Pi_- + \Pi_+ &= 1. \end{aligned}$$

For  $f$  analytic near  $\lambda_{\pm}$ :

$$\begin{aligned} Z &= \lambda_- \Pi_- + \lambda_+ \Pi_+, \\ f(Z) &= f(\lambda_-) \Pi_- + f(\lambda_+) \Pi_+. \end{aligned}$$

## Spacetime from the determinant

For a future-directed massive particle, proper time and four-momentum follow from a real paravector, with four-velocity  $U$ .

$$\begin{aligned} dX &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, & ds^2 &:= \det(dX) = dt^2 - d\mathbf{r}^2, \\ U &:= \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), & \gamma &= (1 - \mathbf{v}^2)^{-1/2}, \quad \mathbf{v} := \frac{d\mathbf{r}}{dt}, \\ P &:= mU = \underbrace{E}_{\text{energy}} + \underbrace{\mathbf{p} \cdot \boldsymbol{\sigma}}_{\text{momentum}}, & \det P &= E^2 - \mathbf{p}^2 = m^2. \end{aligned}$$

## Maxwell's four equations in one

Combine the electric and magnetic fields and use the paravector derivative:

$$F := (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad \partial := \partial_t + \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}, \quad \boxed{\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.}$$

Here  $\rho$  is charge density,  $\mathbf{J}$  is current density, and  $\partial_{\mathbf{r}}$  is the gradient. Its divergence and curl give the four components of Maxwell's equations:

$$\begin{aligned} \text{Real scalar:} \quad & \partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \\ \text{Imaginary scalar:} \quad & \partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \\ \text{Real vector:} \quad & \partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B} = -\mathbf{J}, \\ \text{Imaginary vector:} \quad & \partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E} = 0. \end{aligned}$$

Applying  $\partial^* = \partial_t - \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}$  gives the d'Alembertian and the sourced wave equations:

$$\begin{aligned} \square &:= \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2, \\ \partial^*(\partial F) &= \square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Since  $\square F = (\square \mathbf{E} + i\square \mathbf{B}) \cdot \boldsymbol{\sigma}$ , its components give charge conservation and the field wave equations

$$\begin{aligned} \text{Real scalar:} \quad & 0 = \partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} \quad (\text{charge conservation}), \\ \text{Imaginary scalar:} \quad & 0 = 0, \\ \text{Real vector:} \quad & \square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho, \\ \text{Imaginary vector:} \quad & \square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}. \end{aligned}$$

In vacuum,  $\square F = 0$ .

The same field gives the [Lorentz force](#) and its rate of work on a particle of charge  $q$  and velocity  $\mathbf{v}$ :

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad \frac{dE}{dt} = q\mathbf{v} \cdot \mathbf{E}.$$

## Electromagnetic potentials and gauge invariance

Define the real four-potential from the scalar potential  $V$  and vector potential  $\mathbf{A}$ , with gauge scalar  $S$ :

$$\Phi := V + \mathbf{A} \cdot \boldsymbol{\sigma}, \quad S := \text{sc}(\partial \Phi).$$

**Fields from the potential:**

$$\boxed{\partial \Phi = S + F^*}$$

Its components give

$$\begin{aligned} \text{Real scalar:} \quad & S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}, \\ \text{Imaginary scalar:} \quad & 0 = 0, \\ \text{Real vector:} \quad & \mathbf{E} = -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \\ \text{Imaginary vector:} \quad & \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}. \end{aligned}$$

**Source equations in any gauge:**

$$\square\Phi = \partial^*(\partial\Phi) = \partial^*(S + F^*) = \partial^*S + (\partial F)^*.$$

Conjugating and using Maxwell's equation gives

$$\boxed{\square\Phi^* - \partial S = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}}$$

Its components give

$$\begin{aligned} \text{Real scalar: } \quad & \square V - \partial_t S = \rho, \\ \text{Imaginary scalar: } \quad & 0 = 0, \\ \text{Real vector: } \quad & \square \mathbf{A} + \partial_r S = \mathbf{J}, \\ \text{Imaginary vector: } \quad & \mathbf{0} = \mathbf{0}. \end{aligned}$$

**Gauge invariance:** for real scalar  $\lambda$ ,

$$\begin{aligned} \Phi' &:= \Phi + \partial^* \lambda, \\ S' &= S + \square \lambda, \\ F' &= F. \end{aligned}$$

**Lorenz gauge:**  $S = 0$  gives the potential wave equations

$$\begin{aligned} \text{Real scalar: } \quad & \square V = \rho, \\ \text{Real vector: } \quad & \square \mathbf{A} = \mathbf{J}. \end{aligned}$$

## Dirac as a first-order wave equation

In the Weyl representation, write  $\boldsymbol{\Psi} := (\Psi_+, \Psi_-)^\top$ , where the signs label the entries, each with two complex components. Let  $\mathbf{I}$  be the block identity.

Hats mark named energy, momentum, and Dirac operators; the derivative symbols  $\partial$  and  $\square$  remain unhatted.

For a free particle of mass  $m$ , define

$$\begin{aligned} \hat{E} &:= i\partial_t, \\ \hat{\mathbf{p}} &:= -i\partial_r, \\ \hat{P} &:= \hat{E} + \hat{\mathbf{p}} \cdot \boldsymbol{\sigma} = i\partial^*, \\ \mathbf{W}(\hat{P}) &:= \begin{pmatrix} 0 & \text{adj } \hat{P} \\ \hat{P} & 0 \end{pmatrix}, \\ \hat{D}(m) &:= \mathbf{W}(\hat{P}) - m\mathbf{I}. \end{aligned}$$

The free Dirac equation is

$$\boxed{\hat{D}(m)\boldsymbol{\Psi} = 0.}$$

Equivalently,

$$(\hat{E} \pm \hat{\mathbf{p}} \cdot \boldsymbol{\sigma})\Psi_\pm = m\Psi_\mp.$$

For commuting free operators,

$$\hat{P} \text{adj } \hat{P} = \det \hat{P} = \hat{E}^2 - \hat{\mathbf{p}}^2 = -\square.$$

Thus the spacetime mass shell  $\det P = m^2$  becomes

$$(\det \hat{P} - m^2)\Psi = 0 \iff (\square + m^2)\Psi = 0.$$

Using  $\mathbf{W}(\hat{P})^2 = (\det \hat{P})\mathbf{I}$  gives the Klein–Gordon factorization:

$$\begin{aligned} \hat{D}(m)\hat{D}(-m) &= \hat{D}(-m)\hat{D}(m) \\ &= \mathbf{W}(\hat{P})^2 - m^2\mathbf{I} = -(\square + m^2)\mathbf{I}. \end{aligned}$$

Hence every free Dirac solution satisfies Klein–Gordon.

## Dirac with an electromagnetic potential

For constant mass  $m > 0$  and charge  $q$ , minimal coupling to the real potential  $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$  defines the gauge-covariant energy and momentum operators

$$\begin{aligned} \hat{E}_q &:= \hat{E} - qV = \text{i}\partial_t - qV, \\ \hat{\mathbf{p}}_q &:= \hat{\mathbf{p}} - q\mathbf{A} = -\text{i}\partial_{\mathbf{r}} - q\mathbf{A}, \\ \hat{P}_q &:= \hat{E}_q + \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}. \end{aligned}$$

$$\boxed{\hat{P}_q = \hat{P} - q\Phi = \text{i}\partial^* - q\Phi.}$$

At  $q = 0$ , the coupled operators reduce to the canonical ones:  $\hat{E}_0 = \hat{E}$ ,  $\hat{\mathbf{p}}_0 = \hat{\mathbf{p}}$ , and  $\hat{P}_0 = \hat{P}$ . The coupled Dirac operator is

$$\begin{aligned} \hat{D}_q(m) &:= \mathbf{W}(\hat{P}_q) - m\mathbf{I} \\ &= \begin{pmatrix} -m & \text{adj } \hat{P}_q \\ \hat{P}_q & -m \end{pmatrix}. \end{aligned}$$

The coupled Dirac equation is

$$\boxed{\hat{D}_q(m)\Psi = 0.}$$

Equivalently,

$$(\hat{E}_q \pm \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})\Psi_{\pm} = m\Psi_{\mp}.$$

The coupled operators obey these field identities:

$$\begin{aligned} [\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}, \hat{E}_q] &= -\text{i}q\mathbf{E} \cdot \boldsymbol{\sigma}, \\ (\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})^2 &= \hat{\mathbf{p}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}, \\ \hat{\mathbf{p}}_q^2 + \text{i}qF &= (\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})^2 - [\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}, \hat{E}_q]. \end{aligned}$$

When derivatives act on the potentials as well as the wavefunction, the opposite-mass product becomes

$$\hat{D}_q(-m)\hat{D}_q(m) = (\hat{E}_q^2 - \hat{\mathbf{p}}_q^2 - m^2)\mathbf{I} - \text{i}q \begin{pmatrix} F^* & 0 \\ 0 & F \end{pmatrix}.$$

Every Dirac solution satisfies  $\hat{D}_q(-m)\hat{D}_q(m)\Psi = 0$ . Setting  $q = 0$  recovers the free Klein–Gordon equation.

## Low-energy Klein–Gordon recovers Schrödinger

For a charged scalar field of constant mass  $m$ , remove the rest-energy phase,  $\psi' = e^{-imt}\psi$ . The product rule gives

$$\begin{aligned}\hat{E}_q\psi' &= (\mathrm{i}\partial_t - qV)(e^{-imt}\psi) \\ &= e^{-imt}(m\psi + \mathrm{i}\partial_t\psi - qV\psi) \\ &= e^{-imt}(m + \hat{E}_q)\psi, \\ \hat{\mathbf{p}}_q\psi' &= e^{-imt}\hat{\mathbf{p}}_q\psi.\end{aligned}$$

Substitution into the minimally coupled Klein–Gordon equation gives

$$\begin{aligned}0 &= (\hat{E}_q^2 - \hat{\mathbf{p}}_q^2 - m^2)\psi' \\ &= e^{-imt}((m + \hat{E}_q)^2 - \hat{\mathbf{p}}_q^2 - m^2)\psi \\ &= e^{-imt}(2m\hat{E}_q + \hat{E}_q^2 - \hat{\mathbf{p}}_q^2)\psi.\end{aligned}$$

For any  $q$  and constant  $m > 0$ , canceling the phase gives the exact recursion:

$$\hat{E}_q\psi = \frac{\hat{\mathbf{p}}_q^2 - \hat{E}_q^2}{2m}\psi.$$

If  $\hat{E}_q$  and  $\hat{\mathbf{p}}_q^2$  commute, iteration on the positive-energy branch gives

$$\hat{E}_q\psi = \left( \frac{\hat{\mathbf{p}}_q^2}{2m} - \frac{\hat{\mathbf{p}}_q^4}{8m^3} + \dots \right) \psi.$$

For weak, slowly varying fields, neglect  $\hat{E}_q^2\psi$  relative to  $2m\hat{E}_q\psi$  to obtain Schrödinger’s equation:

$$\boxed{\mathrm{i}\partial_t\psi = \left( \frac{(-\mathrm{i}\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \psi.}$$

With  $\mathbf{A} = 0$  (hence  $\mathbf{B} = 0$ ), this recovers the conventional Schrödinger equation:

$$\mathrm{i}\partial_t\psi = \left( -\frac{\partial_{\mathbf{r}}^2}{2m} + qV \right) \psi.$$

## Low-energy Dirac recovers Pauli

Using the same phase shift,  $\Psi'_\pm := e^{imt}\Psi_\pm$ , gives

$$(m + \hat{E}_q \pm \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})\Psi'_\pm = m\Psi'_\mp.$$

Define the large and small components for the positive-energy branch:

$$\begin{aligned}\phi_+ &:= \frac{\Psi'_+ + \Psi'_-}{\sqrt{2}}, \\ \phi_- &:= \frac{\Psi'_- - \Psi'_+}{\sqrt{2}}.\end{aligned}$$

Adding and subtracting the envelope equations gives

$$\begin{aligned}\hat{E}_q\phi_+ &= (\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})\phi_-, \\ (2m + \hat{E}_q)\phi_- &= (\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})\phi_+.\end{aligned}$$

In the same nonrelativistic limit, neglect  $\hat{E}_q\phi_-$  relative to  $2m\phi_-$ . Substitute the resulting small component into the first equation and use the field identity above:

$$\begin{aligned}\phi_- &\simeq \frac{\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}}{2m}\phi_+, \\ \hat{E}_q\phi_+ &\simeq \frac{(\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})^2}{2m}\phi_+ = \frac{\hat{\mathbf{p}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m}\phi_+.\end{aligned}$$

Define  $\psi := \phi_+$ . Restoring  $\hat{E}_q = i\partial_t - qV$  gives the Pauli equation, retaining the kinetic and spin terms through order  $1/m$ :

$$\boxed{i\partial_t\psi = \left( \frac{\hat{\mathbf{p}}_q^2}{2m} + qV - \frac{q}{2m}\mathbf{B} \cdot \boldsymbol{\sigma} \right)\psi.}$$

The Pauli magnetic term is  $-(q/m)\mathbf{B} \cdot (\boldsymbol{\sigma}/2)$ , giving  $g = 2$  before radiative corrections.

## Spin up and spin down in a uniform magnetic field

For a nonzero, constant magnetic field  $\mathbf{B}$ , define its **unit direction** and **spin projectors**:

$$\begin{aligned}\mathbf{u} &:= \frac{\mathbf{B}}{\|\mathbf{B}\|}, & \Pi_{\pm} &:= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \psi_{\pm} &:= \Pi_{\pm} \psi, & \psi &= \psi_+ + \psi_-.\end{aligned}$$

The projector identity gives **spin**  $\pm \frac{1}{2}$  along  $\mathbf{u}$ :

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi_{\pm} = \pm \Pi_{\pm}, \quad \frac{\mathbf{u} \cdot \boldsymbol{\sigma}}{2}\psi_{\pm} = \pm \frac{1}{2}\psi_{\pm}.$$

Since the projectors are constant, they commute with  $\hat{E}_q$  and  $\hat{\mathbf{p}}_q^2$ . Projecting the Pauli equation gives

$$\begin{aligned}2m\hat{E}_q\psi_{\pm} &= \Pi_{\pm}(\hat{\mathbf{p}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma})\psi \\ &= (\hat{\mathbf{p}}_q^2 \mp q\|\mathbf{B}\|)\psi_{\pm}.\end{aligned}$$

Thus the two spin components evolve independently, with opposite **Zeeman shifts**:

$$\boxed{\mathrm{i}\partial_t\psi_{\pm} = \left( \frac{\hat{\mathbf{p}}_q^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \psi_{\pm}.}$$

The **kinetic momentum**  $\hat{\mathbf{p}}_q = -\mathrm{i}\partial_{\mathbf{r}} - q\mathbf{A}$  retains the orbital coupling.

## Spin measurement probabilities

Let  $\mathbf{u}'$  be the real unit spin direction and  $\mathbf{u}$  the real unit measurement axis. Prepare  $\psi = \Pi'_+ \psi \neq 0$ , with  $\cos \theta := \mathbf{u} \cdot \mathbf{u}'$ :

$$\begin{aligned}\Pi'_+ &:= \frac{1}{2}(1 + \mathbf{u}' \cdot \boldsymbol{\sigma}), & \Pi_{\pm} &:= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi'_+ \Pi_{\pm} \Pi'_+ &= \frac{1 \pm \cos \theta}{2} \Pi'_+.\end{aligned}$$

The Born probabilities for spin  $\pm \frac{1}{2}$  are

$$\text{prob}(\pm) = \frac{\|\Pi_{\pm} \psi\|^2}{\|\psi\|^2} = \frac{1 \pm \cos \theta}{2},$$

$$\boxed{\text{prob}(+) = \cos^2 \frac{\theta}{2}, \quad \text{prob}(-) = \sin^2 \frac{\theta}{2}.}$$

Aligned axes give certain spin up; perpendicular axes give equal odds. Reversing the prepared spin swaps the probabilities.

A rotation through  $\varphi$  about a real unit axis  $\mathbf{n}$  acts on a two-component spinor as  $\psi \mapsto R\psi$ , with  $R = e^{-i\varphi \mathbf{n} \cdot \boldsymbol{\sigma}/2}$ . It changes sign under a  $2\pi$  rotation and returns after  $4\pi$ , while the measurement probabilities are unchanged by the overall sign.

The sections that follow develop the algebra, derive these relationships, and give the physical assumptions and source comparisons. The equation reference in Appendix B provides a compact index.

# Contents

|                                                         |           |
|---------------------------------------------------------|-----------|
| <b>Introduction: results at a glance</b>                | <b>i</b>  |
| <b>1 The algebra at a glance</b>                        | <b>1</b>  |
| <b>2 Hermitian and sigma conjugation</b>                | <b>2</b>  |
| <b>3 Paravectors</b>                                    | <b>3</b>  |
| 3.1 Core operations . . . . .                           | 3         |
| 3.2 Unit vectors and paravectors . . . . .              | 4         |
| 3.3 Products and quadratic forms . . . . .              | 6         |
| 3.4 Quadratic identity and eigenvalues . . . . .        | 6         |
| 3.5 Real products . . . . .                             | 7         |
| 3.6 Complex products . . . . .                          | 7         |
| 3.7 Projection operators . . . . .                      | 8         |
| 3.8 Functions of paravectors . . . . .                  | 9         |
| <b>4 Transformations and invariants</b>                 | <b>15</b> |
| 4.1 Similarity, determinant, and norm . . . . .         | 15        |
| 4.2 Real paravectors and the Lorentz metric . . . . .   | 16        |
| 4.3 Rotation action and Rodrigues' formula . . . . .    | 18        |
| 4.4 Quaternions as paravectors . . . . .                | 18        |
| 4.5 Boost action and its sign convention . . . . .      | 19        |
| <b>5 Differentiated fields</b>                          | <b>20</b> |
| 5.1 Paravector differential operators . . . . .         | 20        |
| 5.2 Components and the product rule . . . . .           | 21        |
| <b>6 Space–time kinematics</b>                          | <b>21</b> |
| 6.1 Intervals, velocity, and momentum . . . . .         | 22        |
| 6.2 Proper acceleration . . . . .                       | 23        |
| 6.3 World lines, clocks, and light cones . . . . .      | 24        |
| 6.4 Free-particle action . . . . .                      | 26        |
| 6.5 Flat spacetime in generalized coordinates . . . . . | 27        |
| 6.6 Varying frames and local bases . . . . .            | 27        |
| <b>7 Electromagnetism</b>                               | <b>28</b> |
| 7.1 Maxwell's equations and wave propagation . . . . .  | 28        |
| 7.2 Potentials and gauge freedom . . . . .              | 29        |
| 7.3 The electromagnetic field action . . . . .          | 30        |
| 7.4 Field invariants . . . . .                          | 30        |
| 7.5 Lorentz force from the action . . . . .             | 31        |
| 7.6 Acceleration and the mass shell . . . . .           | 32        |
| 7.7 Energy and momentum flux . . . . .                  | 33        |

|           |                                                         |           |
|-----------|---------------------------------------------------------|-----------|
| <b>8</b>  | <b>Relativistic wave equations</b>                      | <b>33</b> |
| 8.1       | Amplitudes, densities, and local expectations . . . . . | 33        |
| 8.2       | Energy and momentum as differential operators . . . . . | 35        |
| 8.3       | Klein–Gordon equation . . . . .                         | 37        |
| 8.4       | Dirac equation . . . . .                                | 38        |
| <b>9</b>  | <b>Low-energy limits</b>                                | <b>43</b> |
| 9.1       | Dirac to Pauli . . . . .                                | 43        |
| 9.2       | The same leading order: Schrödinger . . . . .           | 44        |
| 9.3       | The next free order . . . . .                           | 45        |
| <b>10</b> | <b>Spin and probability</b>                             | <b>45</b> |
| 10.1      | Spin one-half and rotations . . . . .                   | 45        |
| 10.2      | Projectors and probabilities . . . . .                  | 46        |
| <b>11</b> | <b>Rotations and Lorentz boosts</b>                     | <b>48</b> |
| 11.1      | Covariance of the physical equations . . . . .          | 48        |
| 11.2      | Moving frames, clock rates, and lengths . . . . .       | 49        |
| 11.3      | Boosted fields and energy flux . . . . .                | 51        |
| <b>A</b>  | <b>Pauli-matrix representation</b>                      | <b>53</b> |
| A.1       | Generators and products . . . . .                       | 53        |
| A.2       | Conjugations and product order . . . . .                | 53        |
| A.3       | Scalar part, determinant, and norm . . . . .            | 54        |
| A.4       | Boost and rotation matrices . . . . .                   | 55        |
| A.5       | Matrix amplitudes and column spinors . . . . .          | 55        |
| A.6       | Dirac columns . . . . .                                 | 56        |
| A.7       | Four components and the standard Dirac basis . . . . .  | 57        |
| A.8       | Quaternion matrix image . . . . .                       | 58        |
| <b>B</b>  | <b>Equation reference</b>                               | <b>59</b> |
| <b>C</b>  | <b>Conventions</b>                                      | <b>63</b> |
|           | <b>References</b>                                       | <b>69</b> |

# 1 The algebra at a glance

The generators  $\sigma_n$ ,  $n = 1, 2, 3$ , obey ordered multiplication rules. Their products generally do not commute. The panel gives their defining relations and the general complex element [1, pp. 4–5]:

$$\begin{array}{ll} \sigma_n^2 = 1, & \sigma_n \sigma_m = -\sigma_m \sigma_n \quad (n \neq m), \\ \mathbf{i} = \sigma_1 \sigma_2 \sigma_3, & \mathbf{i}^2 = -1, \\ \boldsymbol{\sigma} = \{\sigma_1, \sigma_2, \sigma_3\}, & \mathbf{i}\boldsymbol{\sigma} = \{\sigma_2 \sigma_3, \sigma_3 \sigma_1, \sigma_1 \sigma_2\}, \\ Z = (a + b\mathbf{i}) + (\mathbf{A} + \mathbf{B}\mathbf{i}) \cdot \boldsymbol{\sigma}, & a, b \in \mathbb{R}, \quad \mathbf{A}, \mathbf{B} \in \mathbb{R}^3. \end{array}$$

Ordinary three-vector operations use

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= A_1 B_1 + A_2 B_2 + A_3 B_3, \\ \mathbf{A} \times \mathbf{B} &= \{A_2 B_3 - A_3 B_2, A_3 B_1 - A_1 B_3, A_1 B_2 - A_2 B_1\}, \\ \mathbf{A}^2 &:= \mathbf{A} \cdot \mathbf{A}. \end{aligned}$$

Thus the square of a vector always denotes its self dot product. For complex vectors, dot and cross products use the same formulas, without conjugating either factor. The elementary Cartesian vectors used below are the real vectors

$$\mathbf{e}_1 = \{1, 0, 0\}, \quad \mathbf{e}_2 = \{0, 1, 0\}, \quad \mathbf{e}_3 = \{0, 0, 1\}.$$

Thus  $\mathbf{e}_k \cdot \boldsymbol{\sigma} = \sigma_k$ ; the scalar basis element is 1. Unindexed  $e$  denotes only Euler's number.

These eight elements form a complete basis, displayed by type as

$$\underbrace{1}_{\text{scalar}} \quad \underbrace{\sigma_1, \sigma_2, \sigma_3}_{\text{vectors}} \quad \underbrace{\mathbf{i}\sigma_1, \mathbf{i}\sigma_2, \mathbf{i}\sigma_3}_{\text{bivectors}} \quad \underbrace{\mathbf{i}}_{\text{pseudoscalar}}.$$

Anticommutation gives  $\mathbf{i}^2 = -\sigma_1^2 \sigma_2^2 \sigma_3^2 = -1$ . The pseudoscalar is central: moving  $\sigma_n$  through the other two generators produces two sign changes, so

$$\mathbf{i}\sigma_n = \sigma_n \mathbf{i}.$$

Thus its order of multiplication does not matter.

For algebra elements  $X, Y$ , the commutator is

$$\begin{aligned} [X, Y] &:= XY - YX, \\ [Y, X] &= -[X, Y]. \end{aligned}$$

Thus  $[X, Y] = 0$  exactly when  $X$  and  $Y$  commute. For  $n, m, k = 1, 2, 3$ , define the Kronecker delta by

$$\delta_{nm} := \begin{cases} 1 & \text{if } n = m, \\ 0 & \text{if } n \neq m, \end{cases}$$

and the Levi–Civita symbol by

$$\begin{aligned}\epsilon_{nmk} &:= +1 && \text{if } (n, m, k) \text{ is an even permutation of } (1, 2, 3), \\ &:= -1 && \text{if } (n, m, k) \text{ is an odd permutation of } (1, 2, 3), \\ &:= 0 && \text{if two or more indices are equal.}\end{aligned}$$

For  $n, m = 1, 2, 3$ , the ordered multiplication rule and commutator are [1, Eq. (3)] [1, Eq. (10)]

$$\begin{aligned}\sigma_n \sigma_m &= \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k, \\ [\sigma_n, \sigma_m] &= 2 \sum_{k=1}^3 \epsilon_{nmk} \mathbf{i} \sigma_k.\end{aligned}\tag{1.1}$$

## 2 Hermitian and sigma conjugation

Define *Hermitian conjugation* by (2.1) and *sigma conjugation* by (2.2) [2, Eq. (3)]:

$$(\cdot)^{\mathbf{H}} := \text{reverse the order of every product of sigma generators},\tag{2.1}$$

$$(\cdot)^* := \text{replace every } \sigma_n \text{ by } -\sigma_n \text{ without changing product order}.\tag{2.2}$$

Thus, for two generators,

$$\begin{aligned}(\sigma_m \sigma_n)^{\mathbf{H}} &= \sigma_n \sigma_m, \\ (\sigma_m \sigma_n)^* &= (-\sigma_m)(-\sigma_n).\end{aligned}$$

Both operations act term by term on sums and leave real coefficients fixed. Their actions on the four basis types are

| type<br>representative   | scalar<br>1 | vector<br>$\sigma_n$ | bivector<br>$\mathbf{i} \sigma_n$ | pseudoscalar<br>$\mathbf{i}$ |
|--------------------------|-------------|----------------------|-----------------------------------|------------------------------|
| $(\cdot)^*$              | +           | −                    | +                                 | −                            |
| $(\cdot)^{\mathbf{H}}$   | +           | +                    | −                                 | −                            |
| $(\cdot)^{* \mathbf{H}}$ | +           | −                    | −                                 | +                            |

For example, on a bivector the signs follow directly:

$$\begin{aligned}(\mathbf{i} \sigma_n)^* &= \mathbf{i}^* \sigma_n^* = \mathbf{i} \sigma_n, \\ (\mathbf{i} \sigma_n)^{\mathbf{H}} &= \sigma_n^{\mathbf{H}} \mathbf{i}^{\mathbf{H}} = -\mathbf{i} \sigma_n.\end{aligned}$$

For arbitrary algebra elements  $Z, Z'$ , the definitions give the product laws [1, p. 4] [2, p. 2]

$$(ZZ')^{\mathbf{H}} = Z'^{\mathbf{H}} Z^{\mathbf{H}},\tag{2.3}$$

$$(ZZ')^* = Z^* Z'^*.\tag{2.4}$$

The operations commute with each other:  $Z^{*\mathbf{H}} = Z^{\mathbf{H}*}$ . Appendix A gives both operations in the Pauli-matrix representation, where  $^{\mathbf{H}}$  is the usual matrix Hermitian conjugation.

### 3 Paravectors

#### 3.1 Core operations

Let  $a, b \in \mathbb{R}$  and  $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$ . The real paravectors  $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$  and  $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$  form the general complex paravector

$$Z = X + \mathrm{i}Y = (a + \mathrm{b}i) + (\mathbf{A} + \mathrm{i}\mathbf{B}) \cdot \boldsymbol{\sigma} = S + \mathbf{V} \cdot \boldsymbol{\sigma}.$$

Here  $S = a + \mathrm{b}i \in \mathbb{C}$  and  $\mathbf{V} = \mathbf{A} + \mathrm{i}\mathbf{B} \in \mathbb{C}^3$  are its complex scalar and vector coefficients. Define their extraction functions by

$$\begin{aligned} \mathrm{sc}(Z) &:= S, \\ \mathrm{vec}(Z) &:= \mathbf{V}. \end{aligned}$$

This notation applies throughout the algebra unless a real specialization is stated. On the coefficients, both conjugations give  $S^* = S^{\mathrm{H}} = a - \mathrm{b}i$  and  $\mathbf{V}^* = \mathbf{V}^{\mathrm{H}} = \mathbf{A} - \mathrm{i}\mathbf{B}$ ; on the paravector, the two conjugations differ in the vector sign [2, Eq. (45)]:

$$Z^* = S^* - \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - \mathrm{b}i) + (-\mathbf{A} + \mathrm{i}\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.1)$$

$$Z^{\mathrm{H}} = S^* + \mathbf{V}^* \cdot \boldsymbol{\sigma} = (a - \mathrm{b}i) + (\mathbf{A} - \mathrm{i}\mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (3.2)$$

Their composition defines the adjugate [2, Eq. (39)]

$$\mathrm{adj}(Z) := Z^{*\mathrm{H}} = S - \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (3.3)$$

The trace is twice the scalar part:

$$\mathrm{tr}(Z) := Z + \mathrm{adj}(Z) = 2\mathrm{sc}(Z) = 2S. \quad (3.4)$$

The determinant is the complex scalar [3, p. 104]

$$\det(Z) := \mathrm{adj}(Z) Z = Z \mathrm{adj}(Z) = S^2 - \mathbf{V}^2. \quad (3.5)$$

For complex  $Z'$ , adjugation reverses product order, giving determinant multiplicativity:

$$\begin{aligned} \mathrm{adj}(ZZ') &= \mathrm{adj}(Z') \mathrm{adj}(Z), \\ \det(ZZ') &= \mathrm{adj}(Z') (\mathrm{adj}(Z) Z) Z' = \det(Z) \det(Z'). \end{aligned}$$

Since the determinants are scalars, the determinant of any product is unchanged by reordering its factors. For a complex scalar  $c$ ,

$$\begin{aligned} \mathrm{adj}(cZ) &= c \mathrm{adj}(Z), \\ \det(cZ) &= c^2 \det(Z). \end{aligned} \quad (3.6)$$

When  $\det(Z) \neq 0$ , the inverse is [2, Eq. (47)]

$$Z^{-1} = \frac{\mathrm{adj}(Z)}{\det(Z)} = \frac{S - \mathbf{V} \cdot \boldsymbol{\sigma}}{S^2 - \mathbf{V}^2}. \quad (3.7)$$

For  $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$ , with complex coefficients, division is order-sensitive. An unambiguous quotient exists when  $Z$  is invertible and the factors commute:

$$\frac{Z'}{Z} := Z'Z^{-1} = Z^{-1}Z' \quad \text{if } \mathbf{V} \times \mathbf{V}' = 0.$$

Differentiating  $Z^{-1}Z = 1$  gives the inverse rule

$$d(Z^{-1}) = -Z^{-1}(dZ)Z^{-1}.$$

For complex  $Z, Z', Z''$ , scalar part obeys the cyclic laws

$$\begin{aligned} \text{sc}(ZZ') &= \text{sc}(Z'Z), \\ \text{sc}(ZZ'Z'') &= \text{sc}(Z'Z''Z) = \text{sc}(Z''ZZ'). \end{aligned} \tag{3.8}$$

Multiplying these identities by 2 gives trace cyclicity. Their similarity invariants are derived in Section 4.

The real and imaginary extractions use Hermitian conjugation [1, Eq. (9)]:

$$\text{re}(Z) := \frac{1}{2}(Z + Z^H) = X, \tag{3.9}$$

$$\text{im}(Z) := \frac{Z - Z^H}{2i} = Y. \tag{3.10}$$

Adjugation evaluates the scalar and vector extractions explicitly [1, Eq. (14)]:

$$\begin{aligned} \text{sc}(Z) &= \frac{1}{2}(Z + \text{adj}(Z)) = S, \\ (\text{vec}(Z)) \cdot \boldsymbol{\sigma} &= \frac{1}{2}(Z - \text{adj}(Z)) = \mathbf{V} \cdot \boldsymbol{\sigma}. \end{aligned}$$

The functions  $\text{re}$  and  $\text{im}$  also act on complex coefficients. Composing extractions recovers all four real parts:

$$\begin{aligned} a &= \text{re}(S) = \text{sc}(\text{re}(Z)), \\ b &= \text{im}(S) = \text{sc}(\text{im}(Z)), \\ \mathbf{A} &= \text{re}(\mathbf{V}) = \text{vec}(\text{re}(Z)), \\ \mathbf{B} &= \text{im}(\mathbf{V}) = \text{vec}(\text{im}(Z)). \end{aligned}$$

Appendix A reproduces these intrinsic operations with Pauli matrices.

### 3.2 Unit vectors and paravectors

Recall  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ , with  $S \in \mathbb{C}$  and  $\mathbf{V} \in \mathbb{C}^3$ . For  $\mathbf{C} \in \mathbb{C}^3$ , distinguish the square from the positive norm:

$$\begin{aligned} \mathbf{C}^2 &:= \mathbf{C} \cdot \mathbf{C}, \\ \|\mathbf{C}\|^2 &:= \mathbf{C} \cdot \mathbf{C}^*. \end{aligned}$$

For  $S \in \mathbb{C}$ , set  $\|S\|^2 := SS^*$ . The positive paravector norm is [2, Eq. (46)]

$$\|Z\|^2 := \|S\|^2 + \|\mathbf{V}\|^2. \tag{3.11}$$

All norms denote nonnegative real square roots. Here  $\mathbf{u} \in \mathbb{C}^3$  and  $Z$  is a complex paravector. Each name below specifies a condition; a letter alone implies none.

- **Vector of square one.**  $\mathbf{u}$  satisfies

$$\mathbf{u}^2 = \mathbf{u} \cdot \mathbf{u} = 1. \quad (3.12)$$

For  $C^2 \neq 0$ , take  $\mathbf{u} = \mathbf{C}/\sqrt{C^2}$  with either square root. Its norm need not be one. A nonzero vector with  $C^2 = 0$  cannot be normalized this way.

- **Vector of unit norm.**  $\mathbf{u}$  satisfies

$$\|\mathbf{u}\|^2 = \mathbf{u} \cdot \mathbf{u}^* = 1.$$

For any  $\mathbf{C} \neq 0$ , take  $\mathbf{u} = \mathbf{C}/\|\mathbf{C}\|$ ; then  $\mathbf{u}^2 = C^2/\|\mathbf{C}\|^2$  need not be one. For real vectors the two conditions coincide: a *real unit vector* has  $\mathbf{u}^2 = \|\mathbf{u}\|^2 = 1$ . For complex vectors, specify which holds.

- **Unit paravector.**  $Z$  has an inverse on both sides:

$$ZZ^{-1} = Z^{-1}Z = 1. \quad (3.13)$$

Equivalently,  $\det(Z) \neq 0$ .

- **Unitary paravector.** Its Hermitian conjugate is its inverse:

$$Z^H Z = 1. \quad (3.14)$$

Then  $Z^H = Z^{-1}$  and  $ZZ^H = 1$ .

- **Paravector of determinant one.**  $Z$  satisfies

$$\det(Z) = 1. \quad (3.15)$$

It is a unit with  $Z^{-1} = \text{adj}(Z)$ ; it need not be unitary. For real  $Z$ ,  $\text{adj}(Z) = Z^*$ , so this condition is also  $Z^*Z = 1$ . This equivalence need not hold for complex  $Z$ .

- **Paravector of unit norm.**  $Z$  satisfies

$$\|Z\|^2 = 1. \quad (3.16)$$

For  $Z \neq 0$ ,  $Z' = Z/\|Z\|$  has unit norm. It need not have an inverse:  $(1 + \mathbf{u} \cdot \boldsymbol{\sigma})/\sqrt{2}$  has norm one and determinant zero for any real unit vector  $\mathbf{u}$ . Unitarity implies unit norm by (3.25).

- **Self-inverse paravector.**  $Z$  is its own inverse:

$$Z^2 = 1. \quad (3.17)$$

Then  $Z^{-1} = Z$ ; it is unitary exactly when  $Z^H = Z$ . Its parts obey  $S^2 + \mathbf{V}^2 = 1$  and  $2S\mathbf{V} = 0$ : thus  $Z = \pm 1$  or  $Z = \mathbf{u} \cdot \boldsymbol{\sigma}$  with  $\mathbf{u}^2 = 1$ .

Two complex vectors illustrate why the normalizations differ:

| $\mathbf{C}$         | $C^2$ | $\ \mathbf{C}\ ^2$ |
|----------------------|-------|--------------------|
| $\{\sqrt{2}, i, 0\}$ | 1     | 3                  |
| $\{1, i, 0\}$        | 0     | 2                  |

Their vectors of unit norm have squares 1/3 and 0, respectively. The physical roles of the paravector conditions are compared in Section 4.

### 3.3 Products and quadratic forms

Here  $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$  are real coefficient vectors. The sigma rule (1.1) gives the dot-cross identity [2, Eq. (38)]

$$\begin{aligned}
 (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \sigma_n \sigma_m \\
 &= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \delta_{nm} + \sum_{n=1}^3 \sum_{m=1}^3 \sum_{k=1}^3 A_n B_m \epsilon_{nmk} i \sigma_k \\
 &= \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.
 \end{aligned} \tag{3.18}$$

Adding the product with  $\mathbf{A}$  and  $\mathbf{B}$  interchanged gives

$$2(\mathbf{A} \cdot \mathbf{B}) = (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}).$$

Consequently, for every nonnegative integer  $n$ ,

$$\begin{aligned}
 (\mathbf{A} \cdot \boldsymbol{\sigma})^2 &= \mathbf{A}^2, \\
 (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{A}^2)^n, \\
 (\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} &= (\mathbf{A}^2)^n \mathbf{A} \cdot \boldsymbol{\sigma}.
 \end{aligned}$$

Also,

$$(\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} = \mathbf{A} - i\mathbf{A} \times \boldsymbol{\sigma}.$$

Subtracting the two orders gives

$$[\mathbf{A} \cdot \boldsymbol{\sigma}, \mathbf{B} \cdot \boldsymbol{\sigma}] = 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.$$

Hence

$$\begin{aligned}
 [\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] &= -2i\mathbf{A} \times \boldsymbol{\sigma}, \\
 [\mathbf{A} \cdot i\boldsymbol{\sigma}, \boldsymbol{\sigma}] &= 2\mathbf{A} \times \boldsymbol{\sigma}, \\
 [\mathbf{A} \cdot \boldsymbol{\sigma}, \boldsymbol{\sigma}] \cdot \mathbf{B} &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.
 \end{aligned}$$

### 3.4 Quadratic identity and eigenvalues

Recall  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ , with  $S \in \mathbb{C}$  and  $\mathbf{V} \in \mathbb{C}^3$ . Since  $Z - S = \mathbf{V} \cdot \boldsymbol{\sigma}$  and  $(\mathbf{V} \cdot \boldsymbol{\sigma})^2 = \mathbf{V}^2$ ,

$$\begin{aligned}
 (Z - S)^2 &= \mathbf{V}^2, \\
 Z^2 - 2SZ + (S^2 - \mathbf{V}^2) &= 0.
 \end{aligned}$$

Using  $\text{tr}(Z) = 2S$  and  $\det(Z) = S^2 - \mathbf{V}^2$  gives the trace-determinant identity

$$Z^2 - \text{tr}(Z)Z + \det(Z) = 0. \tag{3.19}$$

An eigenvalue is a scalar  $\lambda \in \mathbb{C}$  for which  $Z - \lambda$  has no inverse. Thus  $\det(Z - \lambda) = \lambda^2 - 2S\lambda + \det(Z) = 0$ , the scalar quadratic corresponding to (3.19). Its roots are

$$\lambda_{\pm} = \frac{\text{tr}(Z) \pm \sqrt{\text{tr}(Z)^2 - 4\det(Z)}}{2} = S \pm \sqrt{\mathbf{V}^2}.$$

Because  $\mathbf{V}$  is generally complex,  $\mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$  and its square root may also be complex. The quadratic identity therefore factors as

$$(Z - \lambda_-)(Z - \lambda_+) = 0.$$

### 3.5 Real products

For real  $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$  and  $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$ , with  $a, b \in \mathbb{R}$  and  $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$ , (3.18) gives

$$\begin{aligned} XY &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + \mathbf{i}\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ [X, Y] &= 2\mathbf{i}(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ X^2 &= (a^2 + \mathbf{A}^2) + 2a\mathbf{A} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Their scalar and vector parts are read directly from these equations. The real quadratic identity is (3.19) with  $S = a$ . If  $\mathbf{A} \times \mathbf{B} = 0$ , then  $XY = YX$ ; when  $\det(X) \neq 0$ , also  $X^{-1}Y = YX^{-1}$ .

### 3.6 Complex products

Let  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$  and  $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$ , where  $S, S' \in \mathbb{C}$  and  $\mathbf{V}, \mathbf{V}' \in \mathbb{C}^3$ . Complex bilinearity extends (3.18) to the general product

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + \mathbf{i}\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}. \quad (3.20)$$

Thus

$$\begin{aligned} \text{sc}(ZZ') &= SS' + \mathbf{V} \cdot \mathbf{V}', \\ \text{vec}(ZZ') &= S\mathbf{V}' + S'\mathbf{V} + \mathbf{i}\mathbf{V} \times \mathbf{V}'. \end{aligned}$$

Equivalently, write  $Z = X + \mathbf{i}Y$  and  $Z' = X' + \mathbf{i}Y'$  with all four factors real. Their products need not be real, but distributivity gives

$$ZZ' = (XX' - YY') + \mathbf{i}(XY' + YX').$$

Write  $Z = (a + bi) + (\mathbf{A} + \mathbf{i}\mathbf{B}) \cdot \boldsymbol{\sigma}$  with real  $a, b, \mathbf{A}, \mathbf{B}$ , and use primes for the second paravector's real components. The four real components of  $ZZ'$  are

$$\begin{aligned} \text{real scalar: } & aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}', \\ \text{imaginary scalar: } & ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}', \\ \text{real vector: } & a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}', \\ \text{imaginary vector: } & a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'. \end{aligned}$$

Specializing (3.20) gives the quadratic products:

$$\begin{aligned} Z^2 &= (a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2) + 2\mathbf{i}(ab + \mathbf{A} \cdot \mathbf{B}) \\ &\quad + 2(a\mathbf{A} - b\mathbf{B}) \cdot \boldsymbol{\sigma} + 2\mathbf{i}(a\mathbf{B} + b\mathbf{A}) \cdot \boldsymbol{\sigma}, \end{aligned} \quad (3.21)$$

$$ZZ^* = (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) - 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2\mathbf{i}(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.22)$$

$$Z^*Z = (a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2) + 2(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma} + 2\mathbf{i}(a\mathbf{B} - b\mathbf{A}) \cdot \boldsymbol{\sigma}, \quad (3.22)$$

$$ZZ^H = a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.23)$$

$$Z^H Z = a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 + 2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (3.24)$$

$$ZZ^{*H} = \det(Z) = a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2 + 2\mathbf{i}(ab - \mathbf{A} \cdot \mathbf{B}).$$

Equations (3.21) and (3.22) show  $ZZ^* = Z^*Z$  exactly when  $\mathbf{A} \times \mathbf{B} = 0$ . Equations (3.23) and (3.24) give the same condition for  $ZZ^H = Z^H Z$ .

### 3.6.1 Norm and determinant

For the complex coefficients  $S = a + bi$  and  $\mathbf{V} = \mathbf{A} + i\mathbf{B}$ , the norms defined above give

$$\begin{aligned}\|S\|^2 &= a^2 + b^2, \\ \|\mathbf{V}\|^2 &= \mathbf{A}^2 + \mathbf{B}^2.\end{aligned}$$

For the positive norm (3.11), the quadratic products and cyclicity give the Hermitian form [2, Eq. (46)]

$$\begin{aligned}\|Z\|^2 &= \text{sc}(Z^H Z) = \frac{1}{2} \text{tr}(Z^H Z) = \text{sc}(Z Z^H) \\ &= \text{sc}(Z^* Z^{*H}) = \text{sc}(Z^{*H} Z^*).\end{aligned}\tag{3.25}$$

For complex coefficients,  $S^2$  and  $\mathbf{V}^2$  generally differ from these positive norms. Also  $\det(Z)$  may be complex. For a complex scalar  $c = re^{i\theta}$ , with real  $r \geq 0$  and  $\theta$ , scalar scaling gives

$$\begin{aligned}\det(cZ) &= r^2 e^{2i\theta} \det(Z), \\ \|cZ\|^2 &= \|c\|^2 \|Z\|^2 = r^2 \|Z\|^2.\end{aligned}\tag{3.26}$$

The determinant retains the scalar phase by (3.6); the norm loses it by (3.26).

### 3.7 Projection operators

For a [real unit vector](#)  $\mathbf{u}$ , the paravector  $\mathbf{u} \cdot \boldsymbol{\sigma}$  is Hermitian and [self-inverse](#). Define its two orthogonal projectors [3, p. 121]

$$\Pi_{\pm} := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}).\tag{3.27}$$

Since  $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$ ,

$$\begin{aligned}\Pi_{\pm}^2 &= \Pi_{\pm}, \\ \Pi_+ \Pi_- &= 0, \\ \Pi_+ + \Pi_- &= 1, \\ \Pi_{\pm}^H &= \Pi_{\pm}, \\ \Pi_{\pm}^* &= \Pi_{\mp}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma}) \Pi_{\pm} &= \pm \Pi_{\pm}.\end{aligned}$$

Thus  $\mathbf{u} \cdot \boldsymbol{\sigma} = \Pi_+ - \Pi_-$ . For a second [real unit vector](#)  $\mathbf{u}'$ ,

$$\begin{aligned}(\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) &= \mathbf{u} \cdot \mathbf{u}' + i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ [\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}] &= 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= (2(\mathbf{u} \cdot \mathbf{u}')\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}.\end{aligned}$$

Define  $\mathbf{u}'_{\parallel} = (\mathbf{u}' \cdot \mathbf{u})\mathbf{u}$  and  $\mathbf{u}'_{\perp} = \mathbf{u}' - \mathbf{u}'_{\parallel}$ . Then

$$\begin{aligned}\Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\pm} &= \pm(\mathbf{u} \cdot \mathbf{u}')\Pi_{\pm}, \\ \Pi_{\pm}(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi_{\mp} &= \frac{1}{2}(\mathbf{u}'_{\perp} \pm i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}.\end{aligned}$$

For  $\Pi'_{\pm} = (1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma})/2$ , all four products are

$$\begin{aligned}\Pi_{\pm}\Pi'_{\pm} &= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\pm\mathbf{u} \pm \mathbf{u}' + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}, \\ \Pi_{\pm}\Pi'_{\mp} &= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') + \frac{1}{4}(\pm\mathbf{u} \mp \mathbf{u}' - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}.\end{aligned}$$

### 3.8 Functions of paravectors

#### 3.8.1 Spectral rule

Let  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ , with  $S \in \mathbb{C}$ ,  $\mathbf{V} \in \mathbb{C}^3$ , and  $\mathbf{V}^2 \neq 0$ . Define the complex scalars  $w, \lambda_{\pm}$ , complex vector  $\mathbf{u}$ , and paravector projectors  $\Pi_{\pm}$  by

$$\begin{aligned} w &:= +\sqrt{\mathbf{V}^2}, & \lambda_{\pm} &:= S \pm w, \\ \mathbf{u} &:= \mathbf{V}/w, & \Pi_{\pm} &:= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned}$$

The root is the principal scalar root. Thus  $\mathbf{u}$  is a [vector of square one](#);  $\mathbf{u} \cdot \boldsymbol{\sigma}$  is [self-inverse](#). Its square gives

$$\begin{aligned} \Pi_{\pm}^2 &= \frac{1}{4}(1 \pm 2\mathbf{u} \cdot \boldsymbol{\sigma} + 1) = \Pi_{\pm}, \\ \Pi_+ \Pi_- &= \frac{1}{4}(1 - (\mathbf{u} \cdot \boldsymbol{\sigma})^2) = 0, \\ \Pi_+ + \Pi_- &= 1, \\ Z &= S(\Pi_+ + \Pi_-) + w(\Pi_+ - \Pi_-) = \lambda_+ \Pi_+ + \lambda_- \Pi_-. \end{aligned}$$

These projectors need not be Hermitian. The complementary-projector identities give, for every nonnegative integer  $n$ ,

$$Z^n = \lambda_+^n \Pi_+ + \lambda_-^n \Pi_- = \frac{1}{2}(\lambda_+^n + \lambda_-^n) + \frac{1}{2}(\lambda_+^n - \lambda_-^n) \mathbf{u} \cdot \boldsymbol{\sigma}.$$

For a scalar power series  $f(z) = \sum_{n=0}^{\infty} c_n z^n$  converging at both eigenvalues, with  $c_n \in \mathbb{C}$ , inserting these powers into  $f(Z) = \sum_n c_n Z^n$  gives the spectral rule [\[4, p. 3\]](#)

$$f(Z) = f(\lambda_+) \Pi_+ + f(\lambda_-) \Pi_-. \quad (3.28)$$

The same rule defines chosen analytic branches at the two eigenvalues. Its proof uses only complementary idempotents and therefore applies to any decomposition with these projector identities.

Adjugation exchanges the eigenvalues, so

$$\begin{aligned} \text{adj}(Z) &= \lambda_- \Pi_+ + \lambda_+ \Pi_-, \\ \det(Z) &= \lambda_+ \lambda_-, \\ Z^{-1} &= \frac{\text{adj}(Z)}{\det(Z)} = \lambda_+^{-1} \Pi_+ + \lambda_-^{-1} \Pi_-. \end{aligned}$$

The inverse requires both eigenvalues to be nonzero. Independently choosing the signs of their principal square roots gives

$$\sqrt{Z} = \pm \sqrt{\lambda_+} \Pi_+ \pm \sqrt{\lambda_-} \Pi_-.$$

There are four distinct roots when both eigenvalues are nonzero, and two when one vanishes. Changing  $w$  to  $-w$  exchanges the eigenvalue–projector pairs and leaves the function unchanged when its branch choices are exchanged with them.

#### 3.8.2 Real paravectors and direct square roots

For real  $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$ , with  $a \in \mathbb{R}$ ,  $\mathbf{A} \in \mathbb{R}^3$ , and  $\mathbf{A} \neq 0$ , the spectral quantities above reduce to

$$w = \|\mathbf{A}\|, \quad \mathbf{u} = \mathbf{A}/\|\mathbf{A}\|, \quad \lambda_{\pm} = a \pm \|\mathbf{A}\|.$$

The projectors are now Hermitian. In particular, for a **real unit vector**  $\mathbf{u}$ ,

$$\begin{aligned} f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(1)\Pi_+ + f(-1)\Pi_- \\ &= \frac{1}{2}(f(1) + f(-1)) + \frac{1}{2}(f(1) - f(-1))\mathbf{u} \cdot \boldsymbol{\sigma}, \\ \sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} &= \pm\Pi_+ \pm i\Pi_-. \end{aligned}$$

The two root signs are independent. For a fixed analytic branch,  $f(X) \rightarrow f(a)$  as  $\mathbf{A} \rightarrow 0$ ; independent root choices require the scalar case below.

A direct component derivation also determines all *real* roots. Let  $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$  be real, with  $b \in \mathbb{R}$ ,  $\mathbf{B} \in \mathbb{R}^3$ , and seek real  $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$  such that  $X^2 = Y$ . Equating parts gives

$$\begin{aligned} a^2 + \mathbf{A}^2 &= b, \\ 2a\mathbf{A} &= \mathbf{B}. \end{aligned}$$

If  $\mathbf{B} \neq 0$ , then  $a \neq 0$  and  $\mathbf{A} = \mathbf{B}/(2a)$ . Substitution gives

$$\begin{aligned} a^4 - ba^2 + \frac{1}{4}\mathbf{B}^2 &= 0, \\ a^2 &= \frac{1}{2} \left( b \pm \sqrt{b^2 - \mathbf{B}^2} \right). \end{aligned} \tag{3.29}$$

For each nonnegative  $a^2$  in (3.29), take both scalar roots  $a$  and set  $\mathbf{A} = \mathbf{B}/(2a)$ ; then  $\mathbf{A}^2 = b - a^2$ . There are four real roots when  $b > \|\mathbf{B}\|$ , two when  $b = \|\mathbf{B}\| > 0$ , and none when  $b < \|\mathbf{B}\|$ .

If  $\mathbf{B} = 0$ , the equations instead give the scalar roots  $X = \pm\sqrt{b}$  and the pure-vector roots  $X = \mathbf{A} \cdot \boldsymbol{\sigma}$  with  $\mathbf{A}^2 = b$  for  $b > 0$ . For  $b = 0$  only  $X = 0$  is real; for  $b < 0$  there is no real root.

### 3.8.3 Coalescing eigenvalues

For complex  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$  with  $\mathbf{V}^2 = 0$ , define  $N = \mathbf{V} \cdot \boldsymbol{\sigma}$ , so  $N^2 = 0$ . A scalar function analytic at  $S$  has the terminating Taylor expansion [4, p. 4]

$$f(Z) = f(S) + f'(S)N. \tag{3.30}$$

For  $S \neq 0$ , the chosen square root and logarithm are

$$\begin{aligned} \sqrt{Z} &= r + \frac{N}{2r}, \\ \log Z &= \log S + \frac{N}{S}. \end{aligned}$$

Here  $r \in \mathbb{C}$  is a chosen scalar root of  $S$  ( $r^2 = S$ ); use a consistent logarithm branch. A nonzero  $N$  with  $S = 0$  has no square root: such a root would have only zero eigenvalues, and its quadratic identity would force its square to vanish. It has no logarithm because it is not invertible. If  $N = 0$ , a fixed analytic branch gives  $f(S)$ ; non-scalar square roots of a scalar must instead be found from the component equations  $a^2 + \mathbf{A}^2 = S$ ,  $2a\mathbf{A} = 0$ , now allowing  $a$  and  $\mathbf{A}$  to be complex. Independent branch choices need not remain finite as distinct eigenvalues coalesce.

### 3.8.4 Exponential: projection and series derivations

For complex  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$  with  $\mathbf{V}^2 \neq 0$ , recall  $w = +\sqrt{\mathbf{V}^2}$ , the [vector of square one](#)  $\mathbf{u} = \mathbf{V}/w$ , and  $\lambda_{\pm} = S \pm w$ . The spectral rule (3.28) gives

$$\begin{aligned}\exp(Z) &= \tfrac{1}{2}e^{S+w}(1 + \mathbf{u} \cdot \boldsymbol{\sigma}) + \tfrac{1}{2}e^{S-w}(1 - \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= e^S \left( \frac{e^w + e^{-w}}{2} + \frac{e^w - e^{-w}}{2} \mathbf{u} \cdot \boldsymbol{\sigma} \right).\end{aligned}$$

Independently,  $S$  commutes with the vector part and  $(w\mathbf{u} \cdot \boldsymbol{\sigma})^{2n} = w^{2n}$ ,  $(w\mathbf{u} \cdot \boldsymbol{\sigma})^{2n+1} = w^{2n+1}\mathbf{u} \cdot \boldsymbol{\sigma}$ . Separating the defining series therefore gives

$$\begin{aligned}\exp(Z) &= e^S \sum_{n=0}^{\infty} \frac{(w\mathbf{u} \cdot \boldsymbol{\sigma})^n}{n!} \\ &= e^S \left( \sum_{n=0}^{\infty} \frac{w^{2n}}{(2n)!} + \sum_{n=0}^{\infty} \frac{w^{2n+1}}{(2n+1)!} \mathbf{u} \cdot \boldsymbol{\sigma} \right).\end{aligned}$$

Both derivations give [2, Eq. (52)]

$$\exp(Z) = e^S (\cosh w + \sinh w \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.31)$$

This is independent of the choice  $w$  or  $-w$ :  $\cosh$  is even, while both  $\sinh w$  and  $\mathbf{u}$  change sign. When  $\mathbf{V}^2 = 0$ , (3.30) instead gives

$$\exp(S + \mathbf{V} \cdot \boldsymbol{\sigma}) = e^S (1 + \mathbf{V} \cdot \boldsymbol{\sigma}).$$

The defining series also gives

$$\begin{aligned}\exp(Z^*) &= \exp(Z)^*, \\ \exp(Z^H) &= \exp(Z)^H.\end{aligned}$$

For the real components  $a, b, \mathbf{A}, \mathbf{B}$  of  $Z = (a + bi) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ , write

$$w = +\sqrt{\mathbf{A}^2 - \mathbf{B}^2 + 2i\mathbf{A} \cdot \mathbf{B}}.$$

Substituting  $\mathbf{u} = (\mathbf{A} + i\mathbf{B})/w$  into (3.31) gives the component form

$$\exp(Z) = e^a (\cos b + i \sin b) \left( \cosh w + \frac{\sinh w}{w} (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} \right).$$

Its real and imaginary specializations are

$$\exp(a + \mathbf{A} \cdot \boldsymbol{\sigma}) = e^a \left( \cosh \|\mathbf{A}\| + \frac{\sinh \|\mathbf{A}\|}{\|\mathbf{A}\|} \mathbf{A} \cdot \boldsymbol{\sigma} \right), \quad (3.32)$$

$$\exp(i(b + \mathbf{B} \cdot \boldsymbol{\sigma})) = (\cos b + i \sin b) \left( \cos \|\mathbf{B}\| + i \frac{\sin \|\mathbf{B}\|}{\|\mathbf{B}\|} \mathbf{B} \cdot \boldsymbol{\sigma} \right). \quad (3.33)$$

Setting  $a = 0$  in (3.32) or  $b = 0$  in (3.33) gives the pure-vector exponentials. Every quotient uses its analytic limit at a zero argument.

### 3.8.5 Rotation and boost factors

For a general real exponent  $a + \mathbf{A} \cdot \boldsymbol{\sigma}$ , where  $a \in \mathbb{R}$  and  $\mathbf{A} \in \mathbb{R}^3$ , (3.32) gives

$$\det(\exp(a + \mathbf{A} \cdot \boldsymbol{\sigma})) = e^{2a}. \quad (3.34)$$

Thus a real exponential has determinant one exactly when its scalar exponent is zero. Its value still has a nonzero scalar part, supplied by the hyperbolic cosine.

Let  $\theta \in \mathbb{R}$  and let  $\mathbf{u} \in \mathbb{R}^3$  be a **real unit vector**,  $\mathbf{u}^2 = 1$ . The two special exponentials just derived define the complex rotation factor  $R$  and the real boost factor  $L$  [1, Eq. (6)] [5, p. 23]:

$$R(\theta, \mathbf{u}) := \exp\left(-i\frac{\theta}{2}\mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cos\frac{\theta}{2} - i\sin\frac{\theta}{2}\mathbf{u} \cdot \boldsymbol{\sigma}, \quad (3.35)$$

$$L(\theta, \mathbf{u}) := \exp\left(-\frac{\theta}{2}\mathbf{u} \cdot \boldsymbol{\sigma}\right) = \cosh\frac{\theta}{2} - \sinh\frac{\theta}{2}\mathbf{u} \cdot \boldsymbol{\sigma}. \quad (3.36)$$

In (3.35) and (3.36), the parameter is a rotation angle for  $R$  and a rapidity for  $L$ ; both are real dimensionless numbers. Here these names specify algebraic objects. Their actions on a general paravector are derived in Section 4; their physical applications follow in Section 11.

Because  $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$ , reversing the parameter gives the inverses

$$\begin{aligned} R^{-1} &= R(-\theta, \mathbf{u}) = \cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\mathbf{u} \cdot \boldsymbol{\sigma}, \\ L^{-1} &= L(-\theta, \mathbf{u}) = \cosh\frac{\theta}{2} + \sinh\frac{\theta}{2}\mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (3.37)$$

Sigma conjugation changes both  $i$  and  $\boldsymbol{\sigma}$  in sign; Hermitian conjugation changes  $i$  but fixes a real vector. Hence

$$\begin{aligned} R^* &= R, \\ R^H &= R^{-1}, \end{aligned} \quad (3.38)$$

$$\begin{aligned} L^* &= L^{-1}, \\ L^H &= L. \end{aligned} \quad (3.39)$$

Equation (3.38) makes  $R$  **unitary**; (3.39) makes  $L$  Hermitian, so  $L^H L = L^2$ . Their determinants and positive norms follow from the circular and hyperbolic identities:

$$\det(R) = \cos^2(\theta/2) + \sin^2(\theta/2) = 1, \quad (3.40)$$

$$\det(L) = \cosh^2(\theta/2) - \sinh^2(\theta/2) = 1, \quad (3.41)$$

$$\|R\|^2 = 1, \quad (3.42)$$

$$\|L\|^2 = \cosh\theta. \quad (3.43)$$

Equations (3.40) and (3.41) make both **paravectors of determinant one**, so their adjugates equal their inverses. Equations (3.42) and (3.43) distinguish their positive norms; only  $R$  is unitary for nonzero  $\theta$ . The half-angle gives the sign and period:

$$R(\theta + 2\pi, \mathbf{u}) = -R(\theta, \mathbf{u}), \quad (3.44)$$

$$R(\theta + 4\pi, \mathbf{u}) = R(\theta, \mathbf{u}).$$

**Gamma form.** Recall  $L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$  from (3.36). Define its real velocity parameter and gamma factor by

$$\begin{aligned}\beta &:= \tanh \theta, \\ \gamma &:= \cosh \theta = \frac{1}{\sqrt{1 - \beta^2}}.\end{aligned}$$

Here  $-1 < \beta < 1$ ,  $\gamma \geq 1$ , and  $\gamma\beta = \sinh \theta$ . The half-angle identities give

$$\begin{aligned}\cosh \frac{\theta}{2} &= \sqrt{\frac{\gamma + 1}{2}}, \\ \sinh \frac{\theta}{2} &= \frac{\gamma\beta}{\sqrt{2(\gamma + 1)}}.\end{aligned}$$

Hence the boost factor itself is

$$L = \sqrt{\frac{\gamma + 1}{2}} \left( 1 - \frac{\gamma\beta}{\gamma + 1} \mathbf{u} \cdot \boldsymbol{\sigma} \right). \quad (3.45)$$

Squaring doubles the exponent, so the full-rapidity paravector is

$$L^2 = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (3.46)$$

Equation (3.46) has prefactor  $\gamma$ ;  $L$  has the half-angle prefactor in (3.45). Equivalently, for any real vector  $\mathbf{A} \neq 0$ , put  $\mathbf{v} = \tanh \|\mathbf{A}\| \mathbf{A}/\|\mathbf{A}\|$  and  $\gamma = \cosh \|\mathbf{A}\|$ . Then

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}).$$

The zero-vector case follows by continuity. This last real paravector has **determinant one** and becomes proper velocity in Section 6.

### 3.8.6 Cayley transform

Let  $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$  be real, with  $b \in \mathbb{R}$ ,  $\mathbf{B} \in \mathbb{R}^3$ , and  $0 < \|\mathbf{B}\| < b$ . Set the **real unit vector**  $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$  and recall  $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ . Its sigma conjugate  $Y^* = b - \mathbf{B} \cdot \boldsymbol{\sigma} = \text{adj}(Y)$  commutes with  $Y$ , so

$$\frac{Y}{Y^*} = \frac{Y^2}{\det(Y)} = \frac{b^2 + \mathbf{B}^2 + 2b\mathbf{B} \cdot \boldsymbol{\sigma}}{b^2 - \mathbf{B}^2}.$$

Define the real parameter  $\theta = 2 \tanh^{-1}(\|\mathbf{B}\|/b)$ . The double-angle identities give

$$\begin{aligned}\cosh \theta &= \frac{b^2 + \mathbf{B}^2}{b^2 - \mathbf{B}^2}, \\ \sinh \theta &= \frac{2b\|\mathbf{B}\|}{b^2 - \mathbf{B}^2}.\end{aligned}$$

Thus the quotient is the full-rapidity exponential

$$\begin{aligned}\frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}} &= \cosh \theta + \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = e^{\theta} \Pi_+ + e^{-\theta} \Pi_-.\end{aligned} \quad (3.47)$$

Equation (3.47) is  $L(\theta, \mathbf{u})^{-2}$  from (3.46). Its real generator vector is  $\mathbf{A} = \theta \mathbf{u}$ , while the Cayley vector obeys

$$\mathbf{B} = b \tanh\left(\frac{\|\mathbf{A}\|}{2}\right) \frac{\mathbf{A}}{\|\mathbf{A}\|}.$$

Thus  $\|\mathbf{B}\|/b = \tanh(\theta/2)$ , the half-rapidity parameter. The corresponding real gamma factor and vector are

$$\begin{aligned} \gamma &= \frac{b^2 + \mathbf{B}^2}{b^2 - \mathbf{B}^2}, \\ \mathbf{v} &= \frac{2b\mathbf{B}}{b^2 + \mathbf{B}^2}. \end{aligned}$$

They give the equivalent forms

$$\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}) = \frac{b + \mathbf{B} \cdot \boldsymbol{\sigma}}{b - \mathbf{B} \cdot \boldsymbol{\sigma}} = \frac{Y}{Y^*}.$$

Choosing  $b = 1$  normalizes the scalar part of  $Y$  to one;  $\mathbf{B} = 0$  follows by continuity.

### 3.8.7 Composing the factors

Recall  $L(\theta, \mathbf{u}) = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$  and  $R(\theta, \mathbf{u}) = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$  from (3.36) and (3.35). For a fixed [real unit direction](#)  $\mathbf{u}$  and real parameters  $\theta, \theta'$ , the generators commute, so their parameters add:

$$L(\theta, \mathbf{u})L(\theta', \mathbf{u}) = L(\theta + \theta', \mathbf{u}), \quad (3.48)$$

$$\begin{aligned} R(\theta, \mathbf{u})R(\theta', \mathbf{u}) &= R(\theta + \theta', \mathbf{u}), \\ L(\theta, \mathbf{u})R(\theta', \mathbf{u}) &= R(\theta', \mathbf{u})L(\theta, \mathbf{u}). \end{aligned}$$

For the aligned real and imaginary generator, put  $w = -(\theta + i\theta')/2 \in \mathbb{C}$ . Its exponential is

$$\exp(w \mathbf{u} \cdot \boldsymbol{\sigma}) = L(\theta, \mathbf{u})R(\theta', \mathbf{u}) = \cosh w + \sinh w \mathbf{u} \cdot \boldsymbol{\sigma}.$$

Expanding the complex scalar coefficients gives

$$\begin{aligned} \cosh w &= \cosh \frac{\theta}{2} \cos \frac{\theta'}{2} + i \sinh \frac{\theta}{2} \sin \frac{\theta'}{2}, \\ \sinh w &= -\sinh \frac{\theta}{2} \cos \frac{\theta'}{2} - i \cosh \frac{\theta}{2} \sin \frac{\theta'}{2}. \end{aligned}$$

This aligned factorization requires parallel real and imaginary generator vectors. For different [real unit directions](#)  $\mathbf{u}, \mathbf{v}$ , the vector product instead gives

$$\left[-\frac{1}{2}\mathbf{u} \cdot \boldsymbol{\sigma}, -\frac{1}{2}\mathbf{v} \cdot \boldsymbol{\sigma}\right] = \frac{i}{2}(\mathbf{u} \times \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (3.49)$$

This commutator is a rotation generator: boosts along nonparallel directions do not commute.

### 3.8.8 Logarithm

For complex  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$  with  $\mathbf{V}^2 \neq 0$ , recall  $w = +\sqrt{\mathbf{V}^2}$ , the **vector of square one**  $\mathbf{u} = \mathbf{V}/w$ ,  $\lambda_{\pm} = S \pm w$ , and  $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$ . Assume both eigenvalues are nonzero. For a nonzero complex scalar  $\lambda$  and any integer  $n$ , the scalar branches are

$$\log \lambda = \log \|\lambda\| + i(\arg \lambda + 2n\pi).$$

Choose one branch for each eigenvalue. The spectral rule gives

$$\begin{aligned} \log Z &= \log(\lambda_+) \Pi_+ + \log(\lambda_-) \Pi_- \\ &= \frac{1}{2}(\log \lambda_+ + \log \lambda_-) + \frac{1}{2}(\log \lambda_+ - \log \lambda_-) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (3.50)$$

Exchanging  $w$  and  $-w$  exchanges both eigenvalue–projector pairs and preserves the same chosen logarithm. Independent scalar branches give the different logarithms.

For a **real unit vector**  $\mathbf{u}$ , (3.50) uses eigenvalues 1 and  $-1$ , giving, for integers  $n, m$ ,

$$\log(\mathbf{u} \cdot \boldsymbol{\sigma}) = 2n\pi i \Pi_+ + (2m+1)\pi i \Pi_-.$$

For real  $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$ , with  $b \in \mathbb{R}$ ,  $\mathbf{B} \in \mathbb{R}^3$ , and  $b > \|\mathbf{B}\|$ , both eigenvalues are positive. Combining their real logarithms yields

$$\log Y = \frac{1}{2} \log(b^2 - \mathbf{B}^2) + \tanh^{-1} \left( \frac{\|\mathbf{B}\|}{b} \right) \frac{\mathbf{B}}{\|\mathbf{B}\|} \cdot \boldsymbol{\sigma}.$$

The vector term vanishes at  $\mathbf{B} = 0$ ; the general coalescing case was treated above.

## 4 Transformations and invariants

Here  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$  is a complex paravector:  $S \in \mathbb{C}$  and  $\mathbf{V} \in \mathbb{C}^3$ . Transformation factors are constant complex paravectors. Distinguish the scalar part  $\text{sc}(Z) = S = \frac{1}{2} \text{tr} Z$  from (3.4), the complex determinant  $S^2 - \mathbf{V}^2$  from (3.5), and the positive norm  $\text{sc}(Z^H Z)$  from (3.25). Scalar part and trace carry exactly the same information; we use  $\text{sc}$  below.

### 4.1 Similarity, determinant, and norm

For a complex **unit paravector**  $T$ , similarity means  $Z' = T Z T^{-1}$ . Scalar-part cyclicity gives

$$\text{sc}(Z') = \text{sc}(T Z T^{-1}) = \text{sc}(Z T^{-1} T) = \text{sc}(Z).$$

Thus trace is also preserved. Multiplicativity gives

$$\det(Z') = \det(T) \det(Z) \det(T^{-1}) = \det(Z). \quad (4.1)$$

The opposite similarity  $Z' = T^{-1} Z T$  has the same invariants. For independent complex factors  $T, T'$ , the law  $Z' = T' Z T$  instead gives

$$\det(Z') = \det(T') \det(T) \det(Z).$$

This preserves the determinant for every  $Z$  exactly when  $\det(T') \det(T) = 1$ . This alone does not preserve scalar part, norm, or the real paravector subspace.

If the factor  $T$  is unitary, then  $T^H = T^{-1}$  by (3.14). For  $Z' = TZT^H$ , cancellation and scalar-part cyclicity give

$$\begin{aligned}\|Z'\|^2 &= \text{sc}(TZ^HT^HTZT^H) \\ &= \text{sc}(Z^HT^HTZ) = \|Z\|^2.\end{aligned}$$

The norm is also preserved by  $Z' = TZ$  and  $Z' = ZT$ . The congruence additionally preserves scalar part and determinant; a one-sided unitary product need not.

#### 4.1.1 Unit determinant is not Hermitian unitarity

The word *unit* in (3.13) means invertible; *unitary* means (3.14). A useful third condition is **determinant one**. To see their difference, let  $dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}$  be a real paravector increment, with  $dt > \|d\mathbf{r}\|$ , and define the positive real scalar  $ds = \sqrt{dt^2 - d\mathbf{r}^2}$ . Reserve  $U$  for the real proper velocity  $U = dX/ds$ . Since  $dX^* dX = ds^2$ ,

$$U^*U = \frac{dX^* dX}{ds^2} = 1. \quad (4.2)$$

Thus  $\det(U) = 1$  and  $U^{-1} = U^*$ :  $U$  is a unit paravector with  $UU^{-1} = U^{-1}U = 1$ . Its letter alone implies no algebraic property. The two conjugates are

$$\begin{aligned}U^H &= U = \frac{dt + d\mathbf{r} \cdot \boldsymbol{\sigma}}{ds}, \\ U^* &= \frac{dt - d\mathbf{r} \cdot \boldsymbol{\sigma}}{ds}.\end{aligned}$$

Hermitian conjugation leaves every real paravector fixed; sigma conjugation reverses its vector part. Therefore

$$\begin{aligned}U^H U &= U^2 = \frac{dt^2 + d\mathbf{r}^2 + 2 dt d\mathbf{r} \cdot \boldsymbol{\sigma}}{ds^2}, \\ \|U\|^2 &= \frac{dt^2 + d\mathbf{r}^2}{dt^2 - d\mathbf{r}^2}.\end{aligned}$$

Both expressions equal 1 only when  $d\mathbf{r} = 0$ , its rest frame. Section 6 develops the kinematics of this proper velocity.

## 4.2 Real paravectors and the Lorentz metric

Let  $X = a + \mathbf{A} \cdot \boldsymbol{\sigma}$  be real, with  $a \in \mathbb{R}$  and  $\mathbf{A} \in \mathbb{R}^3$ . The determinant and positive norm give different quadratic forms:

$$\begin{aligned}\det(X) &= a^2 - \mathbf{A}^2, \\ \|X\|^2 &= a^2 + \mathbf{A}^2.\end{aligned}$$

The first is the Lorentz metric; the second measures the positive sum of squared coefficients. A spacetime transformation must preserve reality and the first form. For a constant complex **unit paravector**  $T$  with  $\det(T) = 1$ , use the Hermitian congruence [1, Eq. (17)]

$$\boxed{X' = T X T^H}. \quad (4.3)$$

Hermitian conjugation and determinant multiplication give

$$\begin{aligned} X'^{\mathbf{H}} &= T X^{\mathbf{H}} T^{\mathbf{H}} = X', \\ \det(X') &= \det(T) \det(X) \det(T^{\mathbf{H}}) = \det(X). \end{aligned}$$

For a second real paravector  $Y = b + \mathbf{B} \cdot \boldsymbol{\sigma}$ , with  $b \in \mathbb{R}$  and  $\mathbf{B} \in \mathbb{R}^3$ , the associated metric pairing is [1, Eq. (15)]

$$\text{sc}(X^*Y) = ab - \mathbf{A} \cdot \mathbf{B} = \frac{1}{2}(\det(X+Y) - \det(X) - \det(Y)). \quad (4.4)$$

Linearity and determinant invariance therefore give  $\text{sc}(X'^*Y') = \text{sc}(X^*Y)$ : the full metric is preserved. These maps preserve the future cone as well. For  $a \geq \|\mathbf{A}\|$ , the eigenvalues  $a \pm \|\mathbf{A}\|$  are nonnegative, so  $X$  has a real square root and

$$\text{sc}(X') = \|\sqrt{X} T^{\mathbf{H}}\|^2 \geq 0.$$

Together with determinant invariance, this keeps  $X'$  in the future cone. These are the proper, future-preserving Lorentz transformations;  $T$  and  $-T$  give the same map.

**Why the scalar exponent vanishes.** More generally, the congruence multiplies  $\det(X)$  by  $\|\det(T)\|^2$ . For real  $T = \exp(a + \mathbf{A} \cdot \boldsymbol{\sigma})$ , (3.34) therefore gives

$$\det(TXT^{\mathbf{H}}) = e^{4a} \det(X).$$

A nonzero real scalar exponent dilates the metric; it is excluded by interval invariance. An imaginary scalar exponent contributes a phase  $e^{ib}$ ,  $b \in \mathbb{R}$ , which cancels between  $T$  and  $T^{\mathbf{H}}$ . Requiring  $\det(T) = 1$  removes this continuous phase freedom. Thus a pure boost needs only a real vector exponent.

**Why the positive norm is different.** For  $L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ , with real  $\theta$  and real unit  $\mathbf{u}$ , applying (4.3) to  $X = 1$  gives

$$\begin{aligned} X' &= L^2 = \cosh \theta - \sinh \theta \mathbf{u} \cdot \boldsymbol{\sigma}, \\ \det(X') &= 1, \\ \text{sc}(X') &= \cosh \theta, \\ \|X'\|^2 &= \cosh(2\theta). \end{aligned}$$

A nonzero boost preserves the interval while changing the scalar part and positive norm. In fact, a determinant-one congruence preserves the positive norm for every real  $X$  exactly when  $T$  is unitary. Necessity follows by taking  $X = 1$ : the real, positive  $TT^{\mathbf{H}}$  can have both determinant and norm one only if it is 1, which makes  $T$  unitary. Sufficiency was proved above. Normalizing  $L$  to  $\|L\| = 1$  would change its determinant and lose the Lorentz metric.

**Every factor separates into a boost and a rotation.** For a determinant-one complex  $T$ , the paravector  $TT^{\mathbf{H}}$  is real, has determinant one, and has positive scalar part  $\|T\|^2$ . Both eigenvalues are therefore positive. Using the positive real square root, define

$$B := \sqrt{TT^{\mathbf{H}}}, \quad R := B^{-1}T.$$

Then  $\det(B) = 1$ ,  $R^H R = 1$ ,  $\det(R) = 1$ , and  $T = BR$ . The real logarithm of  $B$  has zero scalar part by (3.34), so  $B$  is a real vector exponential. For  $R$ , the identity  $R^H = R^{-1} = \text{adj}(R)$  makes its scalar coefficient real and its vector coefficient imaginary; determinant one then gives  $R = \cos(\theta/2) - i \sin(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma}$  for some real  $\theta$  and real unit  $\mathbf{u}$ . Consequently  $TXT^H = BRXR^H B$ : the rotation is followed by the boost. Their component actions follow next.

### 4.3 Rotation action and Rodrigues' formula

Recall the unitary factor  $R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$  from (3.35), with real angle  $\theta$  and real unit axis  $\mathbf{u}$ . For the complex paravector  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ , use the Hermitian congruence [1, Eq. (5)]

$$\boxed{Z' = RZR^H.} \quad (4.5)$$

Since  $R^H = R^{-1}$  by (3.38), this keeps  $S' = S$ . The vector product (3.18) gives

$$\begin{aligned} (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{V} \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= 2(\mathbf{u} \cdot \mathbf{V})\mathbf{u} \cdot \boldsymbol{\sigma} - \mathbf{V} \cdot \boldsymbol{\sigma}, \\ [\mathbf{V} \cdot \boldsymbol{\sigma}, \mathbf{u} \cdot \boldsymbol{\sigma}] &= -2i(\mathbf{u} \times \mathbf{V}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Inserting the sine and cosine forms of  $R$  and  $R^H$  therefore yields

$$\begin{aligned} \mathbf{V}' \cdot \boldsymbol{\sigma} &= \cos^2 \frac{\theta}{2} \mathbf{V} \cdot \boldsymbol{\sigma} + 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} (\mathbf{u} \times \mathbf{V}) \cdot \boldsymbol{\sigma} \\ &\quad + \sin^2 \frac{\theta}{2} (2(\mathbf{u} \cdot \mathbf{V})\mathbf{u} \cdot \boldsymbol{\sigma} - \mathbf{V} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Collecting terms gives Rodrigues' formula [5, p. 14]:

$$\boxed{\mathbf{V}' = \mathbf{V} \cos \theta + (\mathbf{u} \times \mathbf{V}) \sin \theta + (\mathbf{u} \cdot \mathbf{V})(1 - \cos \theta)\mathbf{u}.} \quad (4.6)$$

This is the active right-handed rotation through  $+\theta$  about  $\mathbf{u}$ . Rotating the coordinate axes through  $+\theta$  instead uses the inverse map. It preserves sc, determinant, norm, and the real paravector subspace. For complex  $\mathbf{V}$ , it rotates its real and imaginary parts by the same real rotation. The sign change in (3.44) affects both factors in (4.5), so  $Z'$  has period  $2\pi$ .

### 4.4 Quaternions as paravectors

The quaternion units are already present in the complex vector part of our algebra. Define

$$q_1 = -i\sigma_1, \quad q_2 = -i\sigma_2, \quad q_3 = -i\sigma_3, \quad \mathbf{q} = \{q_1, q_2, q_3\}.$$

The sigma product gives  $q_1^2 = q_2^2 = q_3^2 = -1$ ,  $q_1 q_2 = q_3$ ,  $q_2 q_3 = q_1$ , and  $q_3 q_1 = q_2$ ; reversing any of these distinct pairs changes its sign. Thus  $q_1 q_2 q_3 = -1$ . These are the three quaternion units, distinct from the commuting complex scalar  $i$ .

For four real quaternion components  $a, A_1, A_2, A_3$ , put  $\mathbf{A} = \{A_1, A_2, A_3\}$ . Their paravector representation is [3, p. 92]

$$\boxed{Q = a + A_1 q_1 + A_2 q_2 + A_3 q_3 = a - i\mathbf{A} \cdot \boldsymbol{\sigma}, \quad S = a, \quad \mathbf{V} = -i\mathbf{A}.} \quad (4.7)$$

In the decomposition  $Z = (a + bi) + (\mathbf{C} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ , this selects  $b = 0$ ,  $\mathbf{C} = 0$ , and  $\mathbf{B} = -\mathbf{A}$ . Quaternions therefore have a real scalar coefficient and purely imaginary vector coefficients in STA.

Their four real components form a subalgebra of the eight-real-component complex paravectors. Indeed, for  $Q' = a' + \mathbf{A}' \cdot \mathbf{q}$ , the paravector product becomes [6, p. 9]

$$QQ' = (aa' - \mathbf{A} \cdot \mathbf{A}') + (a\mathbf{A}' + a'\mathbf{A} + \mathbf{A} \times \mathbf{A}') \cdot \mathbf{q}. \quad (4.8)$$

Since  $q_k^H = -q_k$  and  $q_k^* = q_k$ , quaternion conjugation is the existing Hermitian conjugation:

$$\begin{aligned} Q^H &= a - \mathbf{A} \cdot \mathbf{q}, \\ Q^* &= Q, \quad \text{adj}(Q) = Q^H, \\ QQ^H &= Q^H Q = a^2 + \mathbf{A}^2 = \det Q = \|Q\|^2. \end{aligned} \quad (4.9)$$

For nonzero  $Q$ , (4.9) gives  $Q^{-1} = Q^H/(a^2 + \mathbf{A}^2)$ . A unit quaternion has  $a^2 + \mathbf{A}^2 = 1$ , so  $Q^{-1} = Q^H$ . To rotate a real vector  $\mathbf{r}$ , represent it by the pure quaternion  $\mathbf{r} \cdot \mathbf{q} = -i\mathbf{r} \cdot \boldsymbol{\sigma}$ . For a unit quaternion  $Q = a + \mathbf{A} \cdot \mathbf{q}$ , use the quaternion sandwich:

$$\mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^H, \quad \mathbf{r}' \cdot \boldsymbol{\sigma} = Q(\mathbf{r} \cdot \boldsymbol{\sigma})Q^H. \quad (4.10)$$

The two forms agree because  $-i$  commutes with every paravector. Applying (4.8) twice gives

$$\begin{aligned} Q(\mathbf{r} \cdot \mathbf{q}) &= -\mathbf{A} \cdot \mathbf{r} + (a\mathbf{r} + \mathbf{A} \times \mathbf{r}) \cdot \mathbf{q}, \\ Q(\mathbf{r} \cdot \mathbf{q})Q^H &= ((a^2 - \mathbf{A}^2)\mathbf{r} + 2(\mathbf{A} \cdot \mathbf{r})\mathbf{A} + 2a\mathbf{A} \times \mathbf{r}) \cdot \mathbf{q}. \end{aligned}$$

The scalar part cancels. For the real unit axis  $\mathbf{u}$  and angle  $\theta$ , choose

$$a = \cos \frac{\theta}{2}, \quad \mathbf{A} = \mathbf{u} \sin \frac{\theta}{2}, \quad Q = \cos \frac{\theta}{2} - i\mathbf{u} \cdot \boldsymbol{\sigma} \sin \frac{\theta}{2} = R.$$

Then  $a^2 - \mathbf{A}^2 = \cos \theta$ ,  $2a\mathbf{A} = \mathbf{u} \sin \theta$ , and  $2(\mathbf{A} \cdot \mathbf{r})\mathbf{A} = (1 - \cos \theta)(\mathbf{u} \cdot \mathbf{r})\mathbf{u}$ . Consequently (4.10) recovers

$$\boxed{\mathbf{r}' = \mathbf{r} \cos \theta + (\mathbf{u} \times \mathbf{r}) \sin \theta + (\mathbf{u} \cdot \mathbf{r})(1 - \cos \theta)\mathbf{u}.}$$

This is precisely Rodrigues' formula (4.6) for  $\mathbf{V} = \mathbf{r}$ . Thus  $R = Q$ , and the quaternion sandwich is exactly  $R(\mathbf{r} \cdot \boldsymbol{\sigma})R^H$  from (4.5). The reversed sandwich  $Q^H(\mathbf{r} \cdot \boldsymbol{\sigma})Q$  gives the inverse rotation.

## 4.5 Boost action and its sign convention

Recall  $L = \exp(-\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$  from (3.36);  $\theta \in \mathbb{R}$ ,  $\mathbf{u} \in \mathbb{R}^3$ ,  $\mathbf{u}^2 = 1$ ,  $\beta = \tanh \theta$ , and  $\gamma = \cosh \theta$ . Use the same congruence with  $T = L$  and  $L^H = L$  from (3.39) [5, Eq. (24)]:

$$\boxed{Z' = LZL.} \quad (4.11)$$

The paravector  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$  may be complex; spacetime uses its real case. Resolve its vector into parallel and perpendicular parts:

$$\begin{aligned} \mathbf{V}_{\parallel} &:= (\mathbf{u} \cdot \mathbf{V})\mathbf{u}, \\ \mathbf{V}_{\perp} &:= \mathbf{V} - \mathbf{V}_{\parallel}. \end{aligned}$$

The parallel term commutes with  $\mathbf{u} \cdot \boldsymbol{\sigma}$ ; the perpendicular term anticommutes with it. Consequently,

$$\begin{aligned} L(S + \mathbf{V}_{\parallel} \cdot \boldsymbol{\sigma})L &= (S + \mathbf{V}_{\parallel} \cdot \boldsymbol{\sigma})L^2, \\ L(\mathbf{V}_{\perp} \cdot \boldsymbol{\sigma})L &= \mathbf{V}_{\perp} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Using  $L^2 = \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma})$  from (3.46), multiplication gives [7, Eq. (5.1)]

$$\boxed{S' = \gamma(S - \beta \mathbf{u} \cdot \mathbf{V})}, \quad (4.12)$$

$$\boxed{\mathbf{V}' = \mathbf{V}_\perp + \gamma(\mathbf{V}_\parallel - \beta S \mathbf{u})}. \quad (4.13)$$

The equivalent longitudinal combinations expose the invariant directly:

$$S' \pm \mathbf{u} \cdot \mathbf{V}' = e^{\mp \theta} (S \pm \mathbf{u} \cdot \mathbf{V}). \quad (4.14)$$

Their product is  $S^2 - (\mathbf{u} \cdot \mathbf{V})^2$ ; the transverse part is unchanged. Thus (4.14) preserves  $S^2 - \mathbf{V}^2$  and, for real  $Z$ , the light cone. For spacetime, (4.11) gives the frame moving with velocity  $+\beta \mathbf{u}$ ; reversing  $\theta$  gives its inverse.

The inverse map is  $Z = L^{-1} Z' L^{-1}$ , obtained by reversing  $\theta$ . Generally  $LX \neq XL$ : the scalar and parallel parts commute with  $L$ , while a nonzero perpendicular part does not for a nonzero boost. For example, with  $\mathbf{u} \cdot \boldsymbol{\sigma} = \sigma_3$  and  $X = \sigma_1$ ,

$$LX = \cosh(\theta/2) \sigma_1 - i \sinh(\theta/2) \sigma_2, \quad LXL = \sigma_1.$$

The second factor cancels the imaginary transverse term. Defining a full-rapidity factor cannot make a single left product boost every real  $X$ ; it would merely replace the two factors by their square roots.

**The field uses similarity.** For  $Z' = LZL^{-1}$ , the same commutation rules give  $S' = S$  and

$$\mathbf{V}' = \mathbf{V}_\parallel + \gamma(\mathbf{V}_\perp - i\beta \mathbf{u} \times \mathbf{V}). \quad (4.15)$$

This preserves scalar part and determinant, but fixes  $Z = 1$  and generally makes a real vector complex. It cannot give the time–space mixing of (4.12)–(4.13). It is the correct action on the complex electromagnetic field: its imaginary part represents the magnetic field, and its zero scalar part must remain zero. Conversely, congruence would give a scalar part to a field parallel to  $\mathbf{u}$ . The physical field law and its electric and magnetic components are derived in Section 11.

## 5 Differentiated fields

### 5.1 Paravector differential operators

Let  $t \in \mathbb{R}$  and  $\mathbf{r} = \{r_1, r_2, r_3\} \in \mathbb{R}^3$  be real coordinates. Define their scalar and vector derivatives by

$$\partial_t := \frac{\partial}{\partial t}, \quad \partial_{\mathbf{r}} := \left\{ \frac{\partial}{\partial r_1}, \frac{\partial}{\partial r_2}, \frac{\partial}{\partial r_3} \right\}.$$

Define the partial paravector and its sigma conjugate by [8, Eq. (176)]

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} = \partial_t + \sum_{n=1}^3 \sigma_n \frac{\partial}{\partial r_n}, \quad (5.1)$$

$$\partial^* = \text{adj}(\partial) = \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}.$$

The sigma generators are constant. Derivatives act from the left on everything to their right unless parentheses specify their scope. For fields with commuting second derivatives, anticommutation cancels all mixed spatial terms:

$$\begin{aligned} (\boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})^2 &= \sum_{n=1}^3 \frac{\partial^2}{\partial r_n^2} + \sum_{n < m} (\sigma_n \sigma_m + \sigma_m \sigma_n) \frac{\partial^2}{\partial r_n \partial r_m} \\ &= \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}}. \end{aligned}$$

The mixed time–space terms likewise cancel. Both orders therefore give the scalar d’Alembertian [8, Eq. (178)]

$$\partial \partial^* = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}} \cdot \partial_{\mathbf{r}} =: \square. \quad (5.2)$$

These are operator identities:  $\partial \partial^* Z$  means  $\partial(\partial^* Z)$ .

## 5.2 Components and the product rule

Let  $a, b : \mathbb{R}^4 \rightarrow \mathbb{R}$  be real scalar functions and  $\mathbf{A}, \mathbf{B} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$  real vector functions. They define the complex paravector field

$$Z(t, \mathbf{r}) = (a + b\mathbf{i}) + (\mathbf{A} + \mathbf{iB}) \cdot \boldsymbol{\sigma}.$$

Ordered vector multiplication gives its four real components:

$$\begin{aligned} \partial Z &= (\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}) + \mathbf{i}(\partial_t b + \partial_{\mathbf{r}} \cdot \mathbf{B}) \\ &\quad + (\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \\ &\quad + \mathbf{i}(\partial_{\mathbf{r}} b + \partial_{\mathbf{r}} \times \mathbf{A} + \partial_t \mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

Multiplication by  $\mathbf{i}$  exchanges real and imaginary parts, with the former imaginary parts acquiring a minus sign.

For a second complex paravector field  $Z'(t, \mathbf{r})$ , the prime labels the field rather than a derivative. Apply the scalar product rule before reordering any factors:

$$\begin{aligned} \partial(ZZ') &= (\partial_t Z)Z' + Z(\partial_t Z') \\ &\quad + \sum_{n=1}^3 \sigma_n \left( \frac{\partial Z}{\partial r_n} Z' + Z \frac{\partial Z'}{\partial r_n} \right) \\ &= (\partial_t Z)Z' + Z(\partial_t Z') + \sum_{n=1}^3 [\sigma_n, Z] \frac{\partial Z'}{\partial r_n}. \end{aligned}$$

## 6 Space–time kinematics

Let  $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$  be a real event, with time  $t \in \mathbb{R}$  and position  $\mathbf{r} \in \mathbb{R}^3$ . A massive particle follows a future-directed time-like path  $\mathbf{r}(t)$ ;  $s$  denotes its proper time.

## 6.1 Intervals, velocity, and momentum

Recall the real proper velocity  $U = dX/ds$ , a [unit paravector](#) with  $UU^{-1} = 1$ . The differential event and its invariant interval are [\[5, Eq. \(7\)\]](#)

$$\begin{aligned} dX &= dt + d\mathbf{r} \cdot \boldsymbol{\sigma}, \\ ds^2 &= \det(dX) = dX dX^* = dt^2 - d\mathbf{r}^2. \end{aligned} \quad (6.1)$$

By [\(4.3\)](#), a constant complex [unit paravector](#)  $T$  with  $\det T = 1$  gives

$$dX' = T dX T^H, \quad \det(dX') = ds^2.$$

Positive, zero, and negative values of [\(6.1\)](#) mean time-like, null, and space-like separation. Massive particles follow time-like paths; light follows null paths; causal signals cannot connect space-like separated events. For two real intervals,

$$\begin{aligned} [dX, dX'] &= 2i(d\mathbf{r} \times d\mathbf{r}') \cdot \boldsymbol{\sigma}, \\ (dX)^{-1} &= \frac{dX^*}{ds^2} \quad (ds^2 \neq 0). \end{aligned}$$

A null interval has no inverse.

Along the massive path, define the real coordinate velocity  $\mathbf{v} = d\mathbf{r}/dt$  and choose  $ds > 0$ . Then  $ds^2 = (1 - \mathbf{v}^2) dt^2$  gives [\[9, Eq. \(7.18\)\]](#)

$$\gamma := \frac{1}{\sqrt{1 - \mathbf{v}^2}} = \frac{dt}{ds}, \quad (6.2)$$

$$\frac{d}{ds} = \gamma \frac{d}{dt}. \quad (6.3)$$

Applying [\(6.3\)](#) with [\(6.2\)](#) gives the real proper velocity [\[5, Eq. \(8\)\]](#)

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (6.4)$$

The normalization [\(4.2\)](#) gives  $U^*U = 1$  and  $U^{-1} = U^* = \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma})$ . Since  $U^H = U$ , this is [determinant one](#), not Hermitian unitarity except at rest. For a [real unit direction](#)  $\mathbf{u}$ , write  $\mathbf{v} = \beta \mathbf{u}$  and  $\theta = \tanh^{-1} \beta$ . Equation [\(3.46\)](#) gives  $U = L^{-2}$  for  $L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ : the direct boost  $LUL = 1$  takes this tangent to rest.

For constant real mass  $m > 0$ , define the real four-momentum [\[9, Eq. \(7.24\)\]](#)

$$P := mU = E + \mathbf{p} \cdot \boldsymbol{\sigma}. \quad (6.5)$$

The energy, momentum, and inverse are

$$\begin{aligned} E &= \text{sc}(P) = m\gamma, \\ \mathbf{p} &= \text{vec}(P) = m\gamma\mathbf{v}, \\ P^{-1} &= \frac{E - \mathbf{p} \cdot \boldsymbol{\sigma}}{m^2}. \end{aligned}$$

Since  $\det U = 1$ , the determinant gives the mass shell [\[9, Eq. \(7.30\)\]](#)

$$E^2 - \mathbf{p}^2 = \det P = m^2. \quad (6.6)$$

For a massive particle of finite energy,

$$\mathbf{v}^2 = \frac{\mathbf{p}^2}{E^2} = 1 - \frac{m^2}{E^2} < 1.$$

Approaching light speed requires unbounded energy. A massless path has no proper-time velocity. Define its momentum directly as a future null paravector  $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$ , with  $E > 0$  and  $E^2 = \mathbf{p}^2$ ; it has no inverse.

## 6.2 Proper acceleration

Recall the real proper velocity  $U = dX/ds$ , a [unit paravector](#) with  $UU^{-1} = 1$ , and the real coordinate velocity  $\mathbf{v}(t)$  and Lorentz factor  $\gamma$ . Define the real coordinate acceleration  $\mathbf{a} = d\mathbf{v}/dt$ . Differentiating gives

$$\begin{aligned} \frac{d\gamma}{dt} &= -\frac{1}{2}(1 - \mathbf{v}^2)^{-3/2}(-2\mathbf{v} \cdot \mathbf{a}) = \gamma^3 \mathbf{v} \cdot \mathbf{a}, \\ \frac{d^2}{ds^2} &= \gamma \frac{d}{dt} \left( \gamma \frac{d}{dt} \right) = \gamma^2 \frac{d^2}{dt^2} + \gamma^4 (\mathbf{v} \cdot \mathbf{a}) \frac{d}{dt}. \end{aligned}$$

Applying this to  $X$  yields the real proper-acceleration paravector [\[9, p. 141\]](#):

$$\frac{dU}{ds} = \gamma^4 (\mathbf{v} \cdot \mathbf{a}) + (\gamma^2 \mathbf{a} + \gamma^4 (\mathbf{v} \cdot \mathbf{a}) \mathbf{v}) \cdot \boldsymbol{\sigma}. \quad (6.7)$$

Differentiating  $U^*U = 1$  gives

$$\left( \frac{dU}{ds} \right)^* U + U^* \frac{dU}{ds} = 0.$$

The two terms have equal real scalar parts, so

$$\text{sc}(U^* dU/ds) = 0.$$

Thus four-velocity and four-acceleration are Minkowski-orthogonal. Multiplication on the right by  $U^*$  also gives

$$\text{adj} \left( \frac{dU}{ds} \right) = -U^* \frac{dU}{ds} U^*.$$

Expanding the commutator and the product gives

$$\begin{aligned} [U, dU/ds] &= 2i\gamma^2 \left( \mathbf{v} \times \frac{d\mathbf{v}}{ds} \right) \cdot \boldsymbol{\sigma}, \\ U^* \frac{dU}{ds} &= \gamma^3 (\mathbf{a} - i\mathbf{v} \times \mathbf{a}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

The real vector records acceleration and the imaginary vector records turning of the velocity. Since  $\mathbf{a} \cdot (\mathbf{v} \times \mathbf{a}) = 0$ ,

$$\begin{aligned} \det \left( \frac{dU}{ds} \right) &= - \left( U^* \frac{dU}{ds} \right)^2 \\ &= -\gamma^6 (\mathbf{a}^2 - (\mathbf{v} \times \mathbf{a})^2) \\ &= -\gamma^4 (\mathbf{a}^2 + \gamma^2 (\mathbf{v} \cdot \mathbf{a})^2). \end{aligned}$$

The last equality uses  $(\mathbf{v} \times \mathbf{a})^2 = \mathbf{v}^2 \mathbf{a}^2 - (\mathbf{v} \cdot \mathbf{a})^2$  and  $1 - \mathbf{v}^2 = \gamma^{-2}$ . Nonzero proper acceleration is therefore space-like; its invariant squared magnitude is the negative determinant. For  $\mathbf{v} \neq 0$ , define  $\mathbf{a}_{\parallel} = (\mathbf{v} \cdot \mathbf{a})\mathbf{v}/\mathbf{v}^2$  and  $\mathbf{a}_{\perp} = \mathbf{a} - \mathbf{a}_{\parallel}$ . Equivalently,

$$-\det\left(\frac{dU}{ds}\right) = \gamma^4 \mathbf{a}_{\perp}^2 + \gamma^6 \mathbf{a}_{\parallel}^2.$$

At rest this reduces to  $\mathbf{a}^2$ .

**Energy and inertia.** Recall  $E = m\gamma$  and  $\mathbf{p} = m\gamma\mathbf{v}$  for invariant real mass  $m > 0$ . Differentiating, using the derivative of  $\gamma$  above, gives

$$\frac{d\mathbf{p}}{dt} = m\gamma\mathbf{a} + m\gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v}.$$

For  $\mathbf{v} \neq 0$ , resolving along and perpendicular to  $\mathbf{v}$  yields

$$\boxed{\frac{d\mathbf{p}}{dt} = m\gamma\mathbf{a}_{\perp} + m\gamma^3\mathbf{a}_{\parallel}.} \quad (6.8)$$

Taking its dot product with  $\mathbf{v}$  gives the work rate

$$\frac{dE}{dt} = m\gamma^3\mathbf{v} \cdot \mathbf{a} = \mathbf{v} \cdot \frac{d\mathbf{p}}{dt}. \quad (6.9)$$

At rest the force is  $m\mathbf{a}$ . Restoring  $c$ ,  $E = \gamma mc^2$ ;  $E/c^2 = \gamma m$  is sometimes called *relativistic mass*. We retain invariant  $m$ : the longitudinal and transverse inertia in (6.8) differ.

## 6.3 World lines, clocks, and light cones

### 6.3.1 Integrating proper velocity

Recall the real path  $X(s) = t(s) + \mathbf{r}(s) \cdot \boldsymbol{\sigma}$  and its tangent  $U = dX/ds$ , a [unit paravector](#) with  $UU^{-1} = 1$ . For real proper times  $s_0 < s_1$ , integration gives

$$\boxed{X(s_1) - X(s_0) = \int_{s_0}^{s_1} U(s) ds.} \quad (6.10)$$

Its real components are

$$\begin{aligned} t(s_1) - t(s_0) &= \int_{s_0}^{s_1} \gamma(s) ds, \\ \mathbf{r}(s_1) - \mathbf{r}(s_0) &= \int_{s_0}^{s_1} \gamma(s)\mathbf{v}(s) ds. \end{aligned}$$

Thus between coordinate times  $t_0$  and  $t$  the clock records [\[9, Eq. \(7.19\)\]](#)

$$\boxed{s(t) - s(t_0) = \int_{t_0}^t \frac{dt'}{\gamma(t')} = \int_{t_0}^t \sqrt{1 - \mathbf{v}(t')^2} dt'.} \quad (6.11)$$

This is the standard world-line proper-time integral. For constant  $U$ , (6.10) is the straight line  $X(s) = X(s_0) + (s - s_0)U$ . For varying  $U$ , integrate the tangent; an endpoint boost alone does not specify the path.

### 6.3.2 Clock comparison

Let  $X_0 = X(s_0)$  and  $X_1 = X(s_1)$  be fixed, time-like separated events. Write  $X_k = t_k + \mathbf{r}_k \cdot \boldsymbol{\sigma}$  for  $k = 0, 1$ . In their separation's rest frame,  $\mathbf{r}_1 = \mathbf{r}_0$  and  $t_1 - t_0 = \sqrt{\det(X_1 - X_0)}$ . Since  $\sqrt{1 - \mathbf{v}^2} \leq 1$ , (6.11) gives [7, p. 122]

$$s_1 - s_0 \leq \sqrt{\det(X_1 - X_0)}. \quad (6.12)$$

Equality requires the straight inertial path. The determinant of a finite displacement therefore does not generally equal the squared elapsed proper time of an accelerated clock.

For a return trip lasting real coordinate time  $T > 0$ , take velocities  $+\beta\mathbf{u}$  and  $-\beta\mathbf{u}$  for  $T/2$  each, with **real unit vector**  $\mathbf{u}$  and  $0 < \beta < 1$ . The ideal instantaneous reversal is the limit of a short smooth turn. The traveling clock records

$$s(T) - s(0) = T\sqrt{1 - \beta^2},$$

while the clock at rest records  $T$ . This is the twin-clock comparison [9, p. 121].

### 6.3.3 Constant proper acceleration

Recall the real proper velocity  $U = dX/ds$ , a **unit paravector** with  $UU^{-1} = 1$ . Let  $a > 0$  be a constant real proper-acceleration magnitude and  $\mathbf{u}$  a fixed **real unit axis**. Starting at  $X(0) = 0$  and  $U(0) = 1$ , take rapidity  $\theta(s) = as$ . Then

$$U(s) = L(as, \mathbf{u})^{-2} = \cosh(as) + \sinh(as)\mathbf{u} \cdot \boldsymbol{\sigma}.$$

Differentiation verifies the stated acceleration:

$$-\det\left(\frac{dU}{ds}\right) = a^2(\cosh^2(as) - \sinh^2(as)) = a^2.$$

Integrating (6.10) yields [9, Eq. (7.33), pp. 142–143]

$$\boxed{X(s) = \frac{\sinh(as)}{a} + \frac{\cosh(as) - 1}{a}\mathbf{u} \cdot \boldsymbol{\sigma}.} \quad (6.13)$$

Its components give

$$\begin{aligned} \mathbf{v}(s) &= \frac{d\mathbf{r}/ds}{dt/ds} = \tanh(as)\mathbf{u}, \\ (1 + a\mathbf{u} \cdot \mathbf{r})^2 - (at)^2 &= 1. \end{aligned}$$

Thus (6.13) is the usual hyperbolic motion, with speed below light speed at every finite  $s$ .

### 6.3.4 Light cones

For a real event separation  $\Delta X = \Delta t + \Delta\mathbf{r} \cdot \boldsymbol{\sigma}$ , the eigenvalues are  $\Delta t \pm \|\Delta\mathbf{r}\|$ . Both are nonnegative precisely in the future causal cone:

$$\Delta t \geq \|\Delta\mathbf{r}\|. \quad (6.14)$$

The boundary has  $\det(\Delta X) = 0$ ; for a **real unit vector**  $\mathbf{u}$ , its nonzero future rays are

$$\Delta X = \Delta t(1 + \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \Delta t > 0, \quad \mathbf{u}^2 = 1.$$

The Hermitian congruence (4.3) preserves nonnegativity of the eigenvalues and the determinant, hence the cone and its boundary. For a future time-like or null path,

$$\|\Delta \mathbf{r}\| = \left\| \int \mathbf{v} dt \right\| \leq \int \|\mathbf{v}\| dt \leq \Delta t.$$

Its endpoints therefore satisfy (6.14); a space-like separation cannot carry a causal signal [9, p. 118].

## 6.4 Free-particle action

Recall real  $m > 0$ ,  $\mathbf{v} = d\mathbf{r}/dt$ , and  $ds = dt/\gamma$ . Here  $L(t, \mathbf{r}, \mathbf{v})$  is a real scalar Lagrangian, not the boost factor;  $\partial_{\mathbf{v}}$  denotes its vector derivative with respect to the three independent velocity components. The free action is the physical choice [7, Eqs. (5.38)–(5.39)]

$$\int L dt = -m \int ds. \quad (6.15)$$

Using  $ds = dt/\gamma$  gives the Lagrangian

$$L = -m\sqrt{1 - \mathbf{v}^2} = -m/\gamma.$$

For endpoint-fixed real path variations  $\delta \mathbf{r}(t)$ , varying the Lagrangian and integrating by parts gives

$$\delta \int L dt = \int \left( \partial_{\mathbf{r}} L - \frac{d}{dt} \partial_{\mathbf{v}} L \right) \cdot \delta \mathbf{r} dt.$$

Stationarity for every variation yields the Euler–Lagrange equation. With real Hamiltonian  $H := \mathbf{v} \cdot \partial_{\mathbf{v}} L - L$ , the chain rule gives

$$\frac{d}{dt} \partial_{\mathbf{v}} L = \partial_{\mathbf{r}} L, \quad (6.16)$$

$$\frac{dH}{dt} = \mathbf{v} \cdot \left( \frac{d}{dt} \partial_{\mathbf{v}} L - \partial_{\mathbf{r}} L \right) - \partial_t L = -\partial_t L. \quad (6.17)$$

By (6.16) and (6.17), spatial and temporal translation invariance conserve canonical momentum and energy. For the free action,

$$\begin{aligned} \partial_{\mathbf{v}} L &= m\gamma \mathbf{v} = \mathbf{p}, \\ H &= m\gamma \mathbf{v}^2 + m/\gamma = m\gamma = E. \end{aligned}$$

Write  $v = \|\mathbf{v}\| \ll 1$ . Here  $O(v^n)$  denotes terms bounded by a constant times  $v^n$  as  $v \rightarrow 0$ . The binomial series gives

$$\begin{aligned} \sqrt{1 - v^2} &= 1 - \frac{1}{2}v^2 + O(v^4), \\ \gamma &= 1 + \frac{1}{2}v^2 + O(v^4), \\ L &= -m + \frac{1}{2}mv^2 + O(mv^4), \\ H &= m + \frac{1}{2}mv^2 + O(mv^4). \end{aligned}$$

The exact kinetic energy above rest is

$$K := H - m = m(\gamma - 1).$$

This and both expansions recover  $mv^2/2$  at low speed after removal of the constant rest term.

## 6.5 Flat spacetime in generalized coordinates

Here  $(t, \mathbf{r})$  denote generalized real chart coordinates. Smooth real functions  $f(t, \mathbf{r})$  and  $\mathbf{g}(t, \mathbf{r})$ , scalar and vector respectively, specify inertial coordinates on a coordinate domain in  $\mathbb{R}^4$  through the real paravector

$$X = f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}.$$

Along a future timelike path with proper time  $s$ , recall the real  $U = dX/ds$ , a [unit paravector](#) with  $UU^{-1} = 1$ . Require the Jacobian of  $(f, \mathbf{g})$  to be nonsingular. Differentiating gives

$$\begin{aligned} dX &= (\partial_t f + \partial_t \mathbf{g} \cdot \boldsymbol{\sigma}) dt + \sum_{k=1}^3 (\partial_{r_k} f + \partial_{r_k} \mathbf{g} \cdot \boldsymbol{\sigma}) dr_k, \\ ds^2 &= dX dX^* = df^2 - d\mathbf{g}^2. \end{aligned}$$

With  $x_0 = t$ ,  $x_k = r_k$  and  $\partial_k = \partial/\partial x_k$ , the interval is  $ds^2 = \sum_{j,k=0}^3 g_{jk} dx_j dx_k$ , with real coefficients

$$g_{jk} = \partial_j f \partial_k f - \partial_j \mathbf{g} \cdot \partial_k \mathbf{g}, \quad j, k = 0, 1, 2, 3.$$

This is the flat Minkowski metric in changed coordinates; varying coefficients do not imply curvature.

Along this future timelike curve,  $df > 0$ . Define its real inertial three-velocity  $\mathbf{v} = d\mathbf{g}/df$  and  $\gamma = (1 - \mathbf{v}^2)^{-1/2}$ , with  $\mathbf{v}^2 < 1$ . Then

$$\begin{aligned} ds &= \sqrt{1 - \mathbf{v}^2} df, \\ U &= \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Apply (6.7) with inertial time  $f$  and coordinate acceleration  $d\mathbf{v}/df$ . For inertial motion, the scalar part gives  $\mathbf{v} \cdot d\mathbf{v}/df = 0$ , and its vector part then gives  $d\mathbf{v}/df = 0$ . Therefore

$$\mathbf{g} = \mathbf{C}_1 f + \mathbf{C}_2.$$

The real constants  $\mathbf{C}_1, \mathbf{C}_2 \in \mathbb{R}^3$  are inertial velocity and position at inertial time  $f = 0$ , with  $\mathbf{C}_1^2 < 1$ . If primes denote total derivatives along this curve with respect to  $t$ , then, wherever  $f' \neq 0$ ,

$$\frac{d\mathbf{v}}{dt} = \frac{\mathbf{g}'' f' - \mathbf{g}' f''}{(f')^2}.$$

## 6.6 Varying frames and local bases

Use a fixed coordinate chart and let the local frame vary along a curve. Let  $T(s)$  be a smooth complex [unit paravector](#) of [determinant one](#) along a real parameter  $s$ . When  $s$  is proper time, recall the real tangent  $U = dX/ds$ , a [unit paravector](#) with  $UU^{-1} = 1$ . Define the complex pure-vector paravector  $\Omega(s) := (dT/ds)T^{-1}$ . Its scalar part vanishes because  $d(\det T)/ds = 2\text{sc}(\Omega) = 0$ . The inverse rule gives

$$\frac{dT^{-1}}{ds} = -T^{-1}\Omega.$$

For a complex paravector  $Z(s)$  transformed by similarity,  $Z' = TZT^{-1}$ , product differentiation yields

$$\frac{dZ'}{ds} = T \frac{dZ}{ds} T^{-1} + [\Omega, Z']. \quad (6.18)$$

Differentiating again and replacing  $T(dZ/ds)T^{-1}$  by its first-derivative expression gives

$$\begin{aligned}\frac{d^2 Z'}{ds^2} &= T \frac{d^2 Z}{ds^2} T^{-1} + 2 \left[ \Omega, \frac{dZ'}{ds} \right] \\ &\quad + \left[ \frac{d\Omega}{ds}, Z' \right] - [\Omega, [\Omega, Z']].\end{aligned}$$

For a real paravector transformed instead by congruence,  $Z' = TZT^H$ , the same rule gives

$$\frac{dZ'}{ds} = T \frac{dZ}{ds} T^H + \Omega Z' + Z' \Omega^H. \quad (6.19)$$

Equations (6.18) and (6.19) include the frame terms omitted by a constant-factor transformation. For a varying rotation  $T = R = \exp(-i\theta(s)\mathbf{u}(s) \cdot \boldsymbol{\sigma}/2)$ , with real  $\theta(s)$  and **real unit vector**  $\mathbf{u}(s)$ , unitarity gives  $\Omega^H = -\Omega$ , so both formulas agree.

When  $s$  is proper time, the physical acceleration in the local frame is  $T(dU/ds)T^H$ . In (6.19),  $dU'/ds$  also differentiates the frame; the two  $\Omega$  terms account for this change. Likewise, differentiating the transformed event  $X'$  adds frame terms to its tangent. The similarity formula also gives the local algebra-basis law  $\sigma'_k = T\sigma_k T^{-1}$ , with

$$\frac{d\sigma'_k}{ds} = [\Omega, \sigma'_k], \quad k = 1, 2, 3.$$

This evolution preserves all generator products. A general complex local basis is nevertheless distinct from a real spacetime frame, whose vectors require the congruence law.

## 7 Electromagnetism

All coefficient fields depend on real inertial coordinates  $(t, \mathbf{r})$ :  $V, \rho : \mathbb{R}^4 \rightarrow \mathbb{R}$  are scalar fields and  $\mathbf{E}, \mathbf{B}, \mathbf{A}, \mathbf{J} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$  are vector fields. Define [5, Eq. (23)]

$$\begin{aligned}F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma} && \text{electric and magnetic fields,} \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma} && \text{scalar and vector potentials,} \\ D &= \rho + \mathbf{J} \cdot \boldsymbol{\sigma} && \text{charge and current densities.}\end{aligned} \quad (7.1)$$

Thus  $\Phi, D$  are real paravectors, while  $F$  is a complex pure vector. Recall  $\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}$  and  $\partial^* \partial = \square = \partial_t^2 - \partial_{\mathbf{r}}^2$ .

The dot/cross product (3.18) combines metric and orientation. With a spatial derivative it gives divergence and curl, packing the six real electromagnetic components into one complex field  $F$ . The product explains the compact form; Maxwell dynamics remains physical input, derived from an action in Section 7.3.

### 7.1 Maxwell's equations and wave propagation

Maxwell's four equations are the single field law [8, Eq. (177)]

$$\boxed{\partial F = D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.} \quad (7.2)$$

The product rule expands its left side into

$$\begin{aligned}\partial F = & \underbrace{\partial_{\mathbf{r}} \cdot \mathbf{E}}_{\text{real scalar}} + \underbrace{i\partial_{\mathbf{r}} \cdot \mathbf{B}}_{\text{imaginary scalar}} \\ & + \underbrace{(\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}}_{\text{real vector}} \\ & + \underbrace{i(\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot \boldsymbol{\sigma}}_{\text{imaginary vector}}.\end{aligned}$$

Equating the four parts gives [7, Eqs. (1.4)–(1.7)]

$$\partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad (7.3)$$

$$\partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (7.4)$$

$$\partial_{\mathbf{r}} \times \mathbf{B} - \partial_t \mathbf{E} = \mathbf{J}, \quad (7.5)$$

$$\partial_{\mathbf{r}} \times \mathbf{E} + \partial_t \mathbf{B} = 0. \quad (7.6)$$

Equations (7.3)–(7.6) are Gauss’ electric and magnetic laws, Ampère–Maxwell’s law, and Faraday’s law. Applying  $\partial^*$  to (7.2) gives [8, Eq. (152)]

$$\begin{aligned}\square F &= \partial^* D^* = (\partial D)^* \\ &= (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) + (-\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho + i\partial_{\mathbf{r}} \times \mathbf{J}) \cdot \boldsymbol{\sigma}.\end{aligned} \quad (7.7)$$

The scalar part vanishes because  $F$  is scalar-free. Thus [7, Eq. (1.1)]

$$\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0, \quad (7.8)$$

$$\square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho, \quad (7.9)$$

$$\square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}. \quad (7.10)$$

Equation (7.8) is charge conservation; (7.9) and (7.10) are the sourced field waves. In vacuum,  $D = 0$  and both fields obey the free wave equation.

## 7.2 Potentials and gauge freedom

Recall the real potential  $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ . Its derivative is

$$\partial \Phi = (\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}) + (\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma} + i(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}.$$

Define the real scalar  $S := \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$ . Identifying the remaining components with  $F^*$  gives [7, p. 105]

$$\partial \Phi = S + F^*, \quad (7.11)$$

$$\mathbf{E} = -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \quad (7.12)$$

$$\mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}. \quad (7.13)$$

Equation (7.11) separates the gauge scalar from the field; (7.12) and (7.13) satisfy (7.4) and (7.6) identically.

Applying  $\partial^*$  gives

$$\square\Phi = \partial^*(\partial\Phi) = \partial^*(S + F^*) = \partial^*S + (\partial F)^*.$$

Conjugating and using (7.2) gives [7, Eq. (6.1)]

$$\boxed{\square\Phi^* - \partial S = \partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.}$$

Its scalar and vector components are

$$\begin{aligned}\rho &= \square V - \partial_t S, \\ \mathbf{J} &= \square \mathbf{A} + \partial_r S.\end{aligned}$$

These source equations hold before gauge fixing. For a real scalar gauge function  $\lambda(t, \mathbf{r})$ , let [7, Eq. (6.2)]

$$\Phi' := \Phi + \partial^* \lambda.$$

Differentiating gives  $\partial\Phi' = \partial\Phi + \square\lambda$ . Only the scalar changes:  $S' = S + \square\lambda$ , while  $F' = F$ . In the Lorenz gauge  $S = 0$ , the source equations reduce to  $\square V = \rho$  and  $\square \mathbf{A} = \mathbf{J}$  [7, Eq. (6.4)].

### 7.3 The electromagnetic field action

For real scalar potential  $V(t, \mathbf{r})$  and vector potential  $\mathbf{A}(t, \mathbf{r})$ , write  $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ . With prescribed conserved current  $D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}$ , Maxwell dynamics follows from the real Lagrangian density [7, Eq. (5.42)] [7, Eq. (5.43)]

$$L = \frac{1}{2} \text{re}(F^2) - \text{sc}(\Phi^* D) = \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) - \rho V + \mathbf{J} \cdot \mathbf{A}. \quad (7.14)$$

In varying the action, boundary-vanishing variations satisfy

$$\begin{aligned}\delta \mathbf{E} &= -\partial_r \delta V - \partial_t \delta \mathbf{A}, \\ \delta \mathbf{B} &= \partial_r \times \delta \mathbf{A}.\end{aligned}$$

Integration by parts gives

$$\delta \int L \, dt \, d^3 \mathbf{r} = \int ((\partial_r \cdot \mathbf{E} - \rho) \delta V + (\partial_t \mathbf{E} - \partial_r \times \mathbf{B} + \mathbf{J}) \cdot \delta \mathbf{A}) \, dt \, d^3 \mathbf{r}.$$

Independent variations give (7.3) and (7.5); the potential identities supply (7.4) and (7.6). Together they recover (7.2).

### 7.4 Field invariants

Since  $\text{sc}(F) = 0$ , its adjugate is  $-F$ . Squaring gives [3, p. 106]

$$F^2 = -\det F = \mathbf{E}^2 - \mathbf{B}^2 + 2i\mathbf{E} \cdot \mathbf{B}. \quad (7.15)$$

For a complex unit paravector  $T$ , the field similarity  $F' = T F T^{-1}$  preserves this scalar by (4.1). Physical rotations and boosts use  $R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$  and  $L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ , respectively, with real unit vector  $\mathbf{u}$  and real angle or rapidity  $\theta$ ; their field laws are derived in Section 11. The real and imaginary parts of (7.15) are therefore observer-independent. At a point with a nonzero field, they give the classification

| Condition                                                      | Consequence                              |
|----------------------------------------------------------------|------------------------------------------|
| $\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 > \mathbf{B}^2$ | Some frame has $\mathbf{B}' = 0$ .       |
| $\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 < \mathbf{B}^2$ | Some frame has $\mathbf{E}' = 0$ .       |
| $\mathbf{E} \cdot \mathbf{B} \neq 0$                           | Neither field can be eliminated.         |
| $\mathbf{E} \cdot \mathbf{B} = 0, \mathbf{E}^2 = \mathbf{B}^2$ | Null field: equal, perpendicular fields. |

For the first two rows, choose respectively

$$\tanh \theta \mathbf{u} = \frac{\mathbf{E} \times \mathbf{B}}{\mathbf{E}^2}, \quad \tanh \theta \mathbf{u} = \frac{\mathbf{E} \times \mathbf{B}}{\mathbf{B}^2}.$$

Both speeds are less than one. Substitution in (11.17) and (11.18) gives  $\mathbf{B}' = 0$  and  $\mathbf{E}' = 0$ , respectively.

## 7.5 Lorentz force from the action

Recall the real proper velocity  $U = dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma})$ , a [unit paravector](#) with  $UU^{-1} = 1$ . Let  $q \in \mathbb{R}$  be charge and  $m > 0$  constant mass. The real momentum is  $P = mU = E + \mathbf{p} \cdot \boldsymbol{\sigma}$ , with  $E = m\gamma$  and  $\mathbf{p} = m\gamma\mathbf{v}$ . The real particle Lagrangian  $L$  is fixed by the action [7, Eq. (5.41)]

$$\int L dt = -m \int ds - q \int \text{sc}(\Phi^* U) ds. \quad (7.16)$$

Since  $\text{sc}(\Phi^* U) = \gamma(V - \mathbf{v} \cdot \mathbf{A})$ , the Lagrangian is

$$L = -m/\gamma - qV + q\mathbf{v} \cdot \mathbf{A}.$$

Its Euler–Lagrange terms are

$$\begin{aligned} \partial_v L &= \mathbf{p} + q\mathbf{A}, \\ \frac{d}{dt} \partial_v L &= \frac{d\mathbf{p}}{dt} + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \partial_r) \mathbf{A}, \\ \partial_r L &= -q\partial_r V + q\partial_r(\mathbf{v} \cdot \mathbf{A}). \end{aligned}$$

Here spatial differentiation holds the independent velocity argument fixed. Using  $\partial_r(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_r) \mathbf{A} = \mathbf{v} \times (\partial_r \times \mathbf{A})$  gives [7, Eq. (5.32)]

$$\frac{d\mathbf{p}}{dt} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (7.17)$$

Here (7.12) and (7.13) turn the potential derivatives into the fields. The Hamiltonian is

$$H = \mathbf{v} \cdot (\mathbf{p} + q\mathbf{A}) - L = E + qV. \quad (7.18)$$

Using (6.17) in (7.18),

$$\frac{dE}{dt} + q\partial_t V + q\mathbf{v} \cdot \partial_r V = -\partial_t L = q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}.$$

Canceling  $q\partial_t V$  and using (7.12) gives [7, p. 116]

$$\frac{dE}{dt} = q\mathbf{v} \cdot \mathbf{E}. \quad (7.19)$$

This is the mechanical work rate. Thus  $\mathbf{p} + q\mathbf{A}$  and  $H$  are canonical momentum and energy;  $\mathbf{p}$ ,  $E$  are mechanical momentum and energy, with  $E$  including rest energy. Canonical energy is conserved for static potentials in the chosen gauge. A time-dependent gauge transformation can change  $H$  while leaving  $E$  and the force unchanged.

Now expand the complex product of the field and proper velocity:

$$\begin{aligned} \frac{FU}{\gamma} &= \mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma} \\ &+ i(\mathbf{v} \cdot \mathbf{B} + (\mathbf{B} - \mathbf{v} \times \mathbf{E}) \cdot \boldsymbol{\sigma}). \end{aligned} \quad (7.20)$$

The real scalar and vector parts reproduce (7.19) and (7.17). Since  $P$  is real and  $d/ds = \gamma d/dt$ , they combine into [10, Eq. (2)]

$$\boxed{\frac{dP}{ds} = q \operatorname{re}(FU) = \frac{q}{2}(FU + UF^H).} \quad (7.21)$$

Hermitian extraction enforces real mechanical momentum and reproduces the action-derived force. The imaginary part need not vanish for an electric charge.

## 7.6 Acceleration and the mass shell

Recall the real proper velocity  $U = dX/ds$ , a [unit paravector](#) with  $UU^{-1} = 1$  and  $U^{-1} = U^*$ , real momentum  $P = mU$ , and coordinate acceleration  $\mathbf{a} = d\mathbf{v}/dt$ . Scalar-part cyclicity proves tangency to the mass shell:

$$\begin{aligned} \operatorname{sc}(U^* dP/ds) &= \frac{q}{2} \operatorname{sc}(U^* FU + F^H) \\ &= \frac{q}{2} (\operatorname{sc}(F) + \operatorname{sc}(F^H)) = 0. \end{aligned}$$

This agrees with  $\operatorname{sc}(U^* dU/ds) = 0$  in Section 6.2. Combining (6.7) with the force components in (7.20) gives

$$\begin{aligned} m\gamma^3 \mathbf{v} \cdot \mathbf{a} &= q\mathbf{v} \cdot \mathbf{E}, \\ m\gamma (\mathbf{a} + \gamma^2(\mathbf{v} \cdot \mathbf{a})\mathbf{v}) &= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \end{aligned}$$

Eliminating  $\mathbf{v} \cdot \mathbf{a}$  yields the coordinate acceleration

$$\boxed{\mathbf{a} = \frac{q}{m\gamma} (\mathbf{E} + \mathbf{v} \times \mathbf{B} - (\mathbf{v} \cdot \mathbf{E})\mathbf{v}).}$$

The magnetic term changes momentum without doing work. Taking the determinant of the real force gives

$$\begin{aligned} -\det\left(\frac{dP}{ds}\right) &= -m^2 \det\left(\frac{dU}{ds}\right) \\ &= q^2 \gamma^2 ((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2). \end{aligned}$$

Dividing by  $m^2$  gives the invariant squared proper acceleration. The factor  $\gamma^2((\mathbf{E} + \mathbf{v} \times \mathbf{B})^2 - (\mathbf{v} \cdot \mathbf{E})^2)$  is the squared electric field in the particle's instantaneous rest frame.

## 7.7 Energy and momentum flux

Recall  $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ . Its Hermitian product is [8, Eq. (180)]

$$\frac{1}{2}FF^H = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.$$

Define the real energy density  $u = (\mathbf{E}^2 + \mathbf{B}^2)/2$  and real Poynting vector  $\mathbf{S} = \mathbf{E} \times \mathbf{B}$ . These coefficients are frame-dependent, whereas  $F^2$  contains the Lorentz invariants. The full stress–energy tensor is represented by the real linear map [8, Eq. (167)]

$$T(N) := \frac{1}{2}FN F^H, \quad (7.22)$$

with real paravector input  $N$ . Its time-direction value is  $T(1) = u + \mathbf{S} \cdot \boldsymbol{\sigma}$ . It is future causal because  $u \geq 0$  and

$$\det(T(1)) = u^2 - \mathbf{S}^2 = \frac{1}{4}\|\det F\|^2 \geq 0.$$

Equality holds for null fields. Reversing the order gives  $F^H F = (FF^H)^*$ : the same energy density and opposite flux. The transformation of the full map is derived in Section 11.

Using Maxwell’s vector equations,

$$\begin{aligned} \partial_t u &= \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B} - \mathbf{J}) - \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) \\ &= -\partial_{\mathbf{r}} \cdot \mathbf{S} - \mathbf{J} \cdot \mathbf{E}, \end{aligned}$$

where  $\partial_{\mathbf{r}} \cdot (\mathbf{E} \times \mathbf{B}) = \mathbf{B} \cdot (\partial_{\mathbf{r}} \times \mathbf{E}) - \mathbf{E} \cdot (\partial_{\mathbf{r}} \times \mathbf{B})$ . Thus Poynting’s energy-balance law is [7, Eq. (5.45)]

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (7.23)$$

Equivalently, the left side is  $\frac{1}{2}\partial_t \text{sc}(FF^H) + \frac{1}{2}\partial_{\mathbf{r}} \cdot \text{vec}(FF^H)$ . The term  $\mathbf{J} \cdot \mathbf{E}$  is the power delivered to matter per unit volume.

## 8 Relativistic wave equations

Here  $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$  has real coordinates  $(t, \mathbf{r}) \in \mathbb{R}^4$ ,  $m > 0$  is mass,  $q \in \mathbb{R}$  is charge, and  $\hbar = c = 1$ . Amplitudes are complex functions of these real coordinates; their scalar or paravector type is specified below.

### 8.1 Amplitudes, densities, and local expectations

All products and norms here act on paravector values at one real event  $X$ . They sum algebra components only; no spatial integration is implied. For complex paravectors  $\varphi, \psi$ , their algebraic inner product is  $\langle \varphi | \psi \rangle := \text{sc}(\varphi^H \psi)$ . Two entries separated by a vertical bar denote this inner product; one entry in angle brackets denotes a normalized expectation below.

For arbitrary constant perpendicular **real unit vectors**  $\mathbf{u}, \mathbf{a}$ , choose a constant complex paravector  $\psi_+$  of unit norm with  $(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_+ = \psi_+$ , and set  $\psi_- = (\mathbf{a} \cdot \boldsymbol{\sigma})\psi_+$ . Then

$$(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_{\pm} = \pm\psi_{\pm}. \quad (8.1)$$

The pair is orthonormal:  $\text{sc}(\psi_a^H \psi_b) = \delta_{ab}$  for  $a, b = +, -$ . A Pauli amplitude or either Dirac entry is [11, Eq. (69a)]

$$\psi(t, \mathbf{r}) = c_+(t, \mathbf{r})\psi_+ + c_-(t, \mathbf{r})\psi_-. \quad (8.2)$$

Here  $c_+, c_- : \mathbb{R}^4 \rightarrow \mathbb{C}$ . Both Dirac entries use the same fixed pair. Left multiplication by  $\mathbf{u} \cdot \boldsymbol{\sigma}$  or  $\mathbf{a} \cdot \boldsymbol{\sigma}$  preserves this two-component form; these generate all paravector left actions. The matrix verification is in Appendix A.5.

The real norm density is [12, Eq. (1.1a)]

$$\rho := \|\psi\|^2 = \text{sc}(\psi^H \psi) \geq 0. \quad (8.3)$$

Orthonormality gives  $\rho = \|c_+\|^2 + \|c_-\|^2$ . For a Pauli wavefunction normalized to unit total probability, this is its position density; Dirac uses the sum over its two entries. Total spatial normalization is separate from the pointwise norm. This probability interpretation does not apply to an arbitrary scalar KG field.

**Nonnegative weighted densities.** Let  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$  be real ( $Z^H = Z$ ), with  $S \in \mathbb{R}$  and  $\mathbf{V} \in \mathbb{R}^3$ . It acts by multiplication, not differentiation. For  $\mathbf{V} \neq 0$ , take the real unit vector  $\mathbf{u} = \mathbf{V}/\|\mathbf{V}\|$  and recall the projectors  $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$  from Section 3.7. Their spectral decomposition gives

$$\text{sc}(\psi^H Z \psi) = (S + \|\mathbf{V}\|)\|\Pi_+ \psi\|^2 + (S - \|\mathbf{V}\|)\|\Pi_- \psi\|^2. \quad (8.4)$$

For  $\mathbf{V} = 0$ , this reduces to  $S\|\psi\|^2$ . Thus the product is nonnegative for every  $\psi$  exactly when  $S \geq \|\mathbf{V}\|$ . For such a *nonnegative weight*, define

$$\boxed{\|\psi\|_Z^2 := \text{sc}(\psi^H Z \psi) = \|\sqrt{Z} \psi\|^2 \geq 0.} \quad (8.5)$$

The square root has nonnegative eigenvalues, as in Section 3.8. For  $Z = 1$ , this is  $\rho$ . It is a true squared norm when  $S > \|\mathbf{V}\|$ ; at equality it can vanish for nonzero  $\psi$ , so the notation then denotes a weighted density. Projectors are an important example. Reality alone is insufficient:  $Z = -1$  gives  $\text{sc}(\psi^H Z \psi) = -\rho$ .

**Local expectations.** For  $\rho > 0$  and any complex multiplying paravector  $Z$ , define [13, Eq. (6.1)]

$$\boxed{\langle Z \rangle_{\psi} := \frac{\text{sc}(\psi^H Z \psi)}{\rho}.} \quad (8.6)$$

For real  $Z$ , this is real and may have either sign. Dividing (8.4) by  $\rho$  shows its meaning: it averages the two eigenvalues with their projected probabilities. For fields this mean is conditional on position; it is not a spatial average. A whole-wavefunction expectation additionally requires spatial weighting. The same formula extends linearly to complex  $Z$ , including commutators. For a nonnegative weight,  $\langle Z \rangle_{\psi} = \|\psi\|_Z^2/\rho$ . For a projector, this ratio is a Born probability.

**Transforming the amplitude.** For a constant complex *unit paravector*  $T$ , set  $\psi' = T\psi$ . For a nonnegative weight  $Z$ , substitution gives

$$\boxed{\|\psi'\|_Z^2 = \text{sc}(\psi^H T^H Z T \psi) = \|\psi\|_{T^H Z T}^2.} \quad (8.7)$$

This also proves that the transformed weight is nonnegative. The same substitution in (8.6) applies to any complex  $Z$ . With  $\rho' = \|\psi'\|^2 > 0$ , it gives

$$\boxed{\langle Z \rangle_{\psi'} = \frac{\rho}{\rho'} \langle T^H Z T \rangle_{\psi}.} \quad (8.8)$$

For  $T = R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ , with real angle  $\theta$  and real unit  $\mathbf{u}$ , unitarity preserves both  $\rho' = \rho$  and  $\langle \varphi' | \psi' \rangle = \langle \varphi | \psi \rangle$ . For  $T = L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ , with real rapidity  $\theta$ , this one-sided law instead gives  $\rho' = \|\psi\|_{L^2}^2$ ; the positive norm and inner product are generally not invariant. The two Dirac entries use opposite boost factors, as derived in Section 11.

## 8.2 Energy and momentum as differential operators

Recall  $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ , with real coordinates  $(t, \mathbf{r}) \in \mathbb{R}^4$ . Temporarily restore Planck's constant  $h > 0$  and  $\hbar = h/(2\pi)$ , keeping  $c = 1$ . For a free wave, let  $\omega \in \mathbb{R}$  be angular frequency ( $2\pi$  times cycles per unit time), and let  $\mathbf{k} \in \mathbb{R}^3$  be its wavenumber vector in cycles per unit length. For  $\mathbf{k} \neq 0$ , the wavelength  $\lambda > 0$  satisfies  $\|\mathbf{k}\| = 1/\lambda$ .

The physical input is the Planck–Einstein energy relation and de Broglie momentum relation [14, p. 248] [14, p. 249]:

$$E = \hbar\omega, \tag{8.9}$$

$$\mathbf{p} = \hbar\mathbf{k} = 2\pi\hbar\mathbf{k}. \tag{8.10}$$

Here  $E \in \mathbb{R}$  is total energy, including rest energy, and  $\mathbf{p} \in \mathbb{R}^3$  is momentum; its magnitude is  $h/\lambda$ . Photoelectric measurements test the energy relation [15, p. 62]; electron diffraction tests the momentum relation [14, p. 255].

For the constant real four-momentum  $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$ , the complex scalar plane wave is

$$\begin{aligned} f(X, P) &:= \exp(-i(\omega t - 2\pi\mathbf{k} \cdot \mathbf{r})) \\ &= \exp\left(-\frac{i}{\hbar}(Et - \mathbf{p} \cdot \mathbf{r})\right). \end{aligned}$$

The second line substitutes (8.9) and (8.10). Differentiating gives

$$\begin{aligned} i\hbar\partial_t f &= Ef, \\ -i\hbar\partial_{\mathbf{r}} f &= \mathbf{p}f. \end{aligned}$$

Thus  $i\hbar\partial_t$  and  $-i\hbar\partial_{\mathbf{r}}$  represent energy and momentum on each mode, and extend to superpositions by linearity. Returning to  $\hbar = 1$  means  $h = 2\pi$ , giving [16, Eq. (4.104)]

$$f(X, P) = \exp(-i(Et - \mathbf{p} \cdot \mathbf{r})). \tag{8.11}$$

The two eigenvalue equations combine into

$$i\partial^* f = Pf. \tag{8.12}$$

Define the canonical energy and momentum operators by

$$\hat{E} := i\partial_t, \quad \hat{\mathbf{p}} := -i\partial_{\mathbf{r}}, \quad \hat{P} := \hat{E} + \hat{\mathbf{p}} \cdot \boldsymbol{\sigma} = i\partial^*. \tag{8.13}$$

In a field, recall the real potential paravector  $\Phi(t, \mathbf{r}) = V(t, \mathbf{r}) + \mathbf{A}(t, \mathbf{r}) \cdot \boldsymbol{\sigma}$ :  $V$  is a real scalar potential and  $\mathbf{A}$  a real vector potential. The real charge is  $q$ . The particle action gives canonical

energy  $E + qV$  and momentum  $\mathbf{p} + q\mathbf{A}$ ; identifying them with translation operators defines the gauge-covariant mechanical operators [12, Eqs. (2.11)–(2.12)]

$$\hat{E}_q := \hat{E} - qV = i\partial_t - qV, \quad (8.14)$$

$$\hat{\mathbf{p}}_q := \hat{\mathbf{p}} - q\mathbf{A} = -i\partial_{\mathbf{r}} - q\mathbf{A}, \quad (8.15)$$

$$\hat{P}_q := \hat{E}_q + \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}.$$

At  $q = 0$ , the coupled operators reduce to the canonical ones:  $\hat{E}_0 = \hat{E}$ ,  $\hat{\mathbf{p}}_0 = \hat{\mathbf{p}}$ , and  $\hat{P}_0 = \hat{P}$ . Here  $\hat{E}_q$  includes rest energy. Hats distinguish named differential operators from physical values and mode eigenvalues  $E, \mathbf{p}$ . The derivative symbols  $\partial$ ,  $\partial_{\mathbf{r}}$ , and  $\square$  remain unhatted. Fields obtained by differentiating potentials are coefficient fields, not differential operators. The operators act to the right and combine as

$$\boxed{\hat{P}_q = \hat{P} - q\Phi = i\partial^* - q\Phi.} \quad (8.16)$$

Its adjugate reverses the vector part without conjugating  $i$ . Operator order matters in a field; an ordinary paravector determinant cannot replace their product. The canonical operators commute, giving

$$\hat{P} \operatorname{adj}(\hat{P}) = \det \hat{P} = \hat{E}^2 - \hat{\mathbf{p}}^2 = -\square. \quad (8.17)$$

Reversing the order gives the same result.

### 8.2.1 Field strength from noncommuting derivatives

Recall the real electric and magnetic fields from (7.12) and (7.13), and  $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$ , the real scalar in (7.11). The real potential is  $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ ;  $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$  is a complex vector paravector. Apply the operators to a differentiable complex amplitude. The product rule cancels all derivatives of the amplitude in the commutators; writing  $\hat{p}_{q,j} := (\hat{\mathbf{p}}_q)_j$  for Cartesian components  $j, k = 1, 2, 3$ ,

$$\begin{aligned} [\hat{E}_q, \hat{p}_{q,j}] &= -iq(\partial_t A_j + \partial_j V) = iqE_j, \\ [\hat{p}_{q,j}, \hat{p}_{q,k}] &= iq(\partial_j A_k - \partial_k A_j), \\ \hat{\mathbf{p}}_q \times \hat{\mathbf{p}}_q &= iq\mathbf{B}. \end{aligned}$$

Sigma multiplication and the energy commutator consequently give [12, Eq. (2.13)] [12, p. 6]

$$(\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})^2 = \hat{\mathbf{p}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}, \quad (8.18)$$

$$[\hat{E}_q, \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}] = iq\mathbf{E} \cdot \boldsymbol{\sigma}. \quad (8.19)$$

Here  $E_j$  is an electric-field component. Using (8.18) and (8.19) in the ordered products gives

$$\begin{aligned} (\hat{E}_q + \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})(\hat{E}_q - \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}) &= \hat{E}_q^2 - (\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})^2 - [\hat{E}_q, \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}] \\ &= \hat{E}_q^2 - \hat{\mathbf{p}}_q^2 - iqF, \end{aligned} \quad (8.20)$$

$$\begin{aligned} (\hat{E}_q - \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})(\hat{E}_q + \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}) &= \hat{E}_q^2 - (\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})^2 + [\hat{E}_q, \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}] \\ &= \hat{E}_q^2 - \hat{\mathbf{p}}_q^2 - iqF^*. \end{aligned} \quad (8.21)$$

Their common scalar part gives KG; the vector terms give the Dirac spin coupling.

### 8.3 Klein–Gordon equation

Let  $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$  be a scalar field and  $m > 0$  its real mass. For a free real four-momentum,  $\det P = m^2$  means  $E^2 - \mathbf{p}^2 = m^2$ . Replacing its coefficients by the translation operators in (8.12) gives

$$((i\partial)(i\partial^*) - m^2)\psi = 0.$$

Using (8.17), this is the Klein–Gordon equation [16, Eq. (1.13)]:

$$\boxed{(\square + m^2)\psi = 0.} \quad (8.22)$$

#### 8.3.1 Positive and negative frequency

Recall the complex scalar field  $\psi(t, \mathbf{r})$  and real mass  $m > 0$ . Define the positive scalar operator

$$\hat{\Omega} := \sqrt{m^2 - \partial_{\mathbf{r}}^2}, \quad \hat{\Omega}^2 = \det(m + \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}).$$

Since  $\hat{\Omega}$  commutes with  $\partial_t$ , KG factors as

$$\square + m^2 = \partial_t^2 + \hat{\Omega}^2 = (\partial_t - i\hat{\Omega})(\partial_t + i\hat{\Omega}).$$

The reverse product is the same. Its two frequency branches obey

$$(\partial_t \pm i\hat{\Omega})\psi_{\pm} = 0, \quad \psi_{\pm}(t, \mathbf{r}) = e^{\mp i\hat{\Omega}t} \psi_{\pm}(0, \mathbf{r}).$$

Their sum gives the general solution:

$$\boxed{\psi(t, \mathbf{r}) = e^{-i\hat{\Omega}t} \psi_+(0, \mathbf{r}) + e^{i\hat{\Omega}t} \psi_-(0, \mathbf{r}).} \quad (8.23)$$

The two initial data determine its complex branch fields by

$$\psi_{\pm}(0, \mathbf{r}) = \frac{1}{2} \left( \psi(0, \mathbf{r}) \pm i\hat{\Omega}^{-1} \partial_t \psi(0, \mathbf{r}) \right).$$

#### 8.3.2 Square roots from Fourier series

For spatially periodic free KG solutions, write the discrete Fourier series

$$\psi = \sum_n c_n f_n, \quad f_n(t, \mathbf{r}) = \exp(i(\omega_n t + 2\pi \mathbf{k}_n \cdot \mathbf{r})).$$

Here  $n$  labels modes,  $c_n \in \mathbb{C}$  are constant coefficients,  $\omega_n \in \mathbb{R}$  are signed angular phase frequencies, and  $\mathbf{k}_n \in \mathbb{R}^3$  are wavenumber vectors in cycles per unit length. For  $\mathbf{k}_n \neq 0$ , its real wavelength  $\lambda_n > 0$  satisfies  $\|\mathbf{k}_n\| = 1/\lambda_n$ . By (8.10), the mode momentum is  $2\pi \mathbf{k}_n$  in our units. The  $+\omega_n t$  convention gives energy eigenvalue  $-\omega_n$ . Differentiation gives

$$\begin{aligned} \square \psi &= \sum_n (4\pi^2 \mathbf{k}_n^2 - \omega_n^2) c_n f_n, \\ (\square + m^2) \psi &= \sum_n (4\pi^2 \mathbf{k}_n^2 + m^2 - \omega_n^2) c_n f_n. \end{aligned}$$

Thus each nonzero KG coefficient requires

$$\omega_n^2 = 4\pi^2 \mathbf{k}_n^2 + m^2, \quad \omega_n = \pm \sqrt{4\pi^2 \mathbf{k}_n^2 + m^2}.$$

Both signs are independent modes [16, pp. 33–34]; no common time period is assumed. A square-root operator acts on the same coefficients by

$$\sqrt{\square} \psi = \sum_n \sqrt{4\pi^2 \mathbf{k}_n^2 - \omega_n^2} c_n f_n.$$

Choose one square-root branch consistently. Squaring restores  $\square$ , giving

$$\square + m^2 = (m - i\sqrt{\square})(m + i\sqrt{\square}).$$

For the positive spatial root, the same series gives

$$\begin{aligned} \hat{\Omega}^2 \psi &= \sum_n (m^2 + 4\pi^2 \mathbf{k}_n^2) c_n f_n, \\ \hat{\Omega} \psi &= \sum_n \sqrt{m^2 + 4\pi^2 \mathbf{k}_n^2} c_n f_n. \end{aligned}$$

Thus  $e^{\mp i \hat{\Omega} t}$  multiplies each spatial mode by  $e^{\mp i \sqrt{m^2 + 4\pi^2 \mathbf{k}_n^2} t}$ . Unlike  $\hat{\Omega}$ ,  $\sqrt{\square}$  has imaginary eigenvalues on KG modes; its branch does not select the frequency sign.

### 8.3.3 Potential

Recall the real potentials  $V, \mathbf{A}$ , charge  $q$ , and kinetic operators  $\hat{E}_q = i\partial_t - qV$ ,  $\hat{\mathbf{p}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$ . The real potential paravector is  $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ , with real scalar  $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$ . The scalar equation in a field is [17, Eq. (299)]

$$\boxed{(-\hat{E}_q^2 + \hat{\mathbf{p}}_q^2 + m^2) \psi = 0.} \quad (8.24)$$

The mass-shell equation uses the scalar part  $\hat{E}_q^2 - \hat{\mathbf{p}}_q^2$  of (8.20) or (8.21). The product rule gives

$$\hat{E}_q^2 = -\partial_t^2 - iq(\partial_t V) - 2iqV\partial_t + q^2 V^2, \quad (8.25)$$

$$\hat{\mathbf{p}}_q^2 = -\partial_{\mathbf{r}}^2 + iq(\partial_{\mathbf{r}} \cdot \mathbf{A}) + 2iq\mathbf{A} \cdot \partial_{\mathbf{r}} + q^2 \mathbf{A}^2. \quad (8.26)$$

Substituting (8.25) and (8.26) into (8.24) gives

$$\boxed{(\square + iqS - q^2 \det \Phi + m^2 + 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})) \psi = 0.}$$

All potential derivatives are retained.

## 8.4 Dirac equation

### 8.4.1 Block form and free factorization

Recall  $m > 0$  and real space–time coordinates  $(t, \mathbf{r})$ . A Dirac field is the two-entry block

$$\Psi(t, \mathbf{r}) = \begin{Bmatrix} \Psi_+(t, \mathbf{r}) \\ \Psi_-(t, \mathbf{r}) \end{Bmatrix}.$$

Each entry is a complex paravector amplitude of (8.2); the block therefore has four independent complex components. On blocks, Hermitian conjugation also transposes entries; sigma conjugation acts entrywise. Summing the two local squared norms gives the nonnegative real density [16, Eqs. (4.97)–(4.99)]

$$\|\Psi\|^2 = \text{sc}(\Psi_+^H \Psi_+) + \text{sc}(\Psi_-^H \Psi_-). \quad (8.27)$$

For a complex paravector  $Z$ , define the complex-linear block map

$$\mathbf{W}(Z) := \begin{Bmatrix} 0 & \text{adj}(Z) \\ Z & 0 \end{Bmatrix}. \quad (8.28)$$

Here  $\mathbf{I}$  is the two-entry block identity. Applying  $\mathbf{W}(Z)$  twice multiplies the first entry by  $\text{adj}(Z) Z$  and the second by  $Z \text{adj}(Z)$ . Both products are the paravector determinant, so

$$\mathbf{W}(Z)^2 = \det(Z) \mathbf{I}.$$

Sigma conjugation on each block entry gives  $\mathbf{W}(Z)^* = \mathbf{W}(Z^*)$ ; Hermitian conjugation also transposes the block:

$$\mathbf{W}(Z)^H = \mathbf{W}(Z^*).$$

Thus  $\mathbf{W}(Z)$  is not generally Hermitian. For operator arguments,  $\text{adj}$  reverses the vector part without reordering derivatives. For the free field ( $q = 0$ ), use the canonical operators in (8.13). The block map and Dirac operator are

$$\begin{aligned} \mathbf{W}(\hat{P}) &= \begin{Bmatrix} 0 & \text{adj}(\hat{P}) \\ \hat{P} & 0 \end{Bmatrix}, \\ \hat{D}(m) &:= \mathbf{W}(\hat{P}) - m\mathbf{I}, \\ \hat{D}(-m) &= \mathbf{W}(\hat{P}) + m\mathbf{I}. \end{aligned} \quad (8.29)$$

The Dirac equation is [16, Eq. (4.51)]

$$\boxed{\hat{D}(m)\Psi = 0.} \quad (8.30)$$

Using  $\mathbf{W}(\hat{P})^2 = (\det \hat{P})\mathbf{I} = -\square\mathbf{I}$  from (8.17), its mass conjugate gives [16, Eqs. (4.53)–(4.54)]

$$\hat{D}(-m)\hat{D}(m) = \mathbf{W}(\hat{P})^2 - m^2\mathbf{I} = -(\square + m^2)\mathbf{I}. \quad (8.31)$$

Thus every free Dirac solution satisfies the wave form of the mass shell  $\det P = E^2 - \mathbf{p}^2 = m^2$  in (6.6):

$$(\det \hat{P} - m^2)\Psi = 0 \quad \Longleftrightarrow \quad (\square + m^2)\Psi = 0.$$

The original first-order constraint is the coupled pair [16, Eq. (4.108)]

$$i\partial\Psi_- = m\Psi_+, \quad (8.32)$$

$$i\partial^*\Psi_+ = m\Psi_-. \quad (8.33)$$

On one Fourier mode, let  $s \in \mathbb{C}$  satisfy  $\hat{D}(m)\Psi = s\Psi$  for a nonzero amplitude. Here  $\square$  denotes its scalar mode eigenvalue. Since  $\hat{D}(m) + m\mathbf{I} = \mathbf{W}(\hat{P})$ , (8.31) gives

$$(\hat{D}(m) + m\mathbf{I})^2\Psi = -\square\Psi \quad \Longrightarrow \quad ((s + m)^2 + \square)\Psi = 0.$$

Thus the eigenvalues are

$$s = -m \pm i\sqrt{\square}.$$

When distinct, each allows two independent complex coefficients in  $\Psi_+$ , with  $\Psi_- = i\partial^*\Psi_+/(s+m)$ . Each root therefore occurs twice, giving the scalar characteristic equation

$$\boxed{((s+m)^2 + \square)^2 = 0.} \quad (8.34)$$

At  $\square = 0$  the roots coalesce with total algebraic multiplicity four. For KG modes,  $\square = -m^2$ , so  $s = 0, -2m$ ; the Dirac equation selects  $s = 0$ . The matrix verification is (A.14).

#### 8.4.2 Reconstructing free solutions from Klein–Gordon

Let  $\psi' : \mathbb{R}^4 \rightarrow \mathbb{C}$  solve KG and recall the constant basis amplitude  $\psi_+$ . For constant complex scalars  $a, b$ , a two-constant construction is

$$\Psi = \frac{1}{m} \begin{Bmatrix} ma + ib\partial \\ ia\partial^* + mb \end{Bmatrix} \psi' \psi_+. \quad (8.35)$$

Substituting (8.35) into (8.32) and (8.33) gives

$$\begin{aligned} i\partial\Psi_- - m\Psi_+ &= -\frac{a}{m}(\square + m^2)\psi'\psi_+ = 0, \\ i\partial^*\Psi_+ - m\Psi_- &= -\frac{b}{m}(\square + m^2)\psi'\psi_+ = 0. \end{aligned}$$

Taking  $a = 1, b = 0$  gives one scalar-seeded solution. To span every Dirac field, instead let  $\psi$  have the form (8.2), with both scalar coefficients solving KG. Then

$$\boxed{\Psi = \begin{Bmatrix} \psi \\ i\partial^*\psi/m \end{Bmatrix}.} \quad (8.36)$$

The second Dirac equation holds by construction; the first is KG. Conversely, every massive free solution has this form with  $\psi = \Psi_+$ . Equation (8.23) evolves its two coefficients.

#### 8.4.3 Potential and paravector components

Recall the real  $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ , real charge  $q$ , and  $\hat{E}_q = i\partial_t - qV$ ,  $\hat{\mathbf{p}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$ . Minimal coupling gives [8, Eq. (191)]

$$\boxed{(\mathbf{W}(\hat{P}_q) - m\mathbf{I})\Psi = 0.} \quad (8.37)$$

Its entries are

$$(\hat{E}_q - \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})\Psi_- = m\Psi_+, \quad (8.38)$$

$$(\hat{E}_q + \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})\Psi_+ = m\Psi_-. \quad (8.39)$$

Isolating the time derivative gives  $i\partial_t\Psi = \hat{H}\Psi$ , where the Hamiltonian is [17, Eq. (412)]

$$\hat{H} := qV\mathbf{I} + \mathbf{W}(1)(m\mathbf{I} - \mathbf{W}(\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})). \quad (8.40)$$

For real potentials, the Dirac equation conserves total probability when the spatial boundary flux vanishes. The pointwise density (8.27) can still vary through probability flux.

To expose each coefficient, write  $\Psi_{\pm} = a_{\pm} + \mathbf{b}_{\pm} \cdot \boldsymbol{\sigma}$ , with complex scalar functions  $a_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C}$  and complex vector functions  $\mathbf{b}_{\pm} : \mathbb{R}^4 \rightarrow \mathbb{C}^3$ . Sigma multiplication gives

$$\begin{aligned}\hat{E}_q a_- - \hat{\mathbf{p}}_q \cdot \mathbf{b}_- &= m a_+, \\ \hat{E}_q \mathbf{b}_- - \hat{\mathbf{p}}_q a_- - \mathbf{i} \hat{\mathbf{p}}_q \times \mathbf{b}_- &= m \mathbf{b}_+, \\ \hat{E}_q a_+ + \hat{\mathbf{p}}_q \cdot \mathbf{b}_+ &= m a_-, \\ \hat{E}_q \mathbf{b}_+ + \hat{\mathbf{p}}_q a_+ + \mathbf{i} \hat{\mathbf{p}}_q \times \mathbf{b}_+ &= m \mathbf{b}_-.\end{aligned}$$

Every operator acts on the following coefficient. Each entry still has the two independent complex components of (8.2). Appendix A.7 converts the system to the conventional four-component Dirac equation.

#### 8.4.4 Interacting factorization and reconstruction

Recall the real potential  $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ , the complex field  $F = (\mathbf{E} + \mathbf{iB}) \cdot \boldsymbol{\sigma}$ , and  $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$ . For any complex paravector field  $\psi(t, \mathbf{r})$ , the product rule gives

$$\partial^*(\Phi^* \psi) + \Phi \partial \psi = (S + F) \psi + 2(V \partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}) \psi, \quad (8.41)$$

$$\partial(\Phi \psi) + \Phi^* \partial^* \psi = (S + F^*) \psi + 2(V \partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}) \psi. \quad (8.42)$$

The difference of (8.41) and (8.42) is  $(F - F^*) \psi = 2(\mathbf{E} \cdot \boldsymbol{\sigma}) \psi$ .

Applying the block map  $\mathbf{W}$  twice gives

$$\begin{aligned}\mathbf{W}(\partial^*) \mathbf{W}(\Phi) \Psi &= \begin{Bmatrix} \partial(\Phi \Psi_+) \\ \partial^*(\Phi^* \Psi_-) \end{Bmatrix}, \\ \mathbf{W}(\Phi) \mathbf{W}(\partial^*) \Psi &= \begin{Bmatrix} \Phi^* \partial^* \Psi_+ \\ \Phi \partial \Psi_- \end{Bmatrix}, \\ \mathbf{W}(\Phi)^2 &= \det \Phi \mathbf{I}.\end{aligned}$$

Differentiating a parenthesized product includes the potential derivative. For the fixed potential  $\Phi$  and charge  $q$ , use the interacting Dirac operator

$$\begin{aligned}\hat{D}_q(m) &:= \mathbf{W}(\hat{P}_q) - m \mathbf{I} \\ &= \begin{Bmatrix} -m & \text{adj}(\hat{P}_q) \\ \hat{P}_q & -m \end{Bmatrix}.\end{aligned} \quad (8.43)$$

The argument  $-m$  changes only the mass sign, keeping  $q$  and  $\Phi$  fixed. Multiplying and using the preceding mixed-derivative identities gives

$$\hat{D}_q(-m) \hat{D}_q(m) \Psi = - \begin{Bmatrix} (-\hat{E}_q^2 + \hat{\mathbf{p}}_q^2 + m^2 + \mathbf{i}qF^*) \Psi_+ \\ (-\hat{E}_q^2 + \hat{\mathbf{p}}_q^2 + m^2 + \mathbf{i}qF) \Psi_- \end{Bmatrix}.$$

This recovers (8.31) at  $q = 0$ . With coupling, its entries obey

$$\begin{aligned}((\partial + \mathbf{i}q\Phi^*)(\partial^* + \mathbf{i}q\Phi) + m^2) \Psi_+ &= 0, \\ ((\partial^* + \mathbf{i}q\Phi)(\partial + \mathbf{i}q\Phi^*) + m^2) \Psi_- &= 0.\end{aligned}$$

Expanding gives the exact second-order equations [18, Eq. (6)]

$$(-\hat{E}_q^2 + \hat{\mathbf{p}}_q^2 + m^2 + iqF^*)\Psi_+ = 0, \quad (8.44)$$

$$(-\hat{E}_q^2 + \hat{\mathbf{p}}_q^2 + m^2 + iqF)\Psi_- = 0. \quad (8.45)$$

For the spin-term components, write  $\mathbf{C} = \mathbf{E} + i\mathbf{B}$  and  $\psi = a + \mathbf{b} \cdot \boldsymbol{\sigma}$ , with complex functions  $a : \mathbb{R}^4 \rightarrow \mathbb{C}$  and  $\mathbf{b}, \mathbf{C} : \mathbb{R}^4 \rightarrow \mathbb{C}^3$ . Here  $\mathbf{C}^*$  conjugates its scalar coefficients. Sigma multiplication gives

$$\begin{aligned} iqF\psi &= iq\mathbf{C} \cdot \mathbf{b} + (iqa\mathbf{C} - q\mathbf{C} \times \mathbf{b}) \cdot \boldsymbol{\sigma}, \\ iqF^*\psi &= -iq\mathbf{C}^* \cdot \mathbf{b} + (-iqa\mathbf{C}^* + q\mathbf{C}^* \times \mathbf{b}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

To construct solutions, take a complex paravector amplitude  $\psi$  of (8.2) satisfying (8.45) with  $\psi$  in place of  $\Psi_-$ . Set

$$\boxed{\Psi = \begin{Bmatrix} (\hat{E}_q - \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})\psi/m \\ \psi \end{Bmatrix}}. \quad (8.46)$$

The first Dirac equation holds by construction; the second is the seed equation. Every massive solution has this form with  $\psi = \Psi_-$ . One must solve that seed equation for the chosen potential: a scalar KG seed lacks the spin term, and independently solved squared entries need not satisfy the first-order coupling.

#### 8.4.5 Gauge phase

Recall the real scalar  $\lambda : \mathbb{R}^4 \rightarrow \mathbb{R}$  and gauge transformation  $\Phi' = \Phi + \partial^*\lambda$ , or  $V' = V + \partial_t\lambda$ ,  $\mathbf{A}' = \mathbf{A} - \partial_r\lambda$ . For  $\hat{E}'_q := i\partial_t - qV'$ ,  $\hat{\mathbf{p}}'_q := -i\partial_r - q\mathbf{A}'$ , and  $\hat{P}'_q := \hat{E}'_q + \hat{\mathbf{p}}'_q \cdot \boldsymbol{\sigma}$ , the product rule gives

$$\begin{aligned} (\hat{E}_q + \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})(e^{iq\lambda}\Psi'_\pm) &= e^{iq\lambda}(\hat{E}_q - q\partial_t\lambda + \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma} + q(\partial_r\lambda) \cdot \boldsymbol{\sigma})\Psi'_\pm \\ &= e^{iq\lambda}(\hat{E}'_q + \hat{\mathbf{p}}'_q \cdot \boldsymbol{\sigma})\Psi'_\pm. \end{aligned}$$

The adjugate block follows with the vector signs reversed. Thus the new amplitude is

$$\Psi' = e^{-iq\lambda}\Psi.$$

Equivalently,

$$\begin{aligned} \hat{E}'_q\Psi' &= e^{-iq\lambda}\hat{E}_q\Psi, \\ \hat{\mathbf{p}}'_q\Psi' &= e^{-iq\lambda}\hat{\mathbf{p}}_q\Psi. \end{aligned}$$

Consequently [16, Eqs. (6.68)–(6.69)]

$$\begin{aligned} (\mathbf{W}(\hat{P}'_q) - m\mathbf{I})\Psi' \\ = e^{-iq\lambda}(\mathbf{W}(\hat{P}_q) - m\mathbf{I})\Psi. \end{aligned} \quad (8.47)$$

Equation (8.47) preserves the Dirac equation; the phase sign is fixed by the potential shift. Its modulus is one, so the norm is unchanged. The same product rules preserve the KG and Pauli equations.

## 9 Low-energy limits

Recall the real mass  $m > 0$ , charge  $q$ , and potentials  $V : \mathbb{R}^4 \rightarrow \mathbb{R}$ ,  $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ . The mechanical operators are  $\hat{E}_q = i\partial_t - qV$  and  $\hat{\mathbf{p}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$ ; the fields are  $\mathbf{E} = -\partial_{\mathbf{r}}V - \partial_t\mathbf{A}$  and  $\mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}$ . Remove the rest oscillation and retain the leading nonrelativistic terms. This gives Pauli from Dirac and Schrödinger from KG.

### 9.1 Dirac to Pauli

Remove the rest-energy phase to define the envelopes [12, Eq. (2.1)]:

$$\Psi'_{\pm} := e^{imt}\Psi_{\pm}. \quad (9.1)$$

The product rule gives

$$\hat{E}_q\Psi_{\pm} = e^{-imt}(m + \hat{E}_q)\Psi'_{\pm}.$$

On the envelopes,  $\hat{E}_q$  measures mechanical energy above rest. The two rows of the Dirac equation, (8.38) and (8.39), become

$$\begin{aligned} (m + \hat{E}_q - \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})\Psi'_- &= m\Psi'_+, \\ (m + \hat{E}_q + \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})\Psi'_+ &= m\Psi'_-. \end{aligned}$$

At free positive-energy rest,  $\Psi'_+ = \Psi'_-$ . Define the large and small components

$$\phi_+ := \frac{\Psi'_+ + \Psi'_-}{\sqrt{2}}, \quad (9.2)$$

$$\phi_- := \frac{\Psi'_- - \Psi'_+}{\sqrt{2}}. \quad (9.3)$$

Adding and subtracting gives the exact pair [12, Eqs. (2.15)–(2.16)]:

$$\hat{E}_q\phi_+ = (\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})\phi_-, \quad (9.4)$$

$$(2m + \hat{E}_q)\phi_- = (\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})\phi_+. \quad (9.5)$$

For nonrelativistic positive-energy states in weak, slowly varying fields, neglect  $\hat{E}_q\phi_-$  relative to  $2m\phi_-$ . Substitute the small component into the first equation and use (8.18):

$$\begin{aligned} \phi_- &\simeq \frac{\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}}{2m}\phi_+, \\ \hat{E}_q\phi_+ &\simeq \frac{(\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})^2}{2m}\phi_+ = \frac{\hat{\mathbf{p}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m}\phi_+. \end{aligned}$$

Define  $\psi := \phi_+$ . Restoring  $\hat{E}_q = i\partial_t - qV$  gives Pauli's equation through order  $1/m$  [12, Eq. (2.13)]:

$$\boxed{i\partial_t\psi = \left( \frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV - \frac{q}{2m}\mathbf{B} \cdot \boldsymbol{\sigma} \right)\psi.} \quad (9.6)$$

The change of entries preserves the density exactly:

$$\|\Psi\|^2 = \|\phi_+\|^2 + \|\phi_-\|^2.$$

The leading probability density is  $\|\psi\|^2$ . Its two complex components are the Pauli amplitudes; see Appendix A.5. This is the amplitude used in the spin discussion. For  $0 < v \ll 1$ , the scaling  $\hat{\mathbf{p}}_q/m = O(v)$ ,  $\hat{E}_q/m = O(v^2)$ ,  $qV/m = O(v^2)$ ,  $q\mathbf{B}/m^2 = O(v^2)$ ,  $q\mathbf{E}/m^2 = O(v^3)$ , with fields varying on spatial scales  $1/(mv)$  and time scales  $1/(mv^2)$  or longer, gives  $\phi_- = O(v\phi_+)$  and omitted energy terms of order  $mv^4$ .

Equation (10.2) identifies spin as  $\boldsymbol{\sigma}/2$ , so the magnetic energy is  $-(q/m)\mathbf{B} \cdot (\boldsymbol{\sigma}/2)$ : magnetic moment equals  $q/m$  times spin, so the gyromagnetic factor is 2 before radiative corrections.

### 9.1.1 Components along a general field direction

For a nonzero static uniform field, let  $\mathbf{u} = \mathbf{B}/\|\mathbf{B}\|$  be its **real unit direction** and  $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$  its spin projectors. The complex paravector fields  $\phi_{+,\pm} = \Pi_{\pm}\phi_+$  obey

$$i\partial_t\phi_{+,\pm} = \left( \frac{\hat{\mathbf{p}}_q^2 \mp q\|\mathbf{B}\|}{2m} + qV \right) \phi_{+,\pm}.$$

For a nonzero field with varying direction, the projectors depend on  $(t, \mathbf{r})$ . Let  $\hat{H}$  denote the Pauli Hamiltonian in (9.6). The product rule gives

$$\begin{aligned} i\partial_t(\Pi_{\pm}\phi_+) &= i(\partial_t\Pi_{\pm})\phi_+ + \Pi_{\pm}\hat{H}\phi_+ \\ &= \hat{H}(\Pi_{\pm}\phi_+) + \left( i\partial_t\Pi_{\pm} + \frac{[\Pi_{\pm}, \hat{\mathbf{p}}_q^2]}{2m} \right) \phi_+. \end{aligned}$$

Only the projector derivatives and kinetic commutator couple the sectors; the magnetic term commutes with its own projectors.

## 9.2 The same leading order: Schrödinger

When  $\mathbf{B} = 0$ , each complex coefficient of the Pauli amplitude obeys the scalar Schrödinger equation. KG gives the same result directly. Keep the operators  $\hat{E}_q, \hat{\mathbf{p}}_q$  defined above. For the full scalar KG field  $\psi' : \mathbb{R}^4 \rightarrow \mathbb{C}$ , remove the rest oscillation as in (9.1), defining its scalar envelope  $\psi : \mathbb{R}^4 \rightarrow \mathbb{C}$  by [16, Eqs. (2.103)–(2.104)]

$$\psi'(t, \mathbf{r}) = e^{-imt}\psi(t, \mathbf{r}). \quad (9.7)$$

The product rule gives  $\hat{E}_q\psi' = e^{-imt}(\hat{E}_q + m)\psi$ . Since  $m$  is constant, applying  $\hat{E}_q$  again gives  $\hat{E}_q^2\psi' = e^{-imt}(\hat{E}_q + m)^2\psi$ ; spatial derivatives leave the phase untouched. Substitution into (8.24) therefore gives

$$\begin{aligned} 0 &= (-\hat{E}_q^2 + \hat{\mathbf{p}}_q^2 + m^2)\psi' \\ &= e^{-imt}(-(\hat{E}_q + m)^2 + \hat{\mathbf{p}}_q^2 + m^2)\psi \\ &= e^{-imt}(-\hat{E}_q^2 - 2m\hat{E}_q + \hat{\mathbf{p}}_q^2)\psi. \end{aligned}$$

The rest terms cancel. Divide by the nonzero phase to obtain

$$\hat{E}_q\psi = \frac{\hat{\mathbf{p}}_q^2 - \hat{E}_q^2}{2m}\psi. \quad (9.8)$$

On a slow positive-energy envelope,  $\hat{E}_q/m = O(v^2)$ , so  $\hat{E}_q^2\psi$  is smaller than  $2m\hat{E}_q\psi$  by  $O(v^2)$ . Dropping it at leading order gives [19, Eq. (21.3)]

$$\boxed{\mathrm{i}\partial_t\psi = \left(\frac{(-\mathrm{i}\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV\right)\psi.} \quad (9.9)$$

With  $\mathbf{A} = 0$ , its kinetic operator is  $-\partial_{\mathbf{r}}^2/(2m)$ . Schrödinger and Pauli therefore occur at the same leading order  $1/m$ ; retaining spin gives the magnetic term in (9.6).

### 9.3 The next free order

For the free complex scalar envelope  $\psi(t, \mathbf{r})$ , set  $qV = 0$  and  $q\mathbf{A} = 0$ , so  $\hat{E}_q = \hat{E}$ ,  $\hat{\mathbf{p}}_q = \hat{\mathbf{p}}$ , and the operators commute. On the positive-frequency KG branch, solving (9.8) gives

$$\hat{E}\psi = \left(\sqrt{m^2 + \hat{\mathbf{p}}^2} - m\right)\psi.$$

The same energy acts on either normalized positive-energy Dirac spin component. Iteration starts with  $\hat{E}\psi \simeq \hat{\mathbf{p}}^2\psi/(2m)$ ; its square supplies the next term [12, Eq. (2.19)]:

$$\hat{E}\psi = \left(\frac{\hat{\mathbf{p}}^2}{2m} - \frac{\hat{\mathbf{p}}^4}{8m^3} + O\left(\frac{\hat{\mathbf{p}}^6}{m^5}\right)\right)\psi.$$

For the charged scalar equation, the same iteration applies whenever  $\hat{E}_q$  commutes with  $\hat{\mathbf{p}}_q^2$ . On the positive-energy branch,

$$\begin{aligned} \hat{E}_q\psi &= \left(\sqrt{m^2 + \hat{\mathbf{p}}_q^2} - m\right)\psi \\ &= \left(\frac{\hat{\mathbf{p}}_q^2}{2m} - \frac{\hat{\mathbf{p}}_q^4}{8m^3} + O\left(\frac{\hat{\mathbf{p}}_q^6}{m^5}\right)\right)\psi. \end{aligned}$$

This expansion applies when the kinetic momenta in the state are small compared with  $m$ .

## 10 Spin and probability

Spin specifies the intrinsic rotation of an amplitude. Here we rotate the state at fixed coordinates. A scalar KG amplitude stays unchanged under this intrinsic action and has spin zero; a two-component amplitude has the one-sided law below. The wave equation alone does not fix spin: free Dirac components also satisfy KG.

### 10.1 Spin one-half and rotations

Recall the complex paravector amplitude  $\psi(t, \mathbf{r})$  of (8.2). For a **real unit axis**  $\mathbf{u}$  and real angle  $\theta$ , use the unitary rotor  $R = \exp(-\mathrm{i}\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$  of (3.35). The spinor law is [11, p. 4]

$$\boxed{\psi' = R\psi.} \quad (10.1)$$

Writing its infinitesimal action as  $\psi' = \psi - \mathrm{i}\theta S(\mathbf{u})\psi + O(\theta^2)$  defines the spin operator along  $\mathbf{u}$  [13, p. 62]:

$$S(\mathbf{u})\psi = \frac{1}{2}(\mathbf{u} \cdot \boldsymbol{\sigma})\psi. \quad (10.2)$$

Since  $(\mathbf{u} \cdot \boldsymbol{\sigma})^2 = 1$ , its two spin projections are  $\pm 1/2$  in units  $\hbar = 1$ . A  $2\pi$  rotation gives  $R = -1$ , reversing the amplitude's sign; a  $4\pi$  rotation restores it. Probabilities are unchanged by the overall sign, while a relative sign can be detected by interference [20, Eq. (6.18)]. Each Dirac entry obeys this same active rotation law.

For a real paravector observable  $Z$ , the state law (10.1) gives, by (8.8),

$$\langle Z \rangle_{\psi'} = \langle R^H Z R \rangle_{\psi}.$$

The observable inside the original state's pairing rotates through  $-\theta$ . If the observable rotates with the state,  $Z' = R Z R^H$ , then  $\langle Z' \rangle_{\psi'} = \langle Z \rangle_{\psi}$ .

## 10.2 Projectors and probabilities

Recall a nonzero Pauli amplitude  $\psi$  of (8.2). For a real unit measurement axis  $\mathbf{u}$ , the real projectors are

$$\Pi_{\pm} = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}).$$

Their algebra in Section 3.7 gives

$$\begin{aligned} \psi &= \Pi_+ \psi + \Pi_- \psi, \\ S(\mathbf{u})(\Pi_{\pm} \psi) &= \pm \frac{1}{2} \Pi_{\pm} \psi, \\ \text{sc}((\Pi_+ \psi)^H (\Pi_- \psi)) &= 0, \\ \|\Pi_{\pm} \psi\|^2 &= \text{sc}(\psi^H \Pi_{\pm} \psi) = \|\psi\|_{\Pi_{\pm}}^2. \end{aligned}$$

By (8.5), these are nonnegative weighted densities. The Born probability is this density divided by  $\rho = \|\psi\|^2$ . In the fixed orthonormal basis (8.1),  $\psi = c_+ \psi_+ + c_- \psi_-$  with complex scalars  $c_{\pm}$ , so  $\rho = \|c_+\|^2 + \|c_-\|^2$ . When the measurement axis is that basis axis, the probabilities are  $\|c_{\pm}\|^2/\rho$ . The same-side and cross-side projector products are already derived in Section 3.7.

Define the real spin-polarization vector  $\mathbf{u}' \in \mathbb{R}^3$  by its projection along every real unit measurement axis  $\mathbf{u}$ :

$$\mathbf{u} \cdot \mathbf{u}' := \frac{\text{sc}(\psi^H (\mathbf{u} \cdot \boldsymbol{\sigma}) \psi)}{\rho}. \quad (10.3)$$

The right side is real and linear in  $\mathbf{u}$ , so it uniquely specifies  $\mathbf{u}'$ . For the arbitrary perpendicular basis axes  $\mathbf{u}, \mathbf{a}$  in (8.2), expanding the same pairing gives

$$\begin{aligned} \rho \mathbf{u}' &= (\|c_+\|^2 - \|c_-\|^2) \mathbf{u} \\ &\quad + 2 \text{re}(c_+^* c_-) \mathbf{a} + 2 \text{im}(c_+^* c_-) (\mathbf{u} \times \mathbf{a}). \end{aligned}$$

These three directions are orthonormal, hence

$$\rho^2 (\mathbf{u}')^2 = (\|c_+\|^2 - \|c_-\|^2)^2 + 4 \|c_+^* c_-\|^2 = \rho^2.$$

Thus  $\mathbf{u}'$  is a real unit vector, with mean spin  $\mathbf{u}'/2$ . For any real unit measurement axis, inserting  $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$  directly into the pairing gives

$$\|\Pi_{\pm} \psi\|^2 = \text{sc}(\psi^H \Pi_{\pm} \psi) = \frac{1}{2} \rho (1 \pm \mathbf{u} \cdot \mathbf{u}').$$

Therefore [11, Eqs. (66)–(67)]

$$\boxed{\text{prob}(\pm | \mathbf{u}) = \frac{\|\Pi_{\pm} \psi\|^2}{\rho} = \frac{1}{2} (1 \pm \mathbf{u} \cdot \mathbf{u}').} \quad (10.4)$$

The mean spin along  $\mathbf{u}$  is  $(\text{prob}(+) - \text{prob}(-))/2 = \mathbf{u} \cdot \mathbf{u}'/2$ . For a spatial wavefunction, (10.4) is the spin probability conditional on position, not a position density.

### 10.2.1 Measurement along a second axis

Use  $\mathbf{u}'$  for the prepared spin direction and  $\mathbf{u}$  for the measurement axis, both real unit vectors. Prepare a nonzero spin-up amplitude  $\psi = \Pi'_+ \psi$ , where  $\Pi'_+ = (1 + \mathbf{u}' \cdot \boldsymbol{\sigma})/2$ . Let  $0 \leq \theta \leq \pi$  be the angle between the axes and write

$$K' := \mathbf{u}' \cdot \boldsymbol{\sigma}, \quad K := \mathbf{u} \cdot \boldsymbol{\sigma}, \quad c := \mathbf{u} \cdot \mathbf{u}' = \cos \theta.$$

The Pauli product gives  $(K')^2 = 1$  and  $K'K + KK' = 2c$ . Multiplication by  $K'$  gives  $K'KK' = 2cK' - K$ , hence

$$\begin{aligned} \Pi'_+ K \Pi'_+ &= \frac{1}{4}(1 + K')K(1 + K') \\ &= \frac{1}{4}(K + K'K + KK' + K'KK') \\ &= \frac{1}{4}(2c + 2cK') = c\Pi'_+. \end{aligned}$$

The measurement projectors  $\Pi_{\pm} = (1 \pm K)/2$  therefore satisfy

$$\Pi'_+ \Pi_{\pm} \Pi'_+ = \frac{1}{2} (\Pi'_+ \pm \Pi'_+ K \Pi'_+) = \frac{1 \pm c}{2} \Pi'_+. \quad (10.5)$$

Use Hermiticity, idempotence, and  $\psi = \Pi'_+ \psi$  to evaluate the projected norm explicitly:

$$\begin{aligned} \|\Pi_{\pm} \psi\|^2 &= \text{sc}((\Pi_{\pm} \psi)^H (\Pi_{\pm} \psi)) \\ &= \text{sc}(\psi^H (\Pi_{\pm})^2 \psi) \\ &= \text{sc}(\psi^H \Pi'_+ \Pi_{\pm} \Pi'_+ \psi) \\ &= \frac{1 \pm c}{2} \text{sc}(\psi^H \Pi'_+ \psi) = \frac{1 \pm c}{2} \|\psi\|^2. \end{aligned}$$

The Born rule divides by the nonzero norm, so no prior normalization is needed:

$$\text{prob}(\pm) := \frac{\|\Pi_{\pm} \psi\|^2}{\|\psi\|^2} = \frac{1 \pm \cos \theta}{2}, \quad (10.6)$$

$$\text{prob}(+) = \cos^2 \frac{\theta}{2}, \quad \text{prob}(-) = \sin^2 \frac{\theta}{2}.$$

Here  $+$  and  $-$  label measured spin  $+1/2$  and  $-1/2$  along  $\mathbf{u}$ . The probabilities sum to one. Aligned axes give spin up with certainty, perpendicular axes give equal probabilities, and opposite axes give spin down with certainty. Preparing  $\psi = \Pi'_- \psi$  interchanges the probabilities. For a spatial wavefunction with the same spin preparation at every position, integrating both norm densities gives the same result.

### 10.2.2 Rotating the polarization

For  $R = \exp(-i\varphi \mathbf{a} \cdot \boldsymbol{\sigma}/2)$ , with real unit rotation axis  $\mathbf{a}$  and real angle  $\varphi$ , let  $\mathbf{u}''$  denote the polarization of the rotated amplitude  $\psi' = R\psi$ . The amplitude law (10.1) gives

$$\mathbf{u} \cdot \mathbf{u}'' = \langle R^H (\mathbf{u} \cdot \boldsymbol{\sigma}) R \rangle_{\psi}.$$

This is the inverse of Rodrigues' rotation (4.6): the measurement axis rotates through  $-\varphi$ , so  $\mathbf{u}''$  rotates through  $+\varphi$  about  $\mathbf{a}$ . Thus the polarization rotates as a vector although its amplitude has spin one-half. Its overlap with the original polarization is

$$\mathbf{u}' \cdot \mathbf{u}'' = (\mathbf{a} \cdot \mathbf{u}')^2 + (1 - (\mathbf{a} \cdot \mathbf{u}')^2) \cos \varphi.$$

For  $\mathbf{a} \cdot \mathbf{u}' = 0$ , the probability to retain the initial spin is  $\cos^2(\varphi/2)$ .

## 11 Rotations and Lorentz boosts

We now apply the algebraic actions of Section 4 to the physical variables already defined. Appendix C compares alternative exponent signs and sandwich orders. Primed fields are evaluated at  $X'$  and unprimed fields at  $X$ . The frame factors here are constant; varying factors are treated in Section 6.6.

### 11.1 Covariance of the physical equations

Let  $T$  be a complex [unit paravector](#) with  $\det T = 1$ . It may be any ordered product of  $R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$  and  $L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ , with [real unit vector](#)  $\mathbf{u}$  and real angle or rapidity  $\theta$ . Recall the real event  $X$ , momentum  $P$ , potential  $\Phi$ , and electric current  $D$ . Each transforms by [5, Eq. (24)]

$$Z' = T Z T^{\mathbf{H}}, \quad Z = X, P, \Phi, D. \quad (11.1)$$

Since  $\det T = 1$ , sigma conjugation gives  $T^* = (T^{-1})^{\mathbf{H}}$ . The derivative and complex field  $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$  use [5, Eq. (33)]

$$\partial' = T^* \partial T^{-1}, \quad (11.2)$$

$$F' = T F T^{-1}. \quad (11.3)$$

Equation (11.2) follows from the chain rule. For a differentiable scalar  $f : \mathbb{R}^4 \rightarrow \mathbb{C}$  with  $f'(X') = f(X)$ ,

$$df = \text{sc}((\partial f) dX) = \text{sc}((T^*(\partial f) T^{-1}) dX').$$

Since this holds for every real increment  $dX'$ , the coefficient is  $\partial' f'$  [7, p. 101].

The field law is fixed by the potential law, rather than chosen separately. The identity (7.11), with real scalar  $S = \text{sc}(\partial \Phi)$ , gives

$$\partial' \Phi' = T^*(S + F^*) T^{\mathbf{H}}.$$

Scalar-part cyclicity gives  $S' = S$ . Since  $T^* = (T^{\mathbf{H}})^{-1}$ , sigma conjugation of the vector part gives (11.3). Maxwell covariance now follows by cancellation:

$$\partial' F' = T^*(\partial F) T^{-1} = T^* D^* T^{-1} = D'^*.$$

Thus the transformed potentials and fields satisfy the same Maxwell equation (7.2); the scalar wave operator also obeys  $\square' = \det(\partial') = \det(\partial) = \square$ .

For real  $P, X$ , the metric pairing (4.4) is the wave phase. Adjunction of  $P'$  and cyclicity give

$$\text{sc}(P'^* X') = \text{sc}(T^* P^* X T^{\mathbf{H}}) = \text{sc}(P^* X).$$

Consequently  $E't' - \mathbf{p}' \cdot \mathbf{r}' = Et - \mathbf{p} \cdot \mathbf{r}$ , and the plane wave (8.11) satisfies  $f(X', P') = f(X, P)$ .

A scalar Klein–Gordon field uses  $\psi'(X') = \psi(X)$ ; the derivative and potential laws above preserve its wave equation. For the two complex paravector Dirac entries, a change of coordinates  $X' = TXT^H$  uses

$$\begin{aligned}\Psi'_+(X') &= T^* \Psi_+(X), \\ \Psi'_-(X') &= T \Psi_-(X).\end{aligned}$$

Primed mechanical operators use the primed coordinates and potentials. The first interacting equation (8.38) transforms by cancellation:

$$(\hat{E}'_q - \hat{\mathbf{p}}'_q \cdot \boldsymbol{\sigma}) \Psi'_- = T^* (\hat{E}_q - \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}) \Psi_- = m \Psi'_+.$$

Equation (8.39) transforms in the same way with left factor  $T$ ;  $q = 0$  gives the free case. For  $T = R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ , both entries transform by  $R$ , consistently with the active rotation (10.1); the spatial argument also rotates. A passive rotation of axes uses the inverse factor  $T = R^{-1}$ . For  $T = L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ , the two entries instead use  $L^{-1}$  and  $L$ , respectively [16, Eq. (4.31)]. The factor  $L$  already contains half the rapidity; no further square root is needed for either entry. Their cross-pairing  $\text{sc}(\Psi_+^H \Psi_-)$  is invariant: the two transformation factors cancel. Their summed positive density (8.27) is observer-dependent under boosts. Use the full Dirac pair for boosts before taking the Pauli low-energy limit.

## 11.2 Moving frames, clock rates, and lengths

Recall  $L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$ , with real rapidity  $\theta$ , **real unit vector**  $\mathbf{u}$ ,  $\beta = \tanh \theta$ , and  $\gamma = \cosh \theta$ . Here  $\gamma$  is the factor for frame velocity  $\beta \mathbf{u}$ . The gamma form (3.45) and its square (3.46) are the same exponential at half and full rapidity. For real  $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ , take coincident inertial origins at  $t = t' = 0$ . Equation (11.1) with  $T = L$  gives  $X' = LXL$ : the primed frame moves at  $+\beta \mathbf{u}$ . Equations (4.12)–(4.13) become [7, Eq. (5.1)]

$$t' = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}), \tag{11.4}$$

$$\mathbf{r}' = \mathbf{r}_\perp + \gamma(\mathbf{r}_\parallel - \beta t \mathbf{u}), \tag{11.5}$$

where  $\mathbf{r}_\parallel = (\mathbf{u} \cdot \mathbf{r}) \mathbf{u}$  and  $\mathbf{r}_\perp = \mathbf{r} - \mathbf{r}_\parallel$ . In particular,  $\mathbf{r} = \beta t \mathbf{u}$  maps to  $\mathbf{r}' = 0$ , fixing the physical meaning of the sign. Reversing  $\theta$  as in (3.37) gives  $X = L^{-1} X' L^{-1}$ .

### 11.2.1 Intervals and light cones

Specializing (4.14) to the real event  $X$  gives [9, p. 154]

$$t' \pm \mathbf{u} \cdot \mathbf{r}' = e^{\mp \theta} (t \pm \mathbf{u} \cdot \mathbf{r}). \tag{11.6}$$

Multiplying the two signs in (11.6) and using  $\mathbf{r}'_\perp = \mathbf{r}_\perp$  gives

$$t'^2 - \mathbf{r}'^2 = (t' + \mathbf{u} \cdot \mathbf{r}')(t' - \mathbf{u} \cdot \mathbf{r}') - \mathbf{r}'_\perp^2 = t^2 - \mathbf{r}^2.$$

Thus the exponential stretches one light-cone coordinate and compresses the other reciprocally. The same identity holds for event differences; in particular,

$$dt'^2 - d\mathbf{r}'^2 = \det(dX') = \det(dX) = dt^2 - d\mathbf{r}^2. \tag{11.7}$$

Equation (11.7) preserves the interval (6.1) and light cones in every direction. Future causal increments stay future because  $dt' = \gamma(dt - \beta \mathbf{u} \cdot d\mathbf{r}) > 0$  when  $dt > 0$  and  $dt \geq \|d\mathbf{r}\|$  [9, p. 118]. In contrast, the positive norm is

$$\|X'\|^2 = \gamma^2((t - \beta \mathbf{u} \cdot \mathbf{r})^2 + (\mathbf{u} \cdot \mathbf{r} - \beta t)^2) + \mathbf{r}_\perp^2.$$

It generally differs from  $t^2 + \mathbf{r}^2$ . The determinant controls proper time and causality; preserving this positive norm instead would not give these Lorentz transformations.

### 11.2.2 Spacetime increments and velocity

Recall the real proper velocity  $U = dX/ds$ , a unit paravector with  $UU^{-1} = 1$ , along a real world line  $X(s)$ . For constant  $L$ , differentiation gives

$$dX' = L dX L. \quad (11.8)$$

Because  $ds$  is invariant,  $U' = LUL$ . At a point where  $U = L^{-2} = \gamma(1 + \beta \mathbf{u} \cdot \boldsymbol{\sigma})$ , this gives  $U' = 1$  and  $dX' = ds$ : the boost takes the particle's instantaneous tangent to rest. For varying velocity, a different  $L$  is needed at each point; Section 6.6 supplies its derivative terms.

Recall the particle velocity  $\mathbf{v} = d\mathbf{r}/dt$  and resolve it along  $\mathbf{u}$ . Dividing the spatial increment by  $dt' = \gamma(1 - \beta \mathbf{u} \cdot \mathbf{v}) dt$  gives [9, Eq. (7.22)]

$$\boxed{\mathbf{v}' = \frac{\mathbf{v}_\perp/\gamma + \mathbf{v}_\parallel - \beta \mathbf{u}}{1 - \beta \mathbf{u} \cdot \mathbf{v}}}. \quad (11.9)$$

Interval invariance also gives

$$1 - \mathbf{v}'^2 = \frac{1 - \mathbf{v}^2}{\gamma^2(1 - \beta \mathbf{u} \cdot \mathbf{v})^2}. \quad (11.10)$$

Thus (11.9) preserves subluminal motion and light speed. For two real proper times  $s_1, s_2$ , the constant-factor law (11.8) also integrates to

$$X'(s_2) - X'(s_1) = L \left( \int_{s_1}^{s_2} U(s) ds \right) L.$$

This is the transformed world-line integral (6.10).

### 11.2.3 Clock rates and measured lengths

For a clock at rest in the primed frame, (11.4) uses  $d\mathbf{r} = \beta \mathbf{u} dt$ , hence [9, p. 119]

$$dt' = \gamma(1 - \beta^2) dt = dt/\gamma. \quad (11.11)$$

Here  $dt' = ds$ ; finite elapsed time follows from (6.11).

For a rod at rest in the unprimed frame, let  $\Delta \mathbf{r} \in \mathbb{R}^3$  be its fixed endpoint separation. A simultaneous measurement in the primed frame requires  $\Delta t' = 0$ , so  $\Delta t = \beta \mathbf{u} \cdot \Delta \mathbf{r}$ . Substituting in (11.5) gives [9, p. 123]

$$\Delta \mathbf{r}'_\parallel = \Delta \mathbf{r}_\parallel / \gamma. \quad (11.12)$$

The transverse separation stays unchanged; thus a rod of any orientation has

$$\|\Delta\mathbf{r}'\|^2 = \|\Delta\mathbf{r}_\perp\|^2 + \|\Delta\mathbf{r}_\parallel\|^2/\gamma^2.$$

Longitudinal contraction (11.12) therefore compares endpoint events at equal primed times. Equal unprimed times instead give  $\Delta t' = -\gamma\beta\mathbf{u} \cdot \Delta\mathbf{r}$ ; simultaneity is frame-dependent. The reciprocal statement follows by exchanging primes and reversing  $\beta$ .

#### 11.2.4 Composing frames

For constant determinant-one factors  $T, T'$ , applying the frame change  $T$  and then  $T'$  gives

$$Z'' = T'(TZT^H)T'^H = (T'T)Z(T'T)^H.$$

Thus the combined factor is  $T'T$ : the rightmost factor acts first.

Collinear boosts add rapidities by (3.48). For a second real rapidity  $\theta'$ , set  $\beta' = \tanh\theta'$  and  $\gamma' = \cosh\theta'$ . Their combined parameters are

$$\begin{aligned}\beta'' &= \tanh(\theta + \theta') = \frac{\beta + \beta'}{1 + \beta\beta'}, \\ \gamma'' &= \gamma\gamma'(1 + \beta\beta').\end{aligned}$$

These are the collinear velocity and clock-factor composition laws. Noncollinear boosts instead have the rotation commutator (3.49), producing the Thomas–Wigner rotation.

#### 11.2.5 Energy and momentum

For the real momentum  $P = E + \mathbf{p} \cdot \boldsymbol{\sigma}$ , the same congruence gives [21, Eq. (17.12)]

$$E' = \gamma(E - \beta\mathbf{u} \cdot \mathbf{p}), \tag{11.13}$$

$$\mathbf{p}' = \mathbf{p}_\perp + \gamma(\mathbf{p}_\parallel - \beta E\mathbf{u}). \tag{11.14}$$

Here the projections are along  $\mathbf{u}$ . Determinant invariance gives  $E'^2 - \mathbf{p}'^2 = E^2 - \mathbf{p}^2 = m^2$  for a particle of rest mass  $m > 0$ . To see the energy increase directly, start with  $P' = m$  in its rest frame and invert the boost:

$$P = L^{-1}mL^{-1} = m\gamma(1 + \beta\mathbf{u} \cdot \boldsymbol{\sigma}).$$

Thus  $E = m\gamma$  and  $\mathbf{p} = m\gamma\beta\mathbf{u}$ , while  $m$  remains unchanged [9, Eqs. (7.25), (7.28)]. The force response is derived in (6.8)–(6.9).

### 11.3 Boosted fields and energy flux

For  $L = \exp(-\theta\mathbf{u} \cdot \boldsymbol{\sigma}/2)$  with the parameters above, (11.2) becomes  $\partial' = L^{-1}\partial L^{-1}$ . Its scalar and vector parts are

$$\begin{aligned}\partial_{t'} &= \gamma(\partial_t + \beta\mathbf{u} \cdot \partial_{\mathbf{r}}), \\ \partial_{\mathbf{r}'} &= \partial_{\mathbf{r}} + (\gamma - 1)\mathbf{u}(\mathbf{u} \cdot \partial_{\mathbf{r}}) + \gamma\beta\mathbf{u}\partial_t.\end{aligned}$$

These inverse signs express the chain rule. The real potential  $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$  and current  $D = \rho + \mathbf{J} \cdot \boldsymbol{\sigma}$  follow the same component formulas as (11.13)–(11.14), with  $(E, \mathbf{p})$  replaced by  $(V, \mathbf{A})$  or  $(\rho, \mathbf{J})$ .

At corresponding events, (11.3) reduces to

$$F' = LFL^{-1}. \quad (11.15)$$

Both the field coefficients and their spacetime arguments transform:

$$F'(X') = LF(L^{-1}X'L^{-1})L^{-1}. \quad (11.16)$$

Here  $X = L^{-1}X'L^{-1}$ . To expand (11.15), apply (4.15) to  $\mathbf{E} + i\mathbf{B}$  and separate its real and imaginary parts [7, Eqs. (5.17)–(5.18)]:

$$\mathbf{E}' = \mathbf{E}_{\parallel} + \gamma(\mathbf{E}_{\perp} + \beta\mathbf{u} \times \mathbf{B}), \quad (11.17)$$

$$\mathbf{B}' = \mathbf{B}_{\parallel} + \gamma(\mathbf{B}_{\perp} - \beta\mathbf{u} \times \mathbf{E}). \quad (11.18)$$

Parallel components are unchanged. These are the usual boosted fields along any [real unit axis](#), matching. The inverse field law is obtained by reversing  $\beta$ .

Similarity preserves  $F'^2 = LF^2L^{-1} = F^2$ , since  $F^2$  is scalar. Consequently (7.15) gives

$$\begin{aligned} \mathbf{E}'^2 - \mathbf{B}'^2 &= \mathbf{E}^2 - \mathbf{B}^2, \\ \mathbf{E}' \cdot \mathbf{B}' &= \mathbf{E} \cdot \mathbf{B}. \end{aligned}$$

In contrast,  $\|F\|^2 = \mathbf{E}^2 + \mathbf{B}^2$  is twice the observer-dependent energy density. For example, if  $\mathbf{B} = 0$  and  $\mathbf{u} \cdot \mathbf{E} = 0$ , (11.17) and (11.18) give  $\|F'\|^2 = \gamma^2(1 + \beta^2)\mathbf{E}^2$ , while  $\mathbf{E}'^2 - \mathbf{B}'^2 = \mathbf{E}^2$ .

Recall the stress map  $T(N) = FNF^{\mathbf{H}}/2$  from (7.22), with real paravector input  $N$ ; here  $T$  denotes this map. Since  $N' = LNL$ ,

$$T'(N') = \frac{1}{2}F'N'F'^{\mathbf{H}} = LT(N)L.$$

Here  $T'$  is the map formed from  $F'$ . To measure the primed energy density and Poynting vector, use  $N' = 1$ , hence  $N = L^{-2}$ :

$$T'(1) = LT(L^{-2})L = \frac{1}{2}F'F'^{\mathbf{H}}.$$

The full map transforms covariantly; its value at 1 depends on the observer's time direction.

Finally, recall the real proper velocity  $U = dX/ds$ , a [unit paravector](#) with  $UU^{-1} = 1$ , and real momentum  $P = mU$ , with real charge  $q$  and constant mass  $m > 0$ . With  $U' = LUL$  and  $P' = LPL$ , the force law (7.21) transforms as

$$\frac{dP'}{ds} = L\frac{q}{2}(FU + UF^{\mathbf{H}})L = \frac{q}{2}(F'U' + U'F'^{\mathbf{H}}).$$

Thus the electric and magnetic components above reproduce the same real four-force in the moving frame; the Hermitian extraction is preserved.

## A Pauli-matrix representation

This appendix verifies the paravector formulas in matrix form. Let  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ , with  $S \in \mathbb{C}$  and  $\mathbf{V} \in \mathbb{C}^3$ . Write  $\mathbf{M} = \text{mat}(Z)$ , a  $2 \times 2$  matrix with entries in  $\mathbb{C}$ . The identity matrices are  $\mathbf{I}_2$  and  $\mathbf{I}_4$  in their stated dimensions.

### A.1 Generators and products

The Pauli images of the generators are [6, p. 11]

$$\text{mat}(\sigma_1) = \begin{Bmatrix} 0 & 1 \\ 1 & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_2) = \begin{Bmatrix} 0 & -i \\ i & 0 \end{Bmatrix}, \quad \text{mat}(\sigma_3) = \begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix}.$$

Consequently,

$$\text{mat}(S + \mathbf{V} \cdot \boldsymbol{\sigma}) = \begin{Bmatrix} S + V_3 & V_1 - iV_2 \\ V_1 + iV_2 & S - V_3 \end{Bmatrix}. \quad (\text{A.1})$$

The generator images obey the sigma product rule and are linearly independent over  $\mathbb{C}$  together with  $\mathbf{I}_2$ . Thus every complex  $2 \times 2$  matrix corresponds to exactly one paravector. The map preserves addition, sends 1 to  $\mathbf{I}_2$ , and satisfies

$$\text{mat}(Z'Z) = \text{mat}(Z') \text{mat}(Z).$$

Here  $Z'$  is any second complex paravector. Thus algebra multiplication is ordinary matrix multiplication.

### A.2 Conjugations and product order

For complex matrices  $\mathbf{M}, \mathbf{N}$ , transpose  $^T$  exchanges row and column indices; entrywise complex conjugation  $^C$  sends each  $a + ib$  to  $a - ib$  for real  $a, b$ . Their composition is  $\mathbf{M}^H = (\mathbf{M}^C)^T$ . Introduce the fixed real matrix

$$\mathbf{S} = \begin{Bmatrix} 0 & 1 \\ -1 & 0 \end{Bmatrix}, \quad \mathbf{S}^{-1} = -\mathbf{S}.$$

The paravector sigma conjugation has the matrix image

$$\text{mat}(Z^*) = \mathbf{S}^{-1} \mathbf{M}^C \mathbf{S} =: \mathbf{M}^*. \quad (\text{A.2})$$

Substituting the basis in (A.2) gives

| Algebra basis         | 1              | i                | $\sigma_k$              | $i\sigma_k$               |
|-----------------------|----------------|------------------|-------------------------|---------------------------|
| Hermitian image       | $\mathbf{I}_2$ | $-i\mathbf{I}_2$ | $\text{mat}(\sigma_k)$  | $-i \text{mat}(\sigma_k)$ |
| Sigma-conjugate image | $\mathbf{I}_2$ | $-i\mathbf{I}_2$ | $-\text{mat}(\sigma_k)$ | $i \text{mat}(\sigma_k)$  |

Here  $k = 1, 2, 3$ . These are exactly the operations of Section 2, so  $\text{mat}(Z^H) = \mathbf{M}^H$ . Their product-order rules are

$$\begin{aligned} (\mathbf{M}\mathbf{N})^T &= \mathbf{N}^T \mathbf{M}^T, \\ (\mathbf{M}\mathbf{N})^C &= \mathbf{M}^C \mathbf{N}^C, \\ (\mathbf{M}\mathbf{N})^H &= \mathbf{N}^H \mathbf{M}^H. \end{aligned}$$

For sigma conjugation the middle factors cancel:

$$(\mathbf{M}\mathbf{N})^* = \mathbf{S}^{-1}\mathbf{M}^{\mathbf{C}}\mathbf{S}\mathbf{S}^{-1}\mathbf{N}^{\mathbf{C}}\mathbf{S}.$$

Thus sigma conjugation preserves product order:

$$(\mathbf{M}\mathbf{N})^* = \mathbf{M}^*\mathbf{N}^*.$$

This verifies (2.4) in matrices. Hermitian and sigma conjugation commute, and applying either twice restores the original matrix. The matrix-only operations can be recovered as

$$\begin{aligned}\mathbf{M}^{\mathbf{C}} &= \mathbf{S}\mathbf{M}^*\mathbf{S}^{-1}, \\ \mathbf{M}^{\mathbf{T}} &= \mathbf{S}(\mathbf{M}^*)^{\mathbf{H}}\mathbf{S}^{-1}.\end{aligned}$$

### A.3 Scalar part, determinant, and norm

For  $\mathbf{M} = (m_{jk})$ , its trace is  $m_{11} + m_{22}$  and its determinant is  $m_{11}m_{22} - m_{12}m_{21}$ . Substituting (A.1) gives

$$\text{sc}(Z) = \frac{1}{2} \text{tr } \mathbf{M} = S, \tag{A.3}$$

$$(\text{vec}(Z))_k = \frac{1}{2} \text{tr}(\text{mat}(\sigma_k)\mathbf{M}) = V_k,$$

$$\det Z = \det \mathbf{M} = S^2 - \mathbf{V}^2. \tag{A.4}$$

By (A.3), matrix-trace cyclicity is precisely the scalar-part cyclicity used in the main text. For  $\mathbf{M}' = \text{mat}(Z')$ ,

$$\text{sc}(Z'Z) = \frac{1}{2} \text{tr}(\mathbf{M}'\mathbf{M}) = \text{sc}(ZZ').$$

The remaining extraction and norm formulas are

$$\text{mat}(\text{vec}(Z) \cdot \boldsymbol{\sigma}) = \mathbf{M} - \text{sc}(Z) \mathbf{I}_2,$$

$$\text{mat}(\text{re}(Z)) = \frac{1}{2}(\mathbf{M} + \mathbf{M}^{\mathbf{H}}), \tag{A.5}$$

$$\text{mat}(\text{im}(Z)) = (\mathbf{M} - \mathbf{M}^{\mathbf{H}})/(2i), \tag{A.6}$$

$$\|Z\|^2 = \frac{1}{2} \text{tr}(\mathbf{M}^{\mathbf{H}}\mathbf{M}) = \|S\|^2 + \|\mathbf{V}\|^2. \tag{A.7}$$

Equations (A.5)–(A.6) extract Hermitian matrices, so a real paravector has a Hermitian image. The squared norm in (A.7) is half the sum of the squared magnitudes of the matrix entries. The determinant in (A.4) may be complex.

The matrix adjugate is

$$\text{adj}(\mathbf{M}) = (\text{tr } \mathbf{M})\mathbf{I}_2 - \mathbf{M} = \text{mat}(\text{adj}(Z)) = (\mathbf{M}^*)^{\mathbf{H}}.$$

It satisfies

$$\text{mat}(\text{adj}(Z) Z) = \det(\mathbf{M})\mathbf{I}_2.$$

Consequently  $\mathbf{M}^{-1} = \text{adj}(\mathbf{M}) / \det \mathbf{M}$  when the determinant is nonzero. The determinant product and adjugate order rules are the same in both representations. The quadratic identity (3.19) is precisely Cayley–Hamilton for a  $2 \times 2$  matrix.

## A.4 Boost and rotation matrices

Recall the real unit axis  $\mathbf{u} = \{u_1, u_2, u_3\}$  and real rapidity  $\theta$ . The matrix image of (3.36) is

$$\text{mat}(L) = \cosh(\theta/2)\mathbf{I}_2 - \sinh(\theta/2) \begin{Bmatrix} u_3 & u_1 - \mathrm{i}u_2 \\ u_1 + \mathrm{i}u_2 & -u_3 \end{Bmatrix}. \quad (\text{A.8})$$

The displayed axis matrix is Hermitian, has square  $\mathbf{I}_2$ , and has eigenvalues  $\pm 1$ . Thus (A.8) has eigenvalues  $e^{\mp\theta/2}$ , determinant one, and squared norm  $\frac{1}{2} \text{tr}(\text{mat}(L)^\text{H} \text{mat}(L)) = \cosh \theta$ . For a real rotation angle  $\theta$ , replacing the hyperbolic coefficients by  $\cos(\theta/2)$  and  $\mathrm{i} \sin(\theta/2)$  gives  $\text{mat}(R)$ , which is unitary.

Multiplication preserves the distinct spacetime and field laws:

$$\begin{aligned} \text{mat}(X') &= \text{mat}(L) \text{mat}(X) \text{mat}(L), \\ \text{mat}(F') &= \text{mat}(L) \text{mat}(F) \text{mat}(L)^{-1}. \end{aligned}$$

Here  $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$  is real and  $F = (\mathbf{E} + \mathrm{i}\mathbf{B}) \cdot \boldsymbol{\sigma}$  is complex, with  $t \in \mathbb{R}$  and  $\mathbf{r}, \mathbf{E}, \mathbf{B} \in \mathbb{R}^3$ . Extracting scalar and vector coefficients by (A.3) and the preceding component rule gives (11.4)–(11.5) and (11.17)–(11.18) for the same arbitrary axis. The first matrix stays Hermitian; the second stays traceless.

## A.5 Matrix amplitudes and column spinors

Let  $\psi$  be a complex paravector amplitude with matrix image  $\mathbf{M}_\psi$  at one fixed real event  $X$ . For a nonnegative real weight  $Z$ , the local densities (8.3) and (8.5) become

$$\begin{aligned} \|\psi\|^2 &= \frac{1}{2} \text{tr}(\mathbf{M}_\psi^\text{H} \mathbf{M}_\psi), \\ \|\psi\|_Z^2 &= \frac{1}{2} \text{tr}(\mathbf{M}_\psi^\text{H} \text{mat}(Z) \mathbf{M}_\psi). \end{aligned}$$

The first is one half the sum of the squared magnitudes of the matrix entries: the squared Frobenius norm with the factor  $1/2$  fixed by  $\text{sc} = \text{tr}/2$ . The second is its weighted form. Both are pointwise; neither contains a spatial integral. Matrix multiplication gives exactly the congruence identity (8.7).

**An arbitrary spin basis.** Recall the arbitrary constant perpendicular **real unit vectors**  $\mathbf{u}, \mathbf{a} \in \mathbb{R}^3$  of (8.1). Set  $\Pi_+ = (1 + \mathbf{u} \cdot \boldsymbol{\sigma})/2$  and choose a constant complex paravector  $\psi'$  with  $\Pi_+ \psi' \neq 0$ . Then

$$\begin{Bmatrix} \psi_+ \\ \psi_- \end{Bmatrix} := \frac{1}{\|\Pi_+ \psi'\|} \begin{Bmatrix} \Pi_+ \psi' \\ (\mathbf{a} \cdot \boldsymbol{\sigma}) \Pi_+ \psi' \end{Bmatrix}. \quad (\text{A.9})$$

This constructs (8.1) with  $(\mathbf{u} \cdot \boldsymbol{\sigma})\psi_\pm = \pm\psi_\pm$  and  $\text{sc}(\psi_s^\text{H} \psi_{s'}) = \delta_{ss'}$  for  $s, s' = +, -$ . The two complex scalar components give

$$\psi = c_+ \psi_+ + c_- \psi_- = (c_+ + c_- \mathbf{a} \cdot \boldsymbol{\sigma}) \psi_+. \quad (\text{A.10})$$

For fields,  $c_\pm : \mathbb{R}^4 \rightarrow \mathbb{C}$ , and (8.3) is  $\rho = \|c_+\|^2 + \|c_-\|^2$ .

**The two-component column.** Every column of  $\mathbf{M}_{\psi_+}$  has eigenvalue  $+1$  under  $\text{mat}(\mathbf{u} \cdot \boldsymbol{\sigma})$ , whose  $+1$  eigenspace is one-dimensional. Since  $\|\psi_+\| = 1$ , its matrix therefore factors as  $\mathbf{M}_{\psi_+} = \sqrt{2} \psi_+ \mathbf{v}^\text{H}$  for constant columns  $\psi_+, \mathbf{v} \in \mathbb{C}^2$  satisfying  $\psi_+^\text{H} \psi_+ = \mathbf{v}^\text{H} \mathbf{v} = 1$ .

Left multiplication gives  $\mathbf{M}_{\psi_-} = \sqrt{2} \psi_- \mathbf{v}^H$ , where  $\psi_- = \text{mat}(\mathbf{a} \cdot \boldsymbol{\sigma}) \psi_+$ . The columns  $\psi_+, \psi_-$  are orthonormal. Thus every amplitude in (A.10) has the conventional Pauli column

$$\boldsymbol{\psi} := \frac{1}{\sqrt{2}} \mathbf{M}_{\psi} \mathbf{v} = c_+ \psi_+ + c_- \psi_-.$$

Conversely,  $\mathbf{M}_{\psi} = \sqrt{2} \boldsymbol{\psi} \mathbf{v}^H$ , so this correspondence is one-to-one. For two amplitudes in this fixed basis and any complex multiplying paravector  $Z$ , substitution gives

$$\begin{aligned} \text{sc}(\boldsymbol{\psi}^H \boldsymbol{\psi}) &= \boldsymbol{\psi}^H \boldsymbol{\psi}, \\ \text{sc}(\boldsymbol{\psi}^H Z \boldsymbol{\psi}) &= \boldsymbol{\psi}^H \text{mat}(Z) \boldsymbol{\psi}. \end{aligned}$$

For nonnegative  $Z$ , the second line is  $\|\boldsymbol{\psi}\|_Z^2$ . Dividing by  $\rho = \boldsymbol{\psi}^H \boldsymbol{\psi} > 0$  gives  $\langle Z \rangle_{\boldsymbol{\psi}}$ , the ordinary finite-component expectation. Left action  $T\boldsymbol{\psi}$  maps to  $\text{mat}(T)\boldsymbol{\psi}$  for any complex paravector  $T$ .

Let  $\mathbf{C}$  be the  $2 \times 2$  matrix with columns  $\psi_+, \psi_-$ . Then  $\mathbf{C}^H \mathbf{C} = \mathbf{I}_2$  and  $\boldsymbol{\psi} = \mathbf{C} \{c_+, c_-\}^T$ . For a complex paravector  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ , its matrix in this basis is

$$\text{mat}'(Z) := \mathbf{C}^H \text{mat}(Z) \mathbf{C} = \begin{Bmatrix} S + \mathbf{u} \cdot \mathbf{V} & (\mathbf{a} - i\mathbf{u} \times \mathbf{a}) \cdot \mathbf{V} \\ (\mathbf{a} + i\mathbf{u} \times \mathbf{a}) \cdot \mathbf{V} & S - \mathbf{u} \cdot \mathbf{V} \end{Bmatrix}. \quad (\text{A.11})$$

Here the prime denotes only this change of matrix basis. For nonzero norm  $\rho = \boldsymbol{\psi}^H \boldsymbol{\psi}$ , the polarization (10.3) and probabilities for  $\Pi_{\pm} = (1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})/2$  become

$$\mathbf{u} \cdot \mathbf{u}' = \boldsymbol{\psi}^H \text{mat}(\mathbf{u} \cdot \boldsymbol{\sigma}) \boldsymbol{\psi} / \rho, \quad (\text{A.12})$$

$$\text{prob}(\pm | \mathbf{u}) = \boldsymbol{\psi}^H \text{mat}(\Pi_{\pm}) \boldsymbol{\psi} / \rho. \quad (\text{A.13})$$

Equations (A.12) and (A.13) are the ordinary spin expectation and Born rule.

## A.6 Dirac columns

For a Dirac block  $\boldsymbol{\Psi} = \{\Psi_+, \Psi_-\}^T$ , let  $\boldsymbol{\Psi}_+, \boldsymbol{\Psi}_- \in \mathbb{C}^2$  be the corresponding columns. Then

$$\mathbf{W}(Z) \boldsymbol{\Psi} \longleftrightarrow \begin{Bmatrix} 0 & \text{mat}(\text{adj}(Z)) \\ \text{mat}(Z) & 0 \end{Bmatrix} \begin{Bmatrix} \boldsymbol{\Psi}_+ \\ \boldsymbol{\Psi}_- \end{Bmatrix}.$$

The block norm in (8.27) becomes the ordinary four-component norm:

$$\rho = \boldsymbol{\Psi}_+^H \boldsymbol{\Psi}_+ + \boldsymbol{\Psi}_-^H \boldsymbol{\Psi}_-.$$

**Characteristic polynomial.** On one Fourier mode,  $s \in \mathbb{C}$  is the spectral parameter and  $\square$  denotes the scalar eigenvalue of the wave operator. The two off-diagonal blocks multiply to  $-\square \mathbf{I}_2$ , so block elimination gives the ordinary matrix determinant

$$\begin{aligned} \det \begin{Bmatrix} -(s+m) \mathbf{I}_2 & \text{mat}(i\partial) \\ \text{mat}(i\partial^*) & -(s+m) \mathbf{I}_2 \end{Bmatrix} &= \det(((s+m)^2 + \square) \mathbf{I}_2) \\ &= ((s+m)^2 + \square)^2. \end{aligned} \quad (\text{A.14})$$

This verifies (8.34), including multiplicities. The polynomial identity also holds at  $s = -m$ .

## A.7 Four components and the standard Dirac basis

In the arbitrary-axis basis (A.9), write  $\Psi_+ = a\psi_+ + b\psi_-$  and  $\Psi_- = c\psi_+ + d\psi_-$ , where  $a, b, c, d : \mathbb{R}^4 \rightarrow \mathbb{C}$  are the four scalar components. Their columns are  $\Psi'_+ = \{a, b\}^\top$  and  $\Psi'_- = \{c, d\}^\top$ ; here the prime means  $\Psi'_\pm = \mathbf{C}^H \Psi_\pm$ . Recall real mass  $m > 0$ , real charge  $q$ , and real potentials  $V : \mathbb{R}^4 \rightarrow \mathbb{R}$  and  $\mathbf{A} : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ . The mechanical energy and momentum operators are  $\hat{E}_q = i\partial_t - qV$  and  $\hat{\mathbf{p}}_q = -i\partial_{\mathbf{r}} - q\mathbf{A}$ . Their momentum components in this frame are

$$\hat{p}_{q,1} = \mathbf{a} \cdot \hat{\mathbf{p}}_q, \quad \hat{p}_{q,2} = (\mathbf{u} \times \mathbf{a}) \cdot \hat{\mathbf{p}}_q, \quad \hat{p}_{q,3} = \mathbf{u} \cdot \hat{\mathbf{p}}_q.$$

Using (A.11), the coupled paravector equations give

$$(\hat{E}_q - \hat{p}_{q,3})c - (\hat{p}_{q,1} - i\hat{p}_{q,2})d = ma, \quad (\text{A.15})$$

$$-(\hat{p}_{q,1} + i\hat{p}_{q,2})c + (\hat{E}_q + \hat{p}_{q,3})d = mb, \quad (\text{A.16})$$

$$(\hat{E}_q + \hat{p}_{q,3})a + (\hat{p}_{q,1} - i\hat{p}_{q,2})b = mc, \quad (\text{A.17})$$

$$(\hat{p}_{q,1} + i\hat{p}_{q,2})a + (\hat{E}_q - \hat{p}_{q,3})b = md. \quad (\text{A.18})$$

Operators in (A.15)–(A.18) act on the scalar immediately to their right. Their Hamiltonian on  $\{a, b, c, d\}^\top$  is

$$\hat{H} = \begin{Bmatrix} qV - \hat{p}_{q,3} & -\hat{p}_{q,1} + i\hat{p}_{q,2} & m & 0 \\ -\hat{p}_{q,1} - i\hat{p}_{q,2} & qV + \hat{p}_{q,3} & 0 & m \\ m & 0 & qV + \hat{p}_{q,3} & \hat{p}_{q,1} - i\hat{p}_{q,2} \\ 0 & m & \hat{p}_{q,1} + i\hat{p}_{q,2} & qV - \hat{p}_{q,3} \end{Bmatrix}. \quad (\text{A.19})$$

This is the component image of (8.40).

To identify conventional gamma matrices, put  $\sigma_k = \text{mat}(\sigma_k)$  for  $k = 1, 2, 3$ . These are also the images under  $\text{mat}'$  of  $\mathbf{a} \cdot \boldsymbol{\sigma}$ ,  $(\mathbf{u} \times \mathbf{a}) \cdot \boldsymbol{\sigma}$ , and  $\mathbf{u} \cdot \boldsymbol{\sigma}$ , respectively. Their numerical entries apply to every choice of  $\mathbf{u}, \mathbf{a}$ . Define [16, Eqs. (4.16)–(4.17)]

$$\gamma^0 = \begin{Bmatrix} 0 & \mathbf{I}_2 \\ \mathbf{I}_2 & 0 \end{Bmatrix}, \quad (\text{A.20})$$

$$\gamma^k = \begin{Bmatrix} 0 & \sigma_k \\ -\sigma_k & 0 \end{Bmatrix}. \quad (\text{A.21})$$

Equations (A.20)–(A.21) obey  $(\gamma^0)^2 = \mathbf{I}_4$ ,  $(\gamma^k)^2 = -\mathbf{I}_4$ , and  $\gamma^j \gamma^k + \gamma^k \gamma^j = 0$  for distinct  $j, k = 0, 1, 2, 3$ . The standard chiral Dirac operator in this basis, with signature  $(+, -, -, -)$ , is

$$\mathbf{W}(\hat{P}_q) \longleftrightarrow \gamma^0 \hat{E}_q - \sum_{k=1}^3 \gamma^k \hat{p}_{q,k}.$$

Here  $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$  is the real potential paravector.

The unitary change to the standard Dirac basis commutes with the scalar rest-phase factor explained in (9.1). Together they define the envelope columns

$$\begin{Bmatrix} \phi_+ \\ \phi_- \end{Bmatrix} = \frac{e^{imt}}{\sqrt{2}} \begin{Bmatrix} \mathbf{I}_2 & \mathbf{I}_2 \\ -\mathbf{I}_2 & \mathbf{I}_2 \end{Bmatrix} \begin{Bmatrix} \Psi'_+ \\ \Psi'_- \end{Bmatrix}. \quad (\text{A.22})$$

Here  $\phi_+, \phi_- \in \mathbb{C}^2$  are the coefficient columns in the same spin basis of the paravector fields  $\phi_+ = e^{imt}(\Psi_+ + \Psi_-)/\sqrt{2}$  and  $\phi_- = e^{imt}(\Psi_- - \Psi_+)/\sqrt{2}$ . Before removing the rest phase, the basis change in (A.22) sends (A.19) to

$$\hat{H}' = \begin{Bmatrix} (qV + m)\mathbf{I}_2 & \sum_k \hat{p}_{q,k} \boldsymbol{\sigma}_k \\ \sum_k \hat{p}_{q,k} \boldsymbol{\sigma}_k & (qV - m)\mathbf{I}_2 \end{Bmatrix}. \quad (\text{A.23})$$

Multiplying the full column by  $e^{imt}$  subtracts  $m$  from its Hamiltonian:  $\mathbf{i}\partial_t(e^{imt}\psi) = e^{imt}(\mathbf{i}\partial_t - m)\psi$ . Thus (A.23) becomes  $\hat{H}' - m\mathbf{I}_4$ : its diagonal entries are  $qV$  and  $qV - 2m$ . Moving  $qV$  into  $\hat{E}_q = \mathbf{i}\partial_t - qV$  gives the exact envelope equations

$$\begin{aligned} \hat{E}_q \phi_+ &= \sum_k \hat{p}_{q,k} \boldsymbol{\sigma}_k \phi_-, \\ (\hat{E}_q + 2m) \phi_- &= \sum_k \hat{p}_{q,k} \boldsymbol{\sigma}_k \phi_+. \end{aligned}$$

These identify the large and small amplitudes used in the Pauli limit.

For real electric and magnetic components  $E_k, B_k$  in the same frame, the squared Dirac field coupling has the explicit image

$$\begin{aligned} \text{mat}'(\mathbf{i}qF) &= q \begin{Bmatrix} \mathbf{i}E_3 - B_3 & \mathbf{i}E_1 + E_2 - B_1 + \mathbf{i}B_2 \\ \mathbf{i}E_1 - E_2 - B_1 - \mathbf{i}B_2 & -\mathbf{i}E_3 + B_3 \end{Bmatrix}, \\ \text{mat}'(\mathbf{i}qF^*) &= \text{mat}'(\mathbf{i}qF)^{\text{H}}. \end{aligned}$$

The indexed  $E_k$  are field components; whereas  $\hat{E}_q$  is the mechanical energy operator. The two couplings are  $-q\mathbf{B} \cdot \boldsymbol{\sigma} \pm \mathbf{i}q\mathbf{E} \cdot \boldsymbol{\sigma}$ .

## A.8 Quaternion matrix image

For the quaternion  $Q = a + \mathbf{A} \cdot \mathbf{q} = a - \mathbf{i}\mathbf{A} \cdot \boldsymbol{\sigma}$  of Section 4.4, with real  $a$  and  $\mathbf{A}$ , the explicit matrix image is [6, p. 10]

$$\text{mat}(Q) = \begin{Bmatrix} a - \mathbf{i}A_3 & -A_2 - \mathbf{i}A_1 \\ A_2 - \mathbf{i}A_1 & a + \mathbf{i}A_3 \end{Bmatrix}. \quad (\text{A.24})$$

Here  $\text{mat}(q_k) = -\mathbf{i} \text{mat}(\sigma_k)$ . The conjugation and norm identities become

$$\begin{aligned} \text{mat}(Q^{\text{H}}) &= \text{mat}(Q)^{\text{H}}, \\ \text{mat}(Q)^{\text{H}} \text{mat}(Q) &= (a^2 + \mathbf{A}^2) \mathbf{I}_2, \\ \det(\text{mat}(Q)) &= a^2 + \mathbf{A}^2. \end{aligned}$$

For a unit quaternion,  $a^2 + \mathbf{A}^2 = 1$ , these identities give a unitary matrix of determinant one.

## B Equation reference

Each displayed formula repeats a numbered result; its right-hand number links to the definition or derivation. Natural units are  $\hbar = c = 1$ . The linked text supplies domains, component expansions, and approximation conditions.

**Algebra.** Recall complex paravectors  $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$  and  $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$ , with complex scalar  $S, S'$  and complex vectors  $\mathbf{V}, \mathbf{V}'$ . The prime here labels a second element. Their product is

$$\begin{aligned} ZZ' &= (SS' + \mathbf{V} \cdot \mathbf{V}') \\ &\quad + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}. \end{aligned} \quad (3.20)$$

The basic operations are

$$Z^H = S^* + \mathbf{V}^* \cdot \boldsymbol{\sigma}, \quad (3.2)$$

$$Z^* = S^* - \mathbf{V}^* \cdot \boldsymbol{\sigma}, \quad (3.1)$$

$$\text{adj}(Z) = Z^{*H} = S - \mathbf{V} \cdot \boldsymbol{\sigma}, \quad (3.3)$$

$$\text{tr } Z = 2\text{sc}(Z), \quad (3.4)$$

$$\text{re}(Z) = \frac{1}{2}(Z + Z^H), \quad (3.9)$$

$$\text{im}(Z) = (Z - Z^H)/(2i), \quad (3.10)$$

$$\det Z = \text{adj}(Z) Z = S^2 - \mathbf{V}^2, \quad (3.5)$$

$$Z^{-1} = \text{adj}(Z) / \det Z \quad (\det Z \neq 0), \quad (3.7)$$

$$\|Z\|^2 = \text{sc}(Z^H Z). \quad (3.25)$$

The coefficient form is (3.11); product-order and cyclicity rules are (2.3), (2.4), and (3.8). Projectors and the spectral rule are (3.27) and (3.28). The quaternion component map is (4.7); matrix images are (A.1) and (A.24).

**Rotations and boosts.** Here  $\mathbf{u}$  is a [real unit axis](#) and  $\theta$  a real angle or rapidity;  $\beta = \tanh \theta$  and  $\gamma = \cosh \theta$  in the boost formulas. The factors and their actions are

$$R = \exp(-i\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = \cos(\theta/2) - i \sin(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (3.35)$$

$$L = \exp(-\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2) = \cosh(\theta/2) - \sinh(\theta/2) \mathbf{u} \cdot \boldsymbol{\sigma}, \quad (3.36)$$

$$L = \sqrt{\frac{\gamma+1}{2}} \left( 1 - \frac{\gamma\beta}{\gamma+1} \mathbf{u} \cdot \boldsymbol{\sigma} \right), \quad (3.45)$$

$$L^2 = \gamma(1 - \beta \mathbf{u} \cdot \boldsymbol{\sigma}), \quad (3.46)$$

$$Z' = RZR^H, \quad (4.5)$$

$$Z' = LZL, \quad (4.11)$$

$$F' = LFL^{-1}. \quad (11.15)$$

Both  $Z$  transformations use  $Z' = TZT^H$ , with  $R^H = R^{-1}$  and  $L^H = L$ . The  $Z$  boost acts on a real four-vector; the complex electromagnetic field  $F$  uses similarity. Both refer to the same frame. Rodrigues' rotation is (4.6); boost components are (4.12)–(4.13). The general determinant-one congruence is (4.3); interval preservation is (11.7). Physical field components are derived in Section 11, with matrix verification in Appendix A.4.

**Spacetime and particles.** Recall real event  $X$ , proper time  $s$ , coordinate velocity  $\mathbf{v}$ , and constant mass  $m > 0$ . The real proper velocity  $U = dX/ds$  is a [unit paravector](#) with  $UU^{-1} = 1$ . The real four-momentum  $P$  has mechanical energy  $E$  (including rest energy) and spatial momentum  $\mathbf{p}$ . The kinematic identities are

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \quad (5.1)$$

$$\square = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2, \quad (5.2)$$

$$ds^2 = \det(dX) = dt^2 - d\mathbf{r}^2, \quad (6.1)$$

$$U = dX/ds = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad (6.4)$$

$$U^*U = 1, \quad (4.2)$$

$$P = mU = E + \mathbf{p} \cdot \boldsymbol{\sigma}, \quad (6.5)$$

$$E^2 - \mathbf{p}^2 = m^2. \quad (6.6)$$

Here  $\gamma = dt/ds = (1 - \mathbf{v}^2)^{-1/2}$ . Proper velocity is real, with [determinant one](#); Hermitian unitarity is the separate condition (3.14). For real proper times  $s_0 < s_1$  and coordinate times  $t_0 < t$ ,

$$X(s_1) - X(s_0) = \int_{s_0}^{s_1} U(s) ds, \quad (6.10)$$

$$s(t) - s(t_0) = \int_{t_0}^t \frac{dt'}{\gamma(t')} = \int_{t_0}^t \sqrt{1 - \mathbf{v}(t')^2} dt'. \quad (6.11)$$

Clock comparison is (6.12); constant proper acceleration gives (6.13); the future cone is (6.14). Proper acceleration, inertia, and particle action are (6.7), (6.8), and (6.15). Moving-frame velocity and its causal bound are (11.9)–(11.10); clock dilation and measured contraction are (11.11) and (11.12); energy and momentum transform by (11.13)–(11.14).

**Electromagnetism.** Recall the real proper velocity  $U = dX/ds$ , a [unit paravector](#) with  $UU^{-1} = 1$ . Recall real potentials  $V, \mathbf{A}$ , real electric and magnetic fields  $\mathbf{E}, \mathbf{B}$ , the complex field  $F$ , real potential paravector  $\Phi$ , and real charge-current paravector  $D$ . The real gauge scalar is  $S$ ;  $q$  is charge. These fields are functions of real spacetime. Their boosted arguments and components are (11.16) and (11.17)–(11.18). The field and particle equations are

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (7.1)$$

$$\partial F = D^*, \quad (7.2)$$

$$\square F = \partial^* D^*, \quad (7.7)$$

$$\partial \Phi = S + F^*, \quad (7.11)$$

$$F^2 = \mathbf{E}^2 - \mathbf{B}^2 + 2i\mathbf{E} \cdot \mathbf{B}, \quad (7.15)$$

$$\frac{dP}{ds} = q \operatorname{re}(FU), \quad (7.21)$$

$$T(N) = \frac{1}{2} F N F^H, \quad (7.22)$$

$$\partial_t u + \partial_{\mathbf{r}} \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}. \quad (7.23)$$

Here  $N$  is a real paravector argument of the stress map,  $u$  field energy density,  $\mathbf{S}$  energy flux, and  $\mathbf{J}$  electric current density. Potential components are (7.12)–(7.13); force components are (7.17) and (7.19). The action derivation uses (7.14) and (7.16).

**Relativistic waves.** Recall real mass  $m > 0$ , charge  $q$ , and potentials  $V, \mathbf{A}$ . Their real potential paravector is  $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$ . Here  $\hat{E}_q, \hat{\mathbf{p}}_q$  are mechanical differential operators,  $\psi$  a complex scalar KG field, and  $\Psi$  a two-entry complex paravector Dirac field. The operators and wave equations are

$$\hat{E}_q = \hat{E} - qV = i\partial_t - qV, \quad (8.14)$$

$$\hat{\mathbf{p}}_q = \hat{\mathbf{p}} - q\mathbf{A} = -i\partial_{\mathbf{r}} - q\mathbf{A}, \quad (8.15)$$

$$\hat{P}_q = \hat{P} - q\Phi = \hat{E}_q + \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}, \quad (8.16)$$

$$(\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma})^2 = \hat{\mathbf{p}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}, \quad (8.18)$$

$$[\hat{E}_q, \hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}] = iq\mathbf{E} \cdot \boldsymbol{\sigma}, \quad (8.19)$$

$$(-\hat{E}_q^2 + \hat{\mathbf{p}}_q^2 + m^2)\psi = 0, \quad (8.24)$$

$$\mathbf{W}(Z) = \begin{Bmatrix} 0 & \text{adj}(Z) \\ Z & 0 \end{Bmatrix}. \quad (8.28)$$

Here  $Z$  is a complex paravector and  $\mathbf{I}$  the block identity. For the free field ( $q = 0$ ),

$$\hat{P} \text{adj}(\hat{P}) = \det \hat{P} = \hat{E}^2 - \hat{\mathbf{p}}^2 = -\square, \quad (8.17)$$

$$\hat{D}(m) = \mathbf{W}(\hat{P}) - m\mathbf{I}, \quad (8.29)$$

$$\hat{D}(m)\Psi = 0, \quad (8.30)$$

$$\hat{D}(-m)\hat{D}(m) = -(\square + m^2)\mathbf{I}. \quad (8.31)$$

For fixed  $q$  and  $\Phi$ , the interacting definition and equation are

$$\hat{D}_q(m) = \mathbf{W}(\hat{P}_q) - m\mathbf{I}, \quad (8.43)$$

$$\hat{D}_q(m)\Psi = (\mathbf{W}(\hat{P}_q) - m\mathbf{I})\Psi = 0. \quad (8.37)$$

Both  $\hat{D}(-m)$  and  $\hat{D}_q(-m)$  change only the sign of the mass term. The free equations are (8.22) and (8.30); the two-branch solution is (8.23). Massive Dirac reconstruction is given by (8.36) and (8.46); the squared interacting entries are (8.44) and (8.45).

**Low-energy limits.** After removing rest energy by (9.1) and (9.7), the complex scalar envelope  $\psi$  obeys Schrödinger's equation to leading order:

$$i\partial_t\psi = \left(qV + \frac{\hat{\mathbf{p}}_q^2}{2m}\right)\psi. \quad (9.9)$$

The complex paravector envelope  $\psi := \phi_+$  obeys Pauli's equation:

$$i\partial_t\psi = \left(qV + \frac{\hat{\mathbf{p}}_q^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m}\right)\psi. \quad (9.6)$$

The derivations and approximation conditions are in Section 9; these are the same leading order for scalar and spinor amplitudes respectively.

**Spin and probabilities.** Recall a nonzero complex paravector amplitude  $\psi$ , real norm density  $\rho > 0$ , real polarization  $\mathbf{u}'$ , and [real unit measurement axis](#)  $\mathbf{u}$ . The real observable  $Z$  acts from the left. A rotation uses

$$\psi' = R\psi. \quad (10.1)$$

It preserves the norm and inner product; the expectation transforms as  $\langle Z \rangle_{\psi'} = \langle R^H Z R \rangle_{\psi}$  by [\(8.8\)](#). All formulas are pointwise. For a nonnegative real weight  $Z$  from [\(8.5\)](#), the densities and measurements are

$$\rho = \|\psi\|^2 = \text{sc}(\psi^H \psi), \quad (8.3)$$

$$\|\psi\|_Z^2 = \text{sc}(\psi^H Z \psi), \quad (8.5)$$

$$\mathbf{u} \cdot \mathbf{u}' = \text{sc}(\psi^H (\mathbf{u} \cdot \boldsymbol{\sigma}) \psi) / \rho, \quad (10.3)$$

$$\text{prob}(\pm \mid \mathbf{u}) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \mathbf{u}'), \quad (10.4)$$

$$\langle Z \rangle_{\psi} = \text{sc}(\psi^H Z \psi) / \rho. \quad (8.6)$$

The expectation formula also applies to signed real observables. The matrix verification is [\(A.12\)](#)–[\(A.13\)](#).

## C Conventions

**Mapping to the literature.** Our algebra is the algebra of physical space (APS),  $\text{Cl}_{3,0}$ , with the conventions of Baylis and Sobczyk [1, pp. 4–6]. The table translates the notation of the [Paravector article](#);  $A$  denotes a general element and  $x$  a real paravector.

| Object or operation      | This paper                                     | APS notation                           |
|--------------------------|------------------------------------------------|----------------------------------------|
| Spatial basis            | $\sigma_k$                                     | $\mathbf{e}_k$                         |
| Central imaginary unit   | $\mathbf{i} = \sigma_1 \sigma_2 \sigma_3$      | $i = \mathbf{e}_{123}$                 |
| Real paravector          | $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ | $x = x^0 + \mathbf{x}$                 |
| General algebra element  | $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ | $A$ (complex paravector)               |
| Reversion                | $Z^{\mathbf{H}}$                               | $A^\dagger$                            |
| Clifford conjugation     | $\text{adj } Z$                                | $\bar{A}$                              |
| Grade involution         | $Z^*$                                          | $\bar{A}^\dagger$                      |
| Complex scalar part      | $\text{sc}(Z)$                                 | $\langle A \rangle_S$                  |
| Complex vector part      | $(\text{vec}(Z)) \cdot \boldsymbol{\sigma}$    | $\langle A \rangle_V$                  |
| Real part                | $\text{re}(Z)$                                 | $\langle A \rangle_R$                  |
| Imaginary part           | $\mathbf{i} \text{im}(Z)$                      | $\langle A \rangle_I$                  |
| Spacetime quadratic form | $\det X = t^2 - \mathbf{r}^2$                  | $x\bar{x}$                             |
| Projectors               | $\Pi_+, \Pi_-$                                 | $P_{\mathbf{u}}, \bar{P}_{\mathbf{u}}$ |

Real paravectors have four real components; complex  $Z$  packages all eight APS components. Here  $\text{sc}(Z)$  includes the pseudoscalar, and  $\text{im}(Z)$  removes the imaginary part’s factor  $\mathbf{i}$ .

In particular, the bar is our adjugate, not our star:

$$\bar{A} \longleftrightarrow S - \mathbf{V} \cdot \boldsymbol{\sigma}, \quad \bar{A}^\dagger \longleftrightarrow S^* - \mathbf{V}^* \cdot \boldsymbol{\sigma}.$$

They coincide only for real paravectors. Our star conjugates complex coefficients but is grade involution on the full algebra. Wikipedia’s [paragradient](#) maps to our starred derivative:

$$\partial_{\text{article}} = \partial_t - \nabla \longleftrightarrow \partial^*, \quad \bar{\partial}_{\text{article}} \longleftrightarrow \partial, \quad \square = \partial \partial^*.$$

Baylis’s rotor action  $x' = GxG^\dagger$  is our  $X' = GXG^{\mathbf{H}}$ ; his field  $\mathbf{E} + i\mathbf{B}$  and law  $F' = GFG^\dagger$  become our  $F = (\mathbf{E} + \mathbf{iB}) \cdot \boldsymbol{\sigma}$  and  $F' = GFG^{-1}$  for  $\det G = 1$  ( $c = 1$ ) [5, Eq. (23)] [5, Eq. (24)] [5, Eq. (33)]. His positively oriented spatial-plane generator is  $\mathbf{e}_2 \mathbf{e}_1 = -i\mathbf{e}_3$ , giving our  $R = e^{-i\theta \sigma_3/2}$  about  $+\sigma_3$  [5, Eqs. (17)–(19)]. His positive-rapidity boost factor, denoted here by  $B$ , satisfies

$$B = e^{+\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2} = L^{-1}, \quad B^2 = \gamma(1 + \beta \mathbf{u} \cdot \boldsymbol{\sigma}), \quad X' = B^{-1}X(B^{-1})^{\mathbf{H}} = LXL^{\mathbf{H}}.$$

Thus our passive boost uses the inverse of his positive active boost [5, p. 23]; the active/passive distinction is explicit in [1, Eq. (38)]. The algebra is standard; exponent signs must be read with the transformation direction and sandwich order.

To translate to Hestenes’s four-dimensional STA, choose an observer axis  $\gamma_0$ , with  $\gamma_0^2 = 1$  and  $\gamma_k^2 = -1$ . The identification with its even subalgebra is [1, Eq. (40)] [1, Eq. (41)]

$$\sigma_k \longleftrightarrow \gamma_k \gamma_0, \quad \text{adj } Z \longleftrightarrow \tilde{Z}_{\text{STA}}, \quad Z^{\mathbf{H}} \longleftrightarrow \gamma_0 \tilde{Z}_{\text{STA}} \gamma_0.$$

Here  $Z_{\text{STA}}$  is the image of  $Z$  and the tilde reverses STA products: reversion in APS and reversion in STA are different operations.

**Source comparisons for the cited equations.** Body citations point to matching definitions or physical component results. The source locators below identify those comparisons; intervening algebra and our block-map packaging are derived in this paper.

| Result here                                                        | Direct source comparison                                                              |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Sigma product, conjugations, and extractions (1.1)–(3.10)          | [1, pp. 4–6]; [2, Eq. (3)].                                                           |
| Positive norm (3.25); inverse (3.7)                                | [2, Eqs. (46)–(47)].                                                                  |
| Quaternions (4.7); projectors (3.27)                               | [3, p. 92]; [3, p. 121].                                                              |
| Spectral rule (3.28); coalescence (3.30)                           | [4, pp. 3–4].                                                                         |
| Interval and proper velocity (6.1)–(6.4)                           | [5, Eqs. (7)–(8)].                                                                    |
| Momentum, acceleration, and world lines (6.5), (6.7), (6.13)       | [9, Eq. (7.24)]; [9, p. 141]; [9, Eq. (7.33), p. 143].                                |
| Maxwell, stress map, and energy flux (7.2), (7.22)                 | [8, Eq. (177)]; [8, Eq. (167)]; [8, Eq. (180)].                                       |
| Potential, gauge, and field action (7.12)–(7.13), (7.14)           | [7, p. 105]; [7, Eq. (5.42)]; [7, Eq. (5.43)]; [7, Eqs. (6.1)–(6.3)]; [7, Eq. (6.4)]. |
| Lorentz force (7.21); components (7.17), (7.19)                    | [10, Eq. (2)]; [7, Eqs. (5.32)–(5.33)].                                               |
| Boosted components (11.5), (11.18)                                 | [7, Eq. (5.1)]; [7, Eqs. (5.17)–(5.18)].                                              |
| KG and minimally coupled Dirac (8.22), (8.24), (8.37)              | [16, Eq. (1.13)]; [17, Eq. (299)]; [8, Eqs. (190)–(191)].                             |
| Chiral squared Dirac (8.44)                                        | [18, Eq. (6)].                                                                        |
| Kinetic operators, small amplitude, and Pauli (8.15), (9.5), (9.6) | [12, Eqs. (2.11)–(2.16)].                                                             |
| KG to Schrödinger (9.9)                                            | [19, Eq. (21.3)]; [16, Eqs. (2.103)–(2.104)]; [16, Eq. (2.105)].                      |
| Spin amplitudes and probabilities (8.2), (10.4)                    | [11, Eqs. (66)–(67)]; [11, Eq. (69a)]; [13, Eq. (6.1)]; [13, p. 62].                  |

**Translating the physical comparisons.** Tong’s component electrodynamics uses SI units; our formulas set  $c = \epsilon_0 = \mu_0 = 1$ . His gauge parameter in Eq. (5.13) is  $\chi = -\lambda$ , whereas Eq. (6.2) uses our plus-sign convention [7, Eq. (5.13)] [7, Eq. (6.2)]. His positive-exponent spatial rotation is a passive axis rotation; his negative-exponent boost agrees with our passive boost [9, Eq. (7.38)] [9, Eq. (7.39)].

Chappell et al.’s scalar  $Z$  is our  $S$ , and their vector–bivector  $F$  is our  $\mathbf{V} \cdot \boldsymbol{\sigma}$ , with  $|F|^2 = -\mathbf{V}^2$ . Thus their trigonometric exponential is our hyperbolic form [2, Eq. (52)].

For the quaternion basis, Baylis identifies the Hamilton units with  $\mathbf{e}_{32}, \mathbf{e}_{13}, \mathbf{e}_{21}$ , respectively [3, p. 92]. With  $\mathbf{e}_k \leftrightarrow \sigma_k$ , these are exactly our  $-\mathrm{i}\sigma_1, -\mathrm{i}\sigma_2, -\mathrm{i}\sigma_3$ . Higham’s spectral and repeated-root formulas apply through the faithful matrix image in Appendix A; no change of the intrinsic product is required [4, pp. 3–4].

Lenz’s scalar potentials  $\phi, \Phi$  are our  $V$ ; set  $\hbar = c = 1$  and  $e = q$ . His Dirac Hamiltonian is the matrix form (A.23) [17, Eq. (412)].

For Hestenes’s Maxwell equation, write  $\nabla_{\text{STA}}$  for his spacetime derivative and  $J_{\text{STA}}$  for his current. Using the STA embedding above,

$$\nabla_{\text{STA}} = \gamma_0 \partial, \quad J_{\text{STA}} = D\gamma_0 = \gamma_0 D^*, \quad \partial F = D^* \iff \nabla_{\text{STA}} F = J_{\text{STA}}.$$

His explicit spacetime split is already our component equation [8, Eqs. (175)–(177)]. For a spacetime vector  $n$ , put  $N = n\gamma_0$ . His stress map  $T_{\text{STA}}(n) = \frac{1}{2}Fn\tilde{F}$ , with  $F^{\text{H}} = \gamma_0\tilde{F}\gamma_0$ , gives

$$T_{\text{STA}}(n)\gamma_0 = \frac{1}{2}FN F^{\text{H}} = T(N),$$

including the energy and flux components in his Eq. (180) [8, Eq. (167)] [8, Eq. (180)].

For Dirac and Pauli, Hestenes and Gurtler–Hestenes represent the complex unit by right multiplication by the spatial bivector  $\mathrm{i}\sigma_3$  on their real spinor; it acts as ordinary  $\mathrm{i}$  in the corresponding column spinor [8, Eqs. (187)–(188)] [8, Eqs. (192)–(193)]. Set  $e = q$  and  $\hbar = c = 1$ . Gurtler–Hestenes’s  $V$  is potential energy (our  $qV$ ); after removal of rest energy, their operators  $K$  and  $p$  correspond to our  $\hat{E}_q$  and  $\hat{\mathbf{p}}_q \cdot \boldsymbol{\sigma}$ . Their large/small equations and Pauli limit then have the same components [12, Eqs. (2.11)–(2.19)].

Feynman–Gell-Mann absorb charge into the potential. Restoring it, their Eq. (6) is our first squared Dirac block because

$$-\mathrm{i}qF^* = q(\mathbf{B} + \mathrm{i}\mathbf{E}) \cdot \boldsymbol{\sigma}, \quad -\mathrm{i}q\tilde{F} = q(\mathbf{B} - \mathrm{i}\mathbf{E}) \cdot \boldsymbol{\sigma}.$$

The two blocks therefore have the same magnetic sign and opposite electric signs [18, Eq. (6)].

Baylis et al.’s  $RP_3$  is a two-component minimal-ideal spinor; its rotations agree with our left amplitude action. Their density-operator formula gives the same  $(1 + \mathbf{u} \cdot \mathbf{u}')/2$  probability after basis normalization, extending to mixed states with  $\|\mathbf{u}'\| \leq 1$  [11, Eqs. (66)–(67)]. Feynman’s spin chapter uses passive basis rotations; its  $2\pi$  sign and probability statements agree, while an active rotor uses the inverse rotation [20, Eq. (6.19)].

Fix the right-handed sigma product and write  $N = \mathbf{u} \cdot \boldsymbol{\sigma}$ , with real  $\mathbf{u}^2 = 1$ . The alternatives below change notation or transformation direction, not the underlying physics. The main text uses  $R = R_-$  and  $L = L_-$ , with

$$R_{\pm} = e^{\pm i\theta N/2}, \quad L_{\pm} = e^{\pm \theta N/2}, \quad R_{\pm}^H = R_{\pm}^{-1} = R_{\mp}, \quad L_{\pm}^H = L_{\pm}, \quad L_{\pm}^{-1} = L_{\mp}.$$

Here  $\theta$  is a rotation angle for  $R_{\pm}$  and rapidity for  $L_{\pm}$ .

**Exponent sign and sandwich order.** For  $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$ , the four choices for each kind of factor are summarized by the two action columns:

| Factor $T$ | $X' = TXT^H$                      | $X' = T^H XT$                     |
|------------|-----------------------------------|-----------------------------------|
| $R_-$      | Active rotation $+\theta$         | Active rotation $-\theta$         |
| $R_+$      | Active rotation $-\theta$         | Active rotation $+\theta$         |
| $L_-$      | Frame velocity $+\beta\mathbf{u}$ | Frame velocity $+\beta\mathbf{u}$ |
| $L_+$      | Frame velocity $-\beta\mathbf{u}$ | Frame velocity $-\beta\mathbf{u}$ |

For rotations  $t' = t$ . Both  $+\theta$  entries give

$$\mathbf{r}' = \mathbf{r} \cos \theta + (\mathbf{u} \times \mathbf{r}) \sin \theta + (1 - \cos \theta)(\mathbf{u} \cdot \mathbf{r})\mathbf{u}.$$

This is the active right-handed Rodrigues formula (4.6); a passive rotation of axes uses the inverse map.

For boosts,  $\beta = \tanh \theta$  and  $\gamma = \cosh \theta$ . Writing  $\mathbf{r}_{\parallel} = (\mathbf{u} \cdot \mathbf{r})\mathbf{u}$  and  $\mathbf{r}_{\perp} = \mathbf{r} - \mathbf{r}_{\parallel}$ , both sandwiches with  $L_{\pm}$  give

$$t' = \gamma(t \pm \beta\mathbf{u} \cdot \mathbf{r}), \quad \mathbf{r}' = \mathbf{r}_{\perp} + \gamma(\mathbf{r}_{\parallel} \pm \beta t\mathbf{u}).$$

The minus choice sends  $\mathbf{r} = \beta t\mathbf{u}$  to  $\mathbf{r}' = 0$ , fixing its frame velocity. Moving the adjoint reverses a rotation but has no effect on a boost; reversing the boost exponent gives the inverse frame transformation. The same  $L_+$  map can instead describe an active boost to velocity  $+\beta\mathbf{u}$  in fixed coordinates.

**Hamilton quaternions fix the basis orientation.** With the sigma orientation held fixed, the two uniform sign choices obey

$$\begin{aligned} q_k &= -i\sigma_k : & q_j q_k &= -\delta_{jk} + \epsilon_{jkl} q_l, \\ \tilde{q}_k &= +i\sigma_k : & \tilde{q}_j \tilde{q}_k &= -\delta_{jk} - \epsilon_{jkl} \tilde{q}_l. \end{aligned}$$

Repeated  $l$  is summed. Thus  $q_1 q_2 = q_3$  is Hamilton's convention, whereas  $\tilde{q}_1 \tilde{q}_2 = -\tilde{q}_3$  has opposite orientation. For the Hamilton basis, Section 4.4 gives

$$Q = e^{\theta \mathbf{u} \cdot \mathbf{q}/2} = R_-, \quad Q^H = Q^{-1}, \quad \mathbf{r}' \cdot \mathbf{q} = Q(\mathbf{r} \cdot \mathbf{q})Q^H.$$

Since  $\mathbf{r} \cdot \mathbf{q} = -i\mathbf{r} \cdot \boldsymbol{\sigma}$  and  $i$  commutes with  $Q$ , the same sandwich rotates  $\mathbf{r} \cdot \boldsymbol{\sigma}$ , with no extra sign. For example,  $Q = (1 - i\sigma_3)/\sqrt{2}$  sends  $\sigma_1$  to  $\sigma_2$ .

Using  $\tilde{q}_k$  with  $\tilde{Q} = e^{\theta \mathbf{u} \cdot \tilde{\mathbf{q}}/2} = R_+$  and the right-adjoint sandwich instead gives  $-\theta$ . Recover  $+\theta$  by reversing either the exponent or the sandwich. Reflecting the basis also restores Hamilton multiplication, but changes its identification with the physical axes. Replacing a factor by its negative leaves both sandwiches unchanged; the spinor acquires an overall minus sign. Consequently rotations are  $2\pi$ -periodic on vectors and  $4\pi$ -periodic on amplitudes.

**Rotating a state versus changing the measured observable.** With the Hamilton basis, the two consistent descriptions of the same active  $+\theta$  rotation are

| Choice    | Quaternion relation | Rotated observable | Rotated amplitude  |
|-----------|---------------------|--------------------|--------------------|
| $R = R_-$ | $R = Q$             | $Z' = RZR^H$       | $\psi' = R\psi$    |
| $R = R_+$ | $R = Q^H$           | $Z' = R^H Z R$     | $\psi' = R^H \psi$ |

Both have physical amplitude factor  $e^{-i\theta N/2}$  and spin generator  $N/2$ . Changing only the vector sandwich would reverse its rotation relative to the spin polarization.

For any invertible factor  $G$  acting as  $\psi' = G\psi$ , the definitions (8.3) and (8.6) give

$$\begin{aligned}\rho &= \|\psi\|^2, & \rho' &= \rho \langle G^H G \rangle_\psi, \\ \langle \varphi' | \psi' \rangle &= \text{sc}(\varphi^H G^H G \psi), \\ \langle Z \rangle_{\psi'} &= \frac{\langle G^H Z G \rangle_\psi}{\langle G^H G \rangle_\psi}.\end{aligned}$$

Here  $\varphi' = G\varphi$ , and  $\rho > 0$ . For a rotation  $G^H G = 1$ :

$$\boxed{\langle Z \rangle_{\psi'} = \langle G^H Z G \rangle_\psi, \quad \langle G Z G^H \rangle_{\psi'} = \langle Z \rangle_\psi.} \quad (\text{C.1})$$

The first identity measures fixed  $Z$  in the rotated state; the second rotates the observable with the state. Thus the opposite sandwich orders in (C.1) are required, not competing conventions. For  $G = L_\pm$ , the denominator is  $\langle L_\pm^2 \rangle_\psi$ : boosts generally change the local positive density and inner product.

**Fields and wave amplitudes under either notation.** Let  $G$  be the constant physical factor of determinant one, so  $X' = G X G^H$ . For notation  $T X T^H$  take  $G = T$ ; for  $T^H X T$  take  $G = T^H$ . At corresponding events the laws are

$$\begin{aligned}Z' &= G Z G^H, & Z &= X, P, \Phi, D, \\ F' &= G F G^{-1}, & \partial' &= G^* \partial G^{-1}, \\ \Psi'_+ &= G^* \Psi_+, & \Psi'_- &= G \Psi_-, \\ \psi'_{\text{KG}}(X') &= \psi_{\text{KG}}(X).\end{aligned} \quad (\text{C.2})$$

These are the covariance laws of Section 11;  $G^* = (G^{-1})^H$ . The field  $F$  uses similarity, not the four-vector sandwich. Equation (C.2) preserves the Maxwell, Klein–Gordon, and Dirac equations.

For either rotation description, both Dirac entries use  $R_-$ . For  $G = L_\pm$ , they use  $L_\pm^{-1}$  and  $L_\pm$ , respectively; the factor already contains half the rapidity. With  $X' = T^H X T$ , the same laws read  $\Psi'_+ = T^{-1} \Psi_+$  and  $\Psi'_- = T^H \Psi_-$ . The scalar paired quantity  $\text{sc}(\Psi_+^H \Psi_-)$  is invariant; the sum of the two positive densities generally changes under boosts. The conserved Dirac current transforms as a four-vector; its time component alone is not invariant. For products, the physical factors compose as  $G_2 G_1$ . Hence the right-adjoint parameters compose as  $T_2 T_1$ , but the left-adjoint parameters compose as  $T_1 T_2$ .

**Preserved quantities.** The sign of the angle or rapidity changes the direction, not these invariance statements. Here  $Z$  is a complex paravector and  $\|Z\|^2 = \text{sc}(Z^H Z)$ ; “no” means not generally preserved.

| Action                      | $\det Z$ | $\text{tr } Z, \text{sc}(Z)$ | $\ Z\ ^2$         |
|-----------------------------|----------|------------------------------|-------------------|
| Rotation $RZR^H$            | Yes      | Yes                          | Yes               |
| Boost $LZL^H$               | Yes      | No                           | No                |
| Field similarity $GZG^{-1}$ | Yes      | Yes                          | If $G$ is unitary |

For real four-vectors transformed together, both rotations and boosts preserve

$$\text{sc}(X'^* Y') = \text{sc}(X^* Y).$$

Similarity also preserves  $\text{sc}(ZY)$  when both factors transform by similarity. The positive amplitude inner product is rotation invariant; boost invariance instead applies to the spacetime metric and the paired Dirac quantity above. Normalized expectations remain equal for a state and observable rotated together, not for an arbitrary fixed observable.

The main convention  $R = Q = R_-$ ,  $X' = R X R^H$ ,  $\psi' = R \psi$  uses the Hamilton quaternion directly. The alternative  $R = Q^H = R_+$ ,  $X' = R^H X R$ ,  $\psi' = R^H \psi$  is equally consistent. Either can use  $L^H X L = L X L^H$  with the boost sign fixed by the chosen frame velocity.

## References

- [1] W. E. Baylis and G. Sobczyk, “Relativity in Clifford’s geometric algebras of space and spacetime,” *Int. J. Theor. Phys.*, vol. 43, pp. 2061–2079, 2004, doi: [10.1023/B:IJTP.0000049010.53558.b7](https://doi.org/10.1023/B:IJTP.0000049010.53558.b7). [Online]. Available: <https://arxiv.org/pdf/math-ph/0405026>.
- [2] J. M. Chappell, A. Iqbal, L. J. Gunn, and D. Abbott, “Functions of multivector variables,” *PLoS ONE*, vol. 10, no. 3, e0116943, 2015, doi: [10.1371/journal.pone.0116943](https://doi.org/10.1371/journal.pone.0116943). [Online]. Available: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0116943>.
- [3] W. E. Baylis, “Applications of Clifford algebras in physics,” in *Lectures on Clifford (Geometric) Algebras and Applications*, R. Ablamowicz and G. Sobczyk, Eds. Boston, MA, USA: Birkhäuser, 2004, pp. 91–133, doi: [10.1007/978-0-8176-8190-6\\_4](https://doi.org/10.1007/978-0-8176-8190-6_4). [Online]. Published volume: <https://ndl.ethernet.edu.et/bitstream/123456789/32989/1/3.pdf.pdf>.
- [4] N. J. Higham, “Functions of matrices,” University of Manchester, MIMS EPrint 2005.21, Jun. 2006. [Online]. Available: [https://eprints.maths.manchester.ac.uk/310/1/fm\\_final.pdf](https://eprints.maths.manchester.ac.uk/310/1/fm_final.pdf).
- [5] W. E. Baylis, “Relativity in introductory physics,” *Can. J. Phys.*, vol. 82, no. 11, pp. 853–873, 2004, doi: [10.1139/p04-058](https://doi.org/10.1139/p04-058). [Online]. Available: <https://arxiv.org/pdf/physics/0406158>.
- [6] P. Lounesto, “Introduction to Clifford algebras,” in *Lectures on Clifford (Geometric) Algebras and Applications*, R. Ablamowicz and G. Sobczyk, Eds. Boston, MA, USA: Birkhäuser, 2004, pp. 1–29, doi: [10.1007/978-0-8176-8190-6\\_1](https://doi.org/10.1007/978-0-8176-8190-6_1). [Online]. Published volume: <https://ndl.ethernet.edu.et/bitstream/123456789/32989/1/3.pdf.pdf>.
- [7] D. Tong, *Electromagnetism*, University of Cambridge, Cambridge, U.K., lecture notes, 2015. [Online]. Available: <https://davidtong.org/pdfs/teaching/electromagnetism/electro.pdf>.
- [8] D. Hestenes, “Spacetime physics with geometric algebra,” *Am. J. Phys.*, vol. 71, no. 7, pp. 691–714, 2003, doi: [10.1119/1.1571836](https://doi.org/10.1119/1.1571836). [Online]. Available: <https://davidhestenes.net/geocalc/pdf/SpacetimePhysics.pdf>.
- [9] D. Tong, *Dynamics and Relativity*, University of Cambridge, Cambridge, U.K., lecture notes, 2013. [Online]. Available: <https://davidtong.org/pdfs/teaching/dynamics-and-relativity/dynrel.pdf>.
- [10] W. E. Baylis, “Surprising symmetries in relativistic charge dynamics,” arXiv:physics/0410197, 2004. [Online]. Available: <https://arxiv.org/pdf/physics/0410197>.
- [11] W. E. Baylis, R. Cabrera, and D. Keselica, “Quantum/Classical interface: Fermion spin,” arXiv:0710.3144, 2007. [Online]. Available: <https://arxiv.org/pdf/0710.3144>.
- [12] R. Gurtler and D. Hestenes, “Consistency in the formulation of the Dirac, Pauli, and Schrödinger theories,” *J. Math. Phys.*, vol. 16, no. 3, pp. 573–584, 1975, doi: [10.1063/1.522555](https://doi.org/10.1063/1.522555). [Online]. Available: <https://davidhestenes.net/geocalc/pdf/Consistency.pdf>.
- [13] J. Sparks, *Quantum Theory*, University of Oxford, Oxford, U.K., lecture notes, 2020. [Online]. Available: <https://users.ox.ac.uk/~math0391/QT2020.pdf>.

- [14] L. de Broglie, “The wave nature of the electron,” Nobel Lecture, Dec. 12, 1929, in *Nobel Lectures, Physics 1922–1941*. Amsterdam, The Netherlands: Elsevier, 1965, pp. 244–256. [Online]. Available: <https://www.nobelprize.org/uploads/2016/04/broglie-lecture.pdf>.
- [15] R. A. Millikan, “The electron and the light-quant from the experimental point of view,” Nobel Lecture, May 23, 1924, in *Nobel Lectures, Physics 1922–1941*. Amsterdam, The Netherlands: Elsevier, 1965. [Online]. Available: <https://www.nobelprize.org/uploads/2018/06/millikan-lecture.pdf>.
- [16] D. Tong, *Quantum Field Theory*, University of Cambridge, Cambridge, U.K., lecture notes, 2006–2007. [Online]. Available: <https://davidtong.org/pdfs/teaching/quantum-field-theory/qft.pdf>.
- [17] A. Lenz, *Theoretical Physics 3: Quantum Mechanics (PHYS3661)*, IPPP, Durham University, lecture notes, Apr. 8, 2015. [Online]. Available: [https://tp.nt.uni-siegen.de/wp-content/uploads/2024/02/TP3-Notes\\_Lenz\\_2015.pdf](https://tp.nt.uni-siegen.de/wp-content/uploads/2024/02/TP3-Notes_Lenz_2015.pdf).
- [18] R. P. Feynman and M. Gell-Mann, “Theory of the Fermi interaction,” *Phys. Rev.*, vol. 109, no. 1, pp. 193–198, 1958, doi: [10.1103/PhysRev.109.193](https://doi.org/10.1103/PhysRev.109.193). [Online]. Available: <https://harvest.aps.org/v2/journals/articles/10.1103/PhysRev.109.193/fulltext>.
- [19] R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, vol. III. Reading, MA, USA: Addison-Wesley, 1965, ch. 21, “The Schrödinger equation in a classical context: A seminar on superconductivity.” [Online]. Available: [https://www.feynmanlectures.caltech.edu/III\\_21.html](https://www.feynmanlectures.caltech.edu/III_21.html).
- [20] R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, vol. III. Reading, MA, USA: Addison-Wesley, 1965, ch. 6, “Spin one-half.” [Online]. Available: [https://www.feynmanlectures.caltech.edu/III\\_06.html](https://www.feynmanlectures.caltech.edu/III_06.html).
- [21] R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, vol. I. Reading, MA, USA: Addison-Wesley, 1963, ch. 17, “Space-time.” [Online]. Available: [https://www.feynmanlectures.caltech.edu/I\\_17.html](https://www.feynmanlectures.caltech.edu/I_17.html).